124.1 Actual Wilson conventions and a weighted D8 contact

Use AI1–8 without changing the measure:

$$S_0(U)=\sum_p\{N-\operatorname{Re}\operatorname{tr}P_p\},\qquad d\mu=Z^{-1}e^{-\beta S_0}\prod_b dU_b,\qquad J_i=a^{-3}X_iS_0,\quad \kappa=(\beta a^3)^{-1}. \tag{CP1}$$

Here $i=(b,A)$ includes the bond and color, and X_i is the actual invariant derivative in AI3. Define

$$\mathscr{E}[V]=a^4\sum_i\{J_iV_i-\kappa X_iV_i\}. \tag{CP2}$$

The exact identity against an arbitrary smooth test G is

$$\int\mathscr{E}[V]\,G\,d\mu=\kappa a^4\sum_i\int V_iX_iG\,d\mu. \tag{CP3}$$

Every vector and scalar below is globally smooth on the compact configuration space. External test values have no field derivative.

Let $f_i=f_b$, $h_j=h_c$ be scalar test values independent of the color index, and put

$$P_h=a^4\sum_j h_jJ_j^2,\qquad C_{f,h}=\kappa a^4\sum_i f_iJ_iX_iP_h. \tag{CP4}$$

This is the actual contact from multiplying the EOM insertion with coefficient fJ by P_h. It is the D8 family tested in C7.28b. No continuum replacement of the current is made.

124.2 An exact two-stage Ward reduction

Set

$$R_j=h_j\sum_i f_iX_i^2J_j,\qquad V_i^{f,h}=\kappa f_iX_iP_h+2\kappa^2a^4R_i. \tag{CP5}$$

Then the following equality is pointwise on the original compact configuration space:

$$\begin{aligned}C_{f,h}=\mathscr{E}[V^{f,h}]&+2\beta^{-2}a^{-4}\sum_{i,j}f_ih_j(X_iX_jS_0)^2\\&+2\beta^{-3}a^{-4}\sum_{i,j}f_ih_jX_jX_i^2X_jS_0.\end{aligned} \tag{CP6}$$

All sums keep their original indices and shifts. In particular $X_jX_i^2X_j$ is an ordered derivative, not a commutative fourth derivative.

To prove the identity, apply CP2 with coefficient $\kappa f_iX_iP_h$:

$$C_{f,h}=\mathscr{E}[\kappa f\,XP_h]+\kappa^2a^4\sum_i f_iX_i^2P_h. \tag{CP7}$$

The exact ordinary product rule gives

$$X_i^2P_h=2a^4\sum_j h_j\{(X_iJ_j)^2+J_jX_i^2J_j\}. \tag{CP8}$$

Apply CP2 a second time to the term containing J_jR_j. The resulting scalar term is $2\kappa^3a^8\sum_j X_jR_j$. Substitution of $J_j = a^{-3}X_jS_0$ and $\kappa = (\beta a^3)^{-1}$ gives the two powers in CP6 exactly.

For example, with constant $f = h = 1$, the full Haar Casimir $\Delta_H = \sum_i X_i^2$ obeys

$$\Delta_H S_0 = 4C_F \sum_p \operatorname{Re}\operatorname{tr} P_p, \qquad \Delta_H J_j = -4C_F J_j. \tag{CP9}$$

The first formula follows because each plaquette has four links. The second uses that the bi-invariant Casimir commutes with each invariant derivative. It is not needed for nonconstant tests. For constant tests CP6 becomes

$$C_{1,1} = \mathscr{E}[\kappa X P_1 - 8C_F\kappa^2 a^4 J] + 2\beta^{-2}a^{-4}\sum_{i,j}(X_iX_jS_0)^2 - 32C_F^2\beta^{-3}a^{-4}\sum_p \operatorname{Re}\operatorname{tr} P_p. \tag{CP10}$$

Thus one scalar term is explicitly an identity-plus-Wilson-action counterterm. The other is a finite local polynomial of link matrices and their inverses.

124.2.1 Locality, invariance, and exact component counts

For fixed bonds b, c, the color sum

$$F_{bc}(U) = \sum_{A,B}(X_b^A X_c^B S_0)^2, \qquad G_{bc}(U) = \sum_{A,B} X_c^B (X_b^A)^2 X_c^B S_0 \tag{CP11}$$

is gauge invariant. Under a fixed gauge transformation each derivative color is rotated by its actual adjoint endpoint matrix; the repeated color contractions preserve the expression. All derivatives have been taken before any source-coordinate description below is introduced.

Both expressions vanish unless the two bonds belong to a common plaquette, apart from repetitions of the same bond already in that statement. Their full determining sets are contained in a fixed union of plaquette stars. The number of such pairs per bond and their word lengths depend on the gauge group and the fixed lattice dimension, not on the lattice spacing or volume. Consequently CP6 can be written using one actual finite incidence list $\mathcal{I}$ of bond pairs:

$$C_{f,h} = \mathscr{E}[V^{f,h}] + 2a^{-4}\sum_{(b,c)\in\mathcal{I}} f_b h_c\{\beta^{-2}F_{bc} + \beta^{-3}G_{bc}\}. \tag{CP12}$$

Duplicated pairs in a chosen plaquette decomposition retain their literal multiplicities. Neither a microscopic sum nor the two different test anchors has been replaced by a current-scale count.

124.3 A globally smooth finite-jet projection on the local loop variables

The next construction is a specified projection on local Wilson functions. It is not an assertion that loop degree equals engineering dimension.

For each pair in CP12 choose a fixed complete contractible cubical carrier Ω_{bc}, containing the determining plaquettes and their links, and one geometric spanning tree. Its number of fine cells is bounded independently of a. For sufficiently fine meshes this carrier is nonwrapping. A wrapping carrier must be treated with its actual holonomy record; the plaquette estimate below is not asserted for a flat noncontractible loop.

Gauge fixing only this finite tree expresses an invariant function F on the carrier as

$$F(U) = \widetilde{F}(W_1(U), \dots, W_s(U)), \tag{CP13}$$

where the W_α are the based fundamental-cycle holonomies. This is a pointwise identity of observables. There is no change of integration variables or Haar measure in CP13. Tree products and inverses are globally smooth, and the only remaining gauge action is simultaneous conjugation at the tree root.

The number s, the cycle lengths, and all word degrees of $\widetilde{F}_{bc}, \widetilde{G}_{bc}$ are bounded constants. For a fixed fundamental loop in this contractible carrier, successively removing plaquettes from a spanning surface gives the exact product formula by transported plaquettes. Since transports are unitary on the real group, telescoping the product gives

$$\max_\alpha \|W_\alpha - I\| \le C_\Omega \sum_{p\subset\Omega} \|P_p - I\|. \tag{CP14}$$

The same estimate on the stated small complex neighborhood has the bounded transport factors retained.

Let

$$L_3(Z) = Z - \tfrac{1}{2}Z^2 + \tfrac{1}{3}Z^3,$$

and, for $W \in \mathrm{SU}(N)$, define a globally smooth Lie-algebra-valued coordinate

$$\widehat{\xi}(W) = \pi_{\mathfrak{su}(N)} L_3(W - I). \tag{CP15}$$

On the complexified group the projection is written algebraically using

$$\tfrac{1}{2}\{L_3(W - I) - L_3(W^{-1} - I)\}$$

and removing its trace. This is the holomorphic extension of CP15. In exponential coordinates,

$$\widehat{\xi}(e^\xi) = \xi + O(\|\xi\|^4). \tag{CP16}$$

Only agreement through degree three is used.

Write $j_{\le 3}\widetilde{F}$ for the Taylor polynomial through total degree three of $\widetilde{F}(e^{\xi_1}, \dots, e^{\xi_s})$ at zero. Define

$$\begin{aligned} (\Pi_{\le 3}\widetilde{F})(W) &= (j_{\le 3}\widetilde{F})(\widehat{\xi}(W_1), \dots, \widehat{\xi}(W_s)), \\ \mathcal{R}_4\widetilde{F} &= \widetilde{F} - \Pi_{\le 3}\widetilde{F}. \end{aligned} \tag{CP17}$$

This is an actual projection:

$$\Pi_{\le 3}^2 = \Pi_{\le 3}. \tag{CP18}$$

Indeed CP16 implies $j_{\le 3}\Pi_{\le 3}\widetilde{F} = j_{\le 3}\widetilde{F}$. This avoids a representation-dependent truncation of a word in separately treated $W - I$ and $W^{-1} - I$.

The projection is gauge invariant when the original function is: its Taylor tensors are invariant under simultaneous adjoint action and $\widehat{\xi}$ transforms by conjugation. Its range is finite dimensional for this fixed carrier and degree. Its coefficients are the actual derivatives of the finite Wilson words in CP11, not unspecified normalization parameters.

The construction can preserve the original finite lattice symmetries. Regard the third jet intrinsically as the quotient of local invariant functions by those vanishing to order four at the flat local gauge orbit. Changing the finite tree or root changes coordinates analytically and preserves that ideal. Each tree extension is therefore a linear right inverse of the same jet map. Average these right inverses over the finite geometric symmetry orbit of the complete carrier, including reflection, or over all its trees and roots. If j is that jet map and $\overline{R}$ the averaged extension, then

$$j\overline{R} = I, \qquad \overline{\Pi} = \overline{R}j, \qquad \overline{\Pi}^2 = \overline{\Pi}. \tag{CP18a}$$

Thus averaging does preserve idempotence here; it is not an arbitrary average of unrelated projections. Use this equivariant projection for the full source family. Its geometric multiplicity is a fixed finite constant. No gauge, hypercubic or reflection symmetry of the complete pointwise identity is lost.

Taylor's theorem on a fixed complex identity neighborhood and compactness away from that neighborhood give, on the real compact group,

$$\begin{aligned} |\mathcal{R}_4\widetilde{F}(W)| &\le C_F\big(\max_\alpha \|W_\alpha - I\|\big)^4, \\ \|\Pi_{\le 3}\widetilde{F}\|_\infty &\le C \max_{0\le r\le 3} \|D^r(\widetilde{F}\circ\exp)(0)\|. \end{aligned} \tag{CP19}$$

For the finite set of functions CP11, all constants are uniform in a and volume. The remainder estimate and fixed-order analytic derivatives also hold on a smaller common complex loop neighborhood. In particular its derivative of order r has the corresponding vanishing order $4-r$ for $r \le 4$; no vanishing order is inferred for arbitrary fine-link derivatives.

The linear Taylor row vanishes for these invariant functions: an invariant linear form on a finite direct sum of $\mathfrak{su}(N)$ is zero. The constant, quadratic, and cubic rows are retained individually. A finite difference among their shifted loop variables can encode higher engineering derivatives. Consequently they are called the **low-loop-degree contact sector**, not a proved D0/D4/D6 normalization block.

124.4 Exact D8 projection and its quantitative scalar remainder

Apply CP17 separately to every original local word in CP12. Define

$$\begin{aligned} L_{f,h} &= 2a^{-4} \sum_{(b,c)\in\mathcal{I}} f_b h_c \{\beta^{-2}\Pi_{\le 3}F_{bc} + \beta^{-3}\Pi_{\le 3}G_{bc}\}, \\ R^{(8)}_{f,h} &= 2a^{-4} \sum_{(b,c)\in\mathcal{I}} f_b h_c \{\beta^{-2}\mathcal{R}_4 F_{bc} + \beta^{-3}\mathcal{R}_4 G_{bc}\}. \end{aligned} \tag{CP20}$$

The exact identity is

$$C_{f,h} = \mathscr{E}[V^{f,h}] + L_{f,h} + R^{(8)}_{f,h}. \tag{CP21}$$

The source tests keep their original different anchors. All finite offsets, powers of a, powers of β, and incidence multiplicities in $L_{f,h}$ remain part of the state.

Let

$$M_\Omega(U) = \max_{p\subset\Omega} a^{-2}\{\|P_p - I\| + \|P_p^{-1} - I\|\}. \tag{CP22}$$

On an actual physical core with this bound, CP14 and CP19 give

$$|R^{(8)}_{f,h}(U)| \le C(\beta^{-2}+\beta^{-3})a^4 \sum_{(b,c)\in\mathcal{I}} |f_b h_c|\, M_{\Omega_{bc}}(U)^4. \tag{CP23}$$

The factors are explicit: a^{-4} from CP20 and $a^8 M_\Omega^4$ from the fourth loop-defect order leave the physical factor a^4 in this sum.

Suppose the actual source carrier has physical volume at most $C\ell^4$, the tests have bounded supremum, and its admitted physical curvature bound is

$$M_\Omega \le \epsilon_\ell \ell^{-2}. \tag{CP24}$$

The literal fine-pair count in CP23 then proves

$$a^4|R^{(8)}_{f,h}| \le C(\beta^{-2}+\beta^{-3})(a/\ell)^4\epsilon_\ell^4\|f\|_\infty\|h\|_\infty. \tag{CP25}$$

The factor a^4 on the left is the own-word-mesh dimension-eight normalization for an integrated insertion. In CP1 this is the original bare mesh. It equals the birth scale of this specified bare contact, but it is not an arbitrary later RG birth length. For $\ell = L^n a$, its gain is L^{-4n}. This is a genuine model-specific regular-core estimate of the projected scalar remainder; it is not obtained by increasing the derivative order in C7.16.

For two canonically normalized dimension-six current-square labels at this specified Wilson scale, the incoming factor is g_{W}^{-4}, where g_{W} is the coupling in CP1, hence the bare coupling in this application. With the original convention $\beta = c_\beta g_{\mathrm{W}}^{-2}$, CP25 retains

$$g_{\mathrm{W}}^{-4}(\beta^{-2} + \beta^{-3}) = c_\beta^{-2} + c_\beta^{-3} g_{\mathrm{W}}^2. \tag{CP26}$$

On the canonical physical core $\epsilon_\ell = C g_\ell P_\ell$, the remaining factor is $(g_\ell P_\ell)^4$. The original coupling and any field-independent old prefactor have not been restarted. Applying CP6 to an effective coarse density would require its full effective current, including inherited interactions and normalization terms. Its Wilson term alone does not give CP3 for that different density.

There is also a real compact energy envelope. For a contractible carrier put $E_\Omega = \beta \sum_{p\subset\Omega} s(P_p)$. Using the exact relation $\|P - I\|_{\mathrm{HS}}^2 = 2s(P)$, CP14 and CP19 yield

$$|\mathcal{R}_4 F_{bc}| + |\mathcal{R}_4 G_{bc}| \le C\beta^{-2} E_{\Omega_{bc}}^2. \tag{CP27}$$

Thus a single coefficient in CP20 has its explicit $a^{-4}(\beta^{-4} + \beta^{-5})E_\Omega^2$ envelope. The original joint positive reserve can pay this fixed polynomial. This statement supplies no inverse reference mass and does not spend a separate reserve for each source label.

124.4.1 What CP25 does not establish

The estimate does not bound the whole original D8 contact by L^{-4n}. The current $\mathscr{E}[V^{f,h}]$ and the exact low-loop-degree scalar $L_{f,h}$ remain. Removing either without its contact and normalization companions changes the original finite-source coefficient.

The moment and symmetry decomposition needed to put $L_{f,h}$ into a closed engineering-grade center is not proved by CP18. In particular a quadratic point-split loop word with coefficient a^{-4} cannot be replaced by one onsite action-density direction with the same coefficient without bounding the exact difference.

For a previously generated coefficient $A(U)C_{f,h}$, the Ward steps also differentiate A. Their terms retain the original saved scale gain only if the required jets of A are controlled on the actual common domain. CP6 does not prove those incoming LM47 estimates.

124.4.2 The physical Ward quotient for this complete contact packet

There is an exact physical projection statement without an inverse Gram matrix for the EOM direction. Suppose the full determining carrier of the projected packet $C_{f,h}, V^{f,h}, L_{f,h}, R_{f,h}^{(8)}$, including every plaquette star, chosen tree/root carrier and symmetry-averaged enlargement, lies strictly in the positive half of a reflection-positive Wilson lattice. Let $\mathcal{A}_+$ be its globally smooth positive-time observable algebra, and let Θ include the actual reflected tensor and tangent conventions. For every $G \in \mathcal{A}_+$, CP3 gives

$$\int \Theta \mathscr{E}[V^{f,h}]\, G\, d\mu = 0. \tag{CP27a}$$

The derivative anchors in the reflected EOM insertion are negative-time bonds; they do not occur in G. The coefficient's dependence on other negative-time bonds stays inside the complete adjoint. In particular the EOM insertion has zero reflected norm. Therefore the exact physical quotient identity is

$$[C_{f,h} - L_{f,h}]_{\mathrm{OS}} = [R^{(8)}_{f,h}]_{\mathrm{OS}}. \tag{CP27b}$$

This removes the EOM direction for this whole contact packet with equality of physical pairings. No nondegeneracy claim for an enlarged operator Gram matrix is needed to establish this equality.

It is a linear physical quotient, not a multiplicative ideal. For a further positive-time observable P, the original exact rule is

$$P\mathscr{E}[V] = \mathscr{E}[PV] + \kappa a^4 \sum_i V_i X_i P. \tag{CP27c}$$

The final term may be nonzero and must be included in the next source coefficient. Thus CP27b applies after forming the specified whole finite-source contact packet. It does not license deleting that packet's current record before a later join.

Nor does CP27b prove a cutoff-uniform physical norm for the quartic remainder in the full quantum measure: CP23–25 are physical-background estimates, and CP27 is a compact energy envelope. The completed quantum and mixed-history transition estimate remains needed.

124.4.3 A weak analytic extension for this explicit quartic remainder

The scalar remainder also supplies a new admitted weak source root. Work on a fixed physical box in one weak potential chart $U = e^{aA}$, with $\|A\|_{W^{1,p}} \le \epsilon$, $p > 8$, and include the complete local carriers used above. WH2–4's exact plaquette-word estimate gives

$$\|\mathcal{Z}\|_{L^p} \le C\epsilon, \qquad \mathcal{Z}_p = a^{-2}(P_p^{\pm 1} - I). \tag{CP27d}$$

The potential supremum controls the finite transport products. In particular a fundamental loop of a fixed microscopic carrier lies in the common complex identity neighborhood: its elementary link-word bound is $\|W - I\| \le Ca\|A\|_\infty$. This does not require a curvature supremum.

Use CP14 before taking the fourth power, and sum the finite carrier overlaps instead of using a pointwise maximum over the whole physical box. CP19 then gives

$$|R^{(8)}_{f,h}(e^{aA})| \le C(\beta^{-2} + \beta^{-3})\|f\|_\infty \|h\|_\infty \|\mathcal{Z}\|^4_{L^4} \le C(\beta^{-2} + \beta^{-3})\|f\|_\infty \|h\|_\infty \epsilon^4. \tag{CP27e}$$

The inclusion $L^p \subset L^4$ uses the fixed physical volume, and the local overlap constant is independent of the fine mesh. The complete scalar remainder is holomorphic on this weak ball, so its fixed-order weak Banach jets have uniform Cauchy bounds on a smaller ball. Its Taylor coefficients at the flat reference start at field degree four.

Thus this particular contact projection produces an actual quartic weak source root. It does not assume that every generated D8 coefficient has such a root. At a nonflat reference the bound retains its full reference-plus-variation strength. Canonical scaling requires all of that strength to be scaled accordingly, as in WH; CP27e does not reset an old source coupling or radius.

124.6 The complete flat constrained Gaussian calculation

The free calculation shows exactly what fluctuation completion does to the same phenomenon. Work on a finite physical transverse or gauge-fixed slice where the flat quadratic operator $H > 0$, with the original physical inner products used for every transpose. Constant/harmonic variables are fixed by their actual constraints or retained separately; no massless inverse is asserted on its kernel.

Let the actual linear coarse average be Q. On its admitted row space set

$$\Delta_c = (QH^{-1}Q^{\mathsf{t}})^{-1}, \qquad K = H^{-1}Q^{\mathsf{t}}\Delta_c, \qquad C = H^{-1} - H^{-1}Q^{\mathsf{t}}\Delta_c QH^{-1}. \tag{CP33}$$

For the Gaussian Wilson quadratic measure with action $\beta\langle A, HA\rangle/2$, conditioning on $QA = B$ gives mean KB, covariance $\beta^{-1}C$, and the original coarse density proportional to $\exp[-\beta\langle B, \Delta_c B\rangle/2]$. All finite-dimensional Gaussian determinants remain in its source-zero normalization.

The flat D8 contact is

$$C_8^0(A) = 2\beta^{-1}\langle A, H^3 A\rangle. \tag{CP34}$$

Keep its explicit vacuum companion $2\beta^{-2}\operatorname{Tr} H^2$, and write $W_8^0 = C_8^0 - 2\beta^{-2}\operatorname{Tr} H^2$. With

$$R_a = QHQ^{\mathsf{t}}, \tag{CP35}$$

the conditional expectation is exactly

$$\mathbb{E}[W_8^0 \mid QA = B] = 2\beta^{-1}\left\{\langle \Delta_c B, R_a \Delta_c B\rangle - \beta^{-1}\operatorname{Tr}(R_a\Delta_c)\right\}. \tag{CP36}$$

Indeed $HK = Q^{\mathsf{t}}\Delta_c$, so the mean term is the first term above. The covariance trace and the vacuum companion differ by $-\beta^{-1}\operatorname{Tr}(R_a\Delta_c)$, by cyclicity of the finite-dimensional trace. This uses the full conditional covariance, not a covariance with the coarse constraint omitted.

Let $\rho_c(B)$ be the actual coarse Gaussian density and define its weighted adjoint by

$$\mathsf{D}^\dagger_{\rho_c} v = \beta\langle \Delta_c B, v(B)\rangle - \operatorname{div} v(B). \tag{CP37}$$

Then CP36 is

$$\mathbb{E}[W_8^0 \mid QA = B] = \mathsf{D}^\dagger_{\rho_c}(2\beta^{-2} R_a \Delta_c B). \tag{CP38}$$

The scalar trace in CP36 is precisely the derivative of this coarse vector coefficient. It cannot be dropped while keeping the quadratic term.

For the actual transverse block average used in CP28, $Q^{\mathsf{t}}\lambda$ is the step field j_a. Consequently

$$\langle \lambda, R_a\lambda\rangle = \langle Q^{\mathsf{t}}\lambda, HQ^{\mathsf{t}}\lambda\rangle = 4\epsilon^2 T^4/a \tag{CP39}$$

with the physical conventions of CP28. The coefficient matrix in CP38 therefore has an interface divergence. Full Gaussian fluctuation completion retains a whole coarse Ward current with that coefficient; it does not make each scalar or vector part uniformly bounded. Its expectation against a constant vanishes exactly, and its products create the original derivative contacts.

This identifies the required cancellation more precisely than a generic higher-jet bound. A physical quotient may use the complete Ward relation. An operation which deletes the current coefficient but keeps arbitrary products loses its contact terms.

124.8 A finite-family extension and its precise limit

There is an algebraic extension of CP6. In any finite compact Wilson contact word, choose an undifferentiated current factor J_i. The component identity

$$J_iA = \{J_iA - \kappa X_iA\} + \kappa X_iA \tag{CP42}$$

replaces it by a complete EOM component and a derivative of the remaining coefficient. If the derivative hits another current, $X_iJ_j = a^{-3}X_iX_jS_0$ is a bounded-length compact Wilson word with its factor a^{-3} retained. If it hits another Wilson word, its derivative is again a finite Wilson word. Applying this operation recursively terminates after a finite number of undifferentiated currents has been removed. The resulting list retains all ordered commutators, source anchors, powers of κ, and finite incidence multiplicities.

For a fixed finite word family, the same local loop construction gives a genuine smooth jet projection $\Pi_{<q}$ at any fixed degree q. One replaces CP15 by the finite logarithm Taylor polynomial through degree $q-1$, followed by the same algebraic Lie-algebra projection. Its exponential jet agrees with the identity to the required order. The remainder has a bound $C\max\|W_\alpha - I\|^q$.

Consequently, for a scalar family written with its actual prefactor as

$$a^{-\Delta}\,a^4\sum_x F_x(U), \tag{CP43}$$

the remainder after $\Pi_{<q}$, on a carrier of physical volume $O(\ell^4)$ satisfying CP24, obeys

$$a^{\Delta-4}\left|a^{-\Delta}a^4\sum_x \mathcal{R}_qF_x(U)\right| \le C(a/\ell)^{\Delta-4}(a/\ell)^{2q-\Delta}\epsilon_\ell^q. \tag{CP44}$$

Here $2q \ge \Delta$. The first factor is the engineering conversion and the second is nonincreasing. The total power is $2q-4$, so a strict decay statement requires $q>2$; the marginal case $\Delta=4, q=2$ has no decay. The normalization is at the actual word mesh a, and original coupling coefficients multiply both sides unchanged. CP25 is the case $\Delta=8, q=4$.

This theorem concerns specified finite compact word families. Previously generated effective coefficients may be nonpolynomial functions on shrinking source domains with retained gauge and reference dependence. Applying a current identity to such a coefficient requires its actual derivatives. Applying the local jet projection requires its actual local representation and domain. Their uniform constants and saved scale gains are not consequences of the finite-family algebraic statement.

125.1 Group normalization, lattice geometry, and the unchanged measure

Use a real basis of anti-Hermitian traceless matrices satisfying

$$-\operatorname{tr}(T_A T_B) = \delta_{AB}, \qquad d_G = N^2 - 1, \qquad \sum_A T_A^2 = -C_F I, \qquad C_F = \frac{N^2-1}{N}, \qquad C_A = 2N. \tag{CQ1}$$

With $[T_A, T_B] = c_{AB}{}^C T_C$, the second Casimir convention is $\sum_{A,B} c_{AB}{}^C c_{AB}{}^D = C_A \delta^{CD}$. Both constants follow from the completeness identity

$$\sum_A (T_A)_{ij}(T_A)_{kl} = -\delta_{il}\delta_{jk} + N^{-1}\delta_{ij}\delta_{kl}. \tag{CQ2}$$

In particular, the formulas below use neither the convention $-\operatorname{tr}(T_A T_B) = \delta_{AB}/2$ nor its Casimir constants.

The four-dimensional periodic lattice has mesh a, one independent positively oriented bond per site and direction, and one elementary plaquette for each site and unordered pair of directions. Assume that each periodic circumference contains at least three lattice spacings. Then each independent bond occurs exactly once in each incident plaquette, each bond has

$$z = 2(4-1) = 6 \tag{CQ3}$$

incident plaquettes, and two distinct independent bonds belong to at most one common elementary plaquette. These facts follow by listing the four edges of the elementary square; a second common square would identify opposite transverse translates, which requires a periodic direction of length one or two. The subsequent single-plaquette off-diagonal formula uses this restriction. Such tiny periodic lattices are not covered by that formula; their repeated incidence terms must instead be retained in CP6. The stated restriction holds eventually along any fixed positive physical torus in the ultraviolet limit.

Write

$$\begin{aligned} s(P) &= N - \operatorname{Re}\operatorname{tr} P, \qquad S_0 = \sum_p s(P_p), \\ d\mu &= Z^{-1} e^{-\beta S_0} \prod_b dU_b, \qquad J_b^A = a^{-3} X_b^A S_0, \\ \kappa &= (\beta a^3)^{-1}, \qquad \mathscr{E}[V] = a^4 \sum_{b,A} \{ J_b^A V_b^A - \kappa X_b^A V_b^A \}. \end{aligned} \tag{CQ4}$$

The derivatives, Haar measure, and original constraint records are those of AI and CP. For color-independent, field-independent test values f_b, h_b, let

$$P_h = a^4 \sum_{b,A} h_b (J_b^A)^2, \qquad C_{f,h} = \kappa a^4 \sum_{b,A} f_b J_b^A X_b^A P_h. \tag{CQ5}$$

The tests may vary at every original bond. All identities are bilinear in them. The real case determines their complex extension. Color-dependent tests would prevent the complete color contractions below and require a different tensor calculation.

125.2 A single-plaquette matrix identity

For $P \in \mathrm{SU}(N)$, define

$$\begin{aligned} R(P) &= I - \frac{P + P^{-1}}{2} = \frac{1}{2}(I - P)(I - P^{-1}), \\ A(P) &= \frac{P - P^{-1}}{2}, \qquad s(P) = \operatorname{tr} R(P), \\ K(P)_{AB} &= -\frac{1}{2} \operatorname{tr}\big(\{T_A, T_B\} R(P) \big). \end{aligned} \tag{CQ6}$$

On the real compact group, R is Hermitian, A is anti-Hermitian, and K is real symmetric. The exact factorization shows that R, and hence K, is second order in the plaquette defect. The color trace is

$$\operatorname{tr}_{\mathfrak{g}} K(P) = C_F s(P). \tag{CQ7}$$

Let $g_A(P) = -\operatorname{Re}\operatorname{tr}(T_A P)$. Completeness applied to the traceless part of $A(P)$ gives

$$\begin{aligned}\sum_A g_A(P)^2 &= -\operatorname{tr} A(P)^2 + \frac{1}{N}\big(\operatorname{tr} A(P)\big)^2 \\ &= 2s(P) - \operatorname{tr} R(P)^2 + \frac{1}{N}\big(\operatorname{tr} A(P)\big)^2.\end{aligned} \tag{CQ8}$$

The second line uses $A^2 = -2R + R^2$. The last squared trace is the algebraic square, which is nonpositive on the real group, rather than its absolute square. The full right side is the squared norm of the traceless anti-Hermitian matrix and is nonnegative.

The matrix $-\operatorname{Re}\operatorname{tr}(T_A T_B P)$ has symmetric part $I - K(P)$. Its antisymmetric part is, up to the sign fixed by the invariant-frame convention, ${c_{AB}}^C g_C(P)/2$. Symmetric and antisymmetric matrices are orthogonal in the Frobenius pairing. Consequently

$$\sum_{A,B}\big(\operatorname{Re}\operatorname{tr}(T_A T_B P)\big)^2 = d_G + c_g s(P) + Q(P), \qquad c_g = -2C_F + \frac{C_A}{2}, \tag{CQ9}$$

where the remainder is the explicit class function

$$Q(P) = \operatorname{tr}_{\mathfrak{g}} K(P)^2 - \frac{C_A}{4}\operatorname{tr} R(P)^2 + \frac{C_A}{4N}\big(\operatorname{tr} A(P)\big)^2. \tag{CQ10}$$

It obeys $Q(P^{-1}) = Q(P)$. This inversion symmetry excludes an odd, in particular cubic, single-plaquette invariant. More quantitatively, on the whole real compact group,

$$|Q(P)| \le C_N \|P - I\|^4. \tag{CQ11}$$

For a direct proof near the identity write $P = e^{\xi}$ with $\operatorname{tr}\xi = 0$. Then $R = O(\xi^2)$, and $\operatorname{tr} A = \operatorname{tr}\sinh\xi = O(\xi^3)$. Thus the three terms in CQ10 have orders at least four, four, and six. Away from a fixed identity neighborhood compactness proves CQ11 with a possibly larger constant. This argument does not infer a source-momentum factor from a field-momentum factor.

The same formulas are holomorphic on the complex group when $\operatorname{Re}\operatorname{tr} P$ is continued as $\frac{1}{2}\operatorname{tr}(P + P^{-1})$. All squares in CQ6–10 are bilinear, without complex conjugation. On a common fixed small complex neighborhood of the identity, the same fourth-order bound holds with complex matrix norms. A common domain must actually be admitted before this estimate can be used in a later history.

125.3 Exact diagonal and off-diagonal Wilson Hessian blocks

For each plaquette incident to a bond b, choose the actual oriented and based plaquette word $P_{p,b}$ which begins with the independent forward bond b. If necessary reverse the plaquette and cyclically change its base point. Its real trace and its action are unchanged. Both its orientation and its prefix transports are retained in the following matrices:

$$K_b = \sum_{p\ni b} K(P_{p,b}), \qquad S_b = \sum_{p\ni b} s(P_p), \qquad g_b^A = X_b^A S_0 = a^3 J_b^A. \tag{CQ12}$$

These color matrices are all expressed in the invariant frame at the same bond. This is why the cross terms in K_b^2 have definite transport factors and cannot be replaced by products of unbased plaquette traces.

Let F_{bc} be CP11. For the same-bond block, its symmetric part is $zI - K_b$; its antisymmetric part is half the invariant-frame commutator applied to S_0. Hence

$$F_{bb} = z^2 d_G - 2zC_F S_b + \operatorname{tr}_{\mathfrak{g}} K_b^2 + \frac{C_A}{4}|g_b|^2. \tag{CQ13}$$

This is an exact identity, including the noncommutation of two same-bond derivatives.

If $b \neq c$ share a plaquette p, differentiation of its actual four-link word puts two color generators at the two specified locations. After both derivatives have been taken, move one generator to the other location by the intervening unitary prefix. Its color basis is rotated by an orthogonal adjoint matrix. Inverting a link can also change an overall sign. The sum over all colors in the squared Hessian is unchanged by these rotations and signs. The block therefore has exactly the squared norm in CQ9:

$$F_{bc} = d_G + c_g s(P_p) + Q(P_p) \quad (b \neq c,\ b, c \in \partial p). \tag{CQ14}$$

No derivative of a field-dependent adjoint rotation has been omitted: the rotation is an algebraic rewriting of the already differentiated Hessian, not a change of frame before differentiation. CQ3 ensures there is just one contributing plaquette. If there is no common plaquette, $F_{bc} = 0$.

For the ordered fourth-derivative term, put $\Delta_b = \sum_A (X_b^A)^2$. This Casimir commutes with all invariant derivatives on its own factor and with every derivative on a different factor. Thus, preserving the original ordered definition,

$$\begin{aligned} G_{bc} &= \sum_B X_c^B \Delta_b X_c^B S_0 = \Delta_b \Delta_c S_0 \\ &= -C_F^2 \sum_{p \ni b,c} \operatorname{Re}\operatorname{tr} P_p. \end{aligned} \tag{CQ15}$$

Each edge is used once in a plaquette and $\Delta_b \operatorname{Re}\operatorname{tr} P_p = -C_F \operatorname{Re}\operatorname{tr} P_p$ when $b \in p$. This proves CQ15 also for $b = c$; it does not replace four noncommuting derivatives by ordinary partial derivatives.

125.4 Preserve the two source anchors and evaluate the full low sector

Define the following exact incidence combinations:

$$\begin{aligned} w_b &= f_b h_b, \\ w_p &= \sum_{b \in \partial p} f_b h_b, \\ t_p &= \sum_{\substack{b,c \in \partial p \\ b \neq c}} f_b h_c, \\ u_p &= \left(\sum_{b \in \partial p} f_b \right) \left(\sum_{c \in \partial p} h_c \right) = w_p + t_p. \end{aligned} \tag{CQ16}$$

The sum defining t_p is over ordered distinct pairs of original bonds. In particular, no $f_b h_c$ has been replaced by a product at the plaquette center. Every summand may retain its original finite source labels, birth record, and paired-carrier record even when the formula is written as a sum over p.

Set

$$\begin{aligned} A_0 &= d_G\left(z^2\sum_b w_b+\sum_p t_p\right),\\ \Lambda_p &= -2zC_F w_p + c_g t_p,\\ \mathcal{Q}_{f,h} &= \sum_b w_b \operatorname{tr}_{\mathfrak{g}} K_b^2 + \sum_p t_p Q(P_p). \end{aligned} \tag{CQ17}$$

Equations CQ13–14 give the complete identity

$$\sum_{b,c} f_b h_c F_{bc} = A_0 + \sum_p \Lambda_p s(P_p) + \frac{C_A}{4}a^2 P_w + \mathcal{Q}_{f,h}. \tag{CQ18}$$

Here $P_w = a^4\sum_{b,A} w_b (J_b^A)^2$. Its further Ward completion is exact:

$$P_w = \mathscr{E}[wJ] + C_F\beta^{-1}a^{-2}\sum_p w_p \operatorname{Re}\operatorname{tr} P_p. \tag{CQ19}$$

Indeed the second term is $\kappa a^4\sum_{b,A} w_b X_b^A J_b^A = \beta^{-1}a^{-2}\sum_b w_b \Delta_b S_0$.

Let $V^{f,h}$ be the actual vector in CP5, and put

$$D_p = \frac{C_A}{2}w_p - 2C_F u_p, \qquad \widetilde{V}^{f,h} = V^{f,h} + \frac{C_A}{2}\beta^{-2}a^{-2}wJ. \tag{CQ20}$$

Substitution of CQ15, CQ18, and CQ19 into CP6 yields the promised complete reduction:

$$\begin{aligned} C_{f,h} = {}& \mathscr{E}[\widetilde{V}^{f,h}]\\ &+ 2\beta^{-2}a^{-4}\left(A_0+\sum_p \Lambda_p s(P_p) + \mathcal{Q}_{f,h}\right)\\ &+ C_F\beta^{-3}a^{-4}\sum_p D_p \operatorname{Re}\operatorname{tr} P_p. \end{aligned} \tag{CQ21}$$

Equivalently, define the explicit scalar and plaquette coefficients

$$\begin{aligned} k_0 &= a^{-4}\left(2\beta^{-2}A_0 + NC_F\beta^{-3}\sum_p D_p\right),\\ k_p &= a^{-4}\left(2\beta^{-2}\Lambda_p - C_F\beta^{-3}D_p\right),\\ \mathcal{R}^{(8)}_{f,h} &= 2\beta^{-2}a^{-4}\mathcal{Q}_{f,h}. \end{aligned} \tag{CQ22}$$

Then, pointwise for the same Wilson measure and before any conditioning,

$$\boxed{C_{f,h} = \mathscr{E}[\widetilde{V}^{f,h}] + k_0 + \sum_p k_p s(P_p) + \mathcal{R}^{(8)}_{f,h}.} \tag{CQ23}$$

This is stronger than identifying a third loop jet: the retained field row is the actual single-plaquette Wilson action with its exact source weight. The quartic remainder still includes the cross terms between different based plaquettes in K_b^2. Such terms are not set to zero by a continuum Bianchi argument.

For arbitrary nonconstant tests the original offsets remain in k_p. Expanding them in spatial differences of f, h, if ever required for a separate test-function norm, is another operation with its own exact remainder. CQ23 does not silently perform it.

125.6 A complete SU(2) constant-test check

For $N = 2$, $d_G = 3$, $C_F = 3/2$, $C_A = 4$, and $\{T_A, T_B\} = -\delta_{AB} I$. Hence

$$K(P) = \tfrac{1}{2} s(P) I_3, \qquad Q(P) = \tfrac{1}{4} s(P)^2. \tag{CQ30}$$

Take $f = h = 1$, and let N_b, N_p be the actual bond and plaquette counts. Then

$$\begin{gathered} w_p = 4, \quad t_p = 12, \quad u_p = 16, \quad A_0 = 108 N_b + 36 N_p, \quad \Lambda_p = -84, \quad D_p = -40, \\ \mathcal{Q}_{1,1} = \tfrac{3}{4} \sum_b S_b^2 + 3 \sum_p s(P_p)^2, \\ \widetilde{V}^{1,1} = \kappa X P_1 - 10 \beta^{-2} a^{-2} J. \end{gathered} \tag{CQ31}$$

CQ23 becomes the explicit identity

$$\begin{aligned} C_{1,1} = {} & \mathscr{E}[\kappa X P_1 - 10 \beta^{-2} a^{-2} J] \\ & + a^{-4} \big\{ (216 N_b + 72 N_p) \beta^{-2} - 120 N_p \beta^{-3} \\ & \qquad + (-168 \beta^{-2} + 60 \beta^{-3}) S_0 + 2 \beta^{-2} \mathcal{Q}_{1,1} \big\}. \end{aligned} \tag{CQ32}$$

The leading constant equals the flat Hessian trace term $2\beta^{-2} a^{-4} \operatorname{Tr} H_0^2$. Indeed for each color the diagonal bond block contributes $z^2 N_b$, and the ordered off-diagonal edge pairs contribute $12 N_p$. Thus $\operatorname{Tr} H_0^2 = 3(36 N_b + 12 N_p)$, which checks the normalization without changing the original counts.

125.7 Physical quotient, constraint transport, and the remaining gap

Suppose the complete determining carrier of every term in CQ23 lies strictly in the positive-time algebra, with its based plaquettes, prefixes, vector coefficients, and any chosen carrier enlargement included. Then the same exact disjoint-support Ward argument as CP27a gives

$$\left[C_{f,h} - k_0 - \sum_p k_p s(P_p) \right]_{\mathrm{OS}} = \left[\mathcal{R}^{(8)}_{f,h} \right]_{\mathrm{OS}}. \tag{CQ33}$$

The identity term is an actual finite-source contact coefficient. Its presence in CQ23 and CQ33 is not optional. Under connected or normalized source extraction it participates in the exact finite-label logarithm and normalization commutators. The same applies to the weighted plaquette row. No absolute scalar increment has been discarded.

The quotient is linear and is not an ideal. For every subsequent insertion G, the exact product identity is

$$G\mathscr{E}[\widetilde{V}] = \mathscr{E}[G\widetilde{V}] + \kappa a^4 \sum_i \widetilde{V}_i X_i G. \tag{CQ34}$$

The rightmost contact must be created with its original finite label union and anchors. The proof has not made it vanish or bounded every possible later descendant.

CQ23 itself is pointwise and can therefore be inserted into the exact joint constrained measure. Dropping its current after conditioning would be wrong. As in CP, its divergence must be pushed through the actual nonlinear map, including the components in every original constraint coordinate. The original selector derivative leaves its cutoff-boundary current. Multiplication by a paired normalized kernel also leaves every derivative of that normalizer. The larger current $\widetilde{V}$, including the explicit last term in CQ20, must be used for all of those operations.

For this bare D8 family, the former unidentified low-loop-degree scalar sector is now an explicitly evaluated identity-plus-Wilson-action row, and the remaining scalar has the required physical-core fourth-power gain and weak analytic root. What is still unproved is uniform creation and transport of all resulting currents and contacts through arbitrary mixed regular and large-field histories, for all previously generated marked coefficients of the same measure. This chapter also does not prove the full inherited source-zero torus comparison S94. Those remaining assertions are not consequences of the pointwise reduction.

In particular, field-dependent inherited weights are outside CQ5. Even with h field independent, a coefficient $f_i(U)$ changes the first completion to

$$C_{f,h} = \mathscr{E}[\kappa f\, X P_h] + \kappa^2 a^4 \sum_i \{f_i X_i^2 P_h + (X_i f_i) X_i P_h\}. \tag{CQ34a}$$

Further completions differentiate these old coefficients again. If the effective density has an inherited action term W, its score is $J + \kappa X W$, so the full Ward current also contains the inherited score derivatives. Neither additional family is bounded by the bare color-Casimir calculation alone.

125.8 An additional exact contact with the physical directional-current square

The preceding calculation also gives a nontrivial contact reduction for the physical quadratic direction which AI35 shows cannot be discarded. For a bond $b = (x, \nu)$, use its three original plane families

$$J_{b;\mu} = a^{-3} X_b S_0^{(\mu\nu)}, \qquad T_{\mu\rho} = \delta_{\mu\rho} - \tfrac{1}{3}, \qquad H_h = a^4 \sum_b h_b \sum_{\mu,\rho\neq\nu} T_{\mu\rho} \langle J_{b;\mu}, J_{b;\rho}\rangle. \tag{CQ35}$$

Thus H_h is exactly the AI33–35 directional-current square, with its original based representatives. The finite hypercubic symmetrization, if used for the primitive, can be applied to the complete identities below, including every source weight.

Let

$$K_{b;\mu} = \sum_{\substack{p \ni b \\ p \text{ in the } (\mu\nu) \text{ plane}}} K(P_{p,b}), \qquad \mathcal{K}_h = \sum_b h_b \sum_{\mu,\rho\neq\nu} T_{\mu\rho} \operatorname{tr}_{\mathfrak{g}}(K_{b;\mu} K_{b;\rho}). \tag{CQ36}$$

There are exactly two plaquettes in each of these sums. This polynomial is fourth order in plaquette defects; its exact transported cross terms are retained. Each partial same-bond Hessian has symmetric part $2I - K_{b;\mu}$. Since $T\mathbf{1} = 0$, the contraction of their constant and linear parts vanishes exactly. The antisymmetric contraction is $(C_A/4)\sum_{\mu,\rho} T_{\mu\rho}\langle a^3 J_{b;\mu}, a^3 J_{b;\rho}\rangle$.

For two distinct bonds in a common plaquette, only one of the target bond's three plane families occurs. Thus the corresponding color-squared block is multiplied by $T_{\mu\mu} = 2/3$. Summing the other three edges of that plaquette gives the factor 2. With $h_p = \sum_{b\in\partial p} h_b$, the full partial-Hessian contraction is therefore

$$\mathcal{F}_H = 2\sum_p h_p\{d_G + c_g s(P_p) + Q(P_p)\} + \mathcal{K}_h + \frac{C_A}{4}a^2 H_h. \tag{CQ37}$$

Here explicitly $\mathcal{F}_H = \sum_{d,b,A,B} h_b \sum_{\mu,\rho} T_{\mu\rho}(X_d^A X_b^B S_0^{(\mu\nu)})(X_d^A X_b^B S_0^{(\rho\nu)})$. Every derivative has the actual original invariant frame.

Consider the contact with constant first external test, $C_{1,H_h} = \kappa a^4 \sum_i J_i X_i H_h$. The total Haar Casimir obeys $\Delta_H J_{b;\mu} = -4C_F J_{b;\mu}$, because every plaquette in that partial action still has four links. The first Ward completion and the ordinary product rule therefore give the exact identity

$$\begin{aligned} C_{1,H_h} = {} & \mathscr{E}[\kappa X H_h] + 4\beta^{-2}a^{-4}\sum_p h_p\{d_G + c_g s(P_p)\} \\ & + \left(\frac{C_A}{2} - 8C_F\right)\beta^{-2}a^{-2}H_h \\ & + 2\beta^{-2}a^{-4}\left\{\mathcal{K}_h + 2\sum_p h_p Q(P_p)\right\}. \end{aligned} \tag{CQ38}$$

There is no unidentified low scalar D8 row in CQ38. There is an explicit inherited H_h row with its coefficient and a quartic remainder. The latter has the same actual-core fourth-power gain, weak analytic root, and compact energy-polynomial estimate as CQ24–29. The H_h row has its own established WH weak extension; its subsequent interacting transfer is still a separate obligation.

The coefficient of this lower row also has an exact coupling power. If the canonical D8 word is multiplied by $g_W^{-4}a^4$, and the own-mesh canonical quadratic primitive is $g_W^{-2}a^2 H_h$, then

$$g_W^{-4}a^4\left(\frac{C_A}{2} - 8C_F\right)\beta^{-2}a^{-2}H_h = \left(\frac{C_A}{2} - 8C_F\right)c_\beta^{-2}g_W^2(g_W^{-2}a^2 H_h). \tag{CQ39}$$

This is an actual bare contact coefficient of order g_W^2, including c_β. It is not an all-history transfer estimate and does not replace the old birth scale by a later mesh.

125.9 Varying the first test creates an explicit source-difference row

One can retain nonconstant f exactly without asserting that the new row is an already normalized scalar center. Put

$$F_p^f = \sum_{d\in\partial p} f_d, \qquad J_{b;\mu}^F = a^{-3}X_b \sum_{p \text{ in } (\mu\nu)} F_p^f\, s(P_p), \qquad W_{b;\mu}^f = J_{b;\mu}^F - 4f_b J_{b;\mu}. \tag{CQ40}$$

The weighted Casimir commutes with X_b, so $\Delta_f J_{b;\mu} = -C_F J_{b;\mu}^F$. For

$$\begin{aligned} \mathcal{B}_{f,h} &= a^4\sum_b h_b \sum_{\mu,\rho\neq\nu} T_{\mu\rho}\langle J_{b;\mu}, W_{b;\rho}^f\rangle, \\ \mathcal{Q}_{f,h}^H &= \mathcal{K}_{fh} + \frac{2}{3}\sum_p t_p Q(P_p), \end{aligned} \tag{CQ41}$$

where t_p is exactly CQ16, the same calculation proves

$$\begin{aligned} C_{f,H_h} &= \mathscr{E}[\kappa f\, X H_h] + \frac{4}{3}\beta^{-2}a^{-4}\sum_p t_p\{d_G + c_g s(P_p)\} \\ &\quad + \left(\frac{C_A}{2} - 8C_F\right)\beta^{-2}a^{-2}H_{fh} - 2C_F\beta^{-2}a^{-2}\mathcal{B}_{f,h} \\ &\quad + 2\beta^{-2}a^{-4}\mathcal{Q}^H_{f,h}. \end{aligned} \tag{CQ42}$$

The same-bond partial-Hessian constant and linear cancellations remain exact with weight $f_b h_b$. Off-diagonal blocks have weight $(2/3)t_p$. The two Laplacian product terms give $-2C_F\beta^{-2}a^{-2}(4H_{fh}+\mathcal{B}_{f,h})$; this accounts for the sign and coefficient of the additional row. For constant $f = 1$, $W^f = 0$, $t_p = 3h_p$, and CQ42 reduces to CQ38.

There is a literal source-difference factor in $\mathcal{B}_{f,h}$; it does not come from field momentum. For an arbitrary bond test the factor is simply the actual difference in CQ40 and can be as large as the bondwise test itself. For the common scalar test $f_b = f(x)$ on all bonds with initial point x, write

$$\begin{aligned} q_+(x;\mu,\nu) &= F^f_{p_{\mu\nu}(x)} - 4f(x) = f(x+a\hat{\mu}) + f(x+a\hat{\nu}) - 2f(x), \\ q_-(x;\mu,\nu) &= F^f_{p_{\mu\nu}(x-a\hat{\mu})} - 4f(x) \\ &= 2f(x-a\hat{\mu}) + f(x-a\hat{\mu}+a\hat{\nu}) - 3f(x). \end{aligned} \tag{CQ43}$$

Using the exact flux and transports of AI4–6 gives

$$W^f_{\nu;\mu}(x) = a^{-1}\{q_-(x;\mu,\nu)T^{-1}_\mu X_{\mu\nu}(x) - q_+(x;\mu,\nu)X_{\mu\nu}(x)\}. \tag{CQ44}$$

This is the weighted two-plaquette current, with both original source anchors kept. On a common chart with bounded actual transports,

$$\|W^f\|_{L^2} \le C\|D_a f\|_\infty \|X\|_{L^2}. \tag{CQ45}$$

For example $q_+/a = D_\mu f + D_\nu f$ and $q_-/a = 3D^*_\mu f + D_\nu f(x-a\hat{\mu})$, with scalar forward and backward differences. There is no fine inverse power remaining in CQ45.

To obtain a uniform weak analytic root of this new row, replace only the directional current $J_{b;\mu}$ in CQ41 by the actual WH17 reconstruction and retain the actual weak-chart flux in CQ44. WH14–17 make this readout equal to $\mathcal{B}_{f,h}$ on the physical compatible KKT graph. The weak bound is

$$|\widehat{\mathcal{B}}_{f,h}| \le C\|h\|_\infty \|D_a f\|_\infty\, \epsilon(\epsilon + \|\bar{\mu}\|_{\mathrm{row}} + \|\mu\|_{\mathrm{row}}). \tag{CQ46}$$

It follows from the physical L^2 pairing, CQ45, and WH16. All reference strengths and the fixed carrier volume are retained. At the flat reference, with $\bar{\mu} = 0$ and b, μ scaled together, the holomorphic extension has valuation at least two and fixed weak Cauchy bounds. A nonflat reference can contribute lower Taylor rows; its recorded strengths remain in CQ46. The estimate is not obtained by bounding a second derivative of an arbitrary $W^{1,p}$ field.

Equations CQ40–46 give an exact additional source-difference row and its own weak analytic root. They do not identify this row with H, with the cubic center, or with a lower scalar without another exact test-derivative/tensor reduction. For arbitrary orientation-dependent bond tests, the uniform derivative bound CQ45 is not asserted. Every Ward current in CQ38 and CQ42 still has the original nonlinear constraint, selector-boundary, and paired-normalization obligations described after CQ34.

126.1 Exact source records

For $b = (x, \nu)$ and $\mu \neq \nu$, write

$$\begin{aligned} F^f_{\mu\nu}(x) &= \sum_{d \in \partial p_{\mu\nu}(x)} f_d, \\ q_{\nu;\mu,-}(x) &= F^f_{\mu\nu}(x - a\hat{\mu}) - 4f_\nu(x), \\ q_{\nu;\mu,+}(x) &= F^f_{\mu\nu}(x) - 4f_\nu(x), \\ \omega_{\nu;\mu}(x) &= \tfrac{1}{2}\{q_{\nu;\mu,-}(x) + q_{\nu;\mu,+}(x)\}. \end{aligned} \tag{CS1}$$

The superscript f distinguishes the scalar plaquette test sum from the gauge flux $X_{\mu\nu}$. Reversing the plaquette orientation does not change its four independent bonds or $F^f_{\mu\nu}$. These definitions are valid for arbitrary color-independent, field-independent bond tests; no interpolation of the test is made.

Use the AI directional current and the CQ40 weighted difference:

$$\begin{aligned} J_{\nu;\mu}(x) &= a^{-1}\{T_\mu^{-1} X_{\mu\nu}(x) - X_{\mu\nu}(x)\}, \\ W^f_{\nu;\mu}(x) &= a^{-1}\{q_{\nu;\mu,-}(x) T_\mu^{-1} X_{\mu\nu}(x) - q_{\nu;\mu,+}(x) X_{\mu\nu}(x)\}, \\ J_\nu &= \sum_{\mu \neq \nu} J_{\nu;\mu}, \qquad W^{f,\mathrm{tot}}_\nu = \sum_{\mu \neq \nu} W^f_{\nu;\mu}. \end{aligned} \tag{CS2}$$

With $T_{\mu\rho} = \delta_{\mu\rho} - 1/3$, the original row is

$$\begin{aligned} \mathcal{B}_{f,h} &= a^4 \sum_{x,\nu} h_\nu(x) \sum_{\mu,\rho \neq \nu} T_{\mu\rho} \langle J_{\nu;\mu}, W^f_{\nu;\rho} \rangle \\ &= a^4 \sum_{x,\nu,\mu \neq \nu} h_\nu \langle J_{\nu;\mu}, W^f_{\nu;\mu} \rangle - \frac{a^4}{3} \sum_{x,\nu} h_\nu \langle J_\nu, W^{f,\mathrm{tot}}_\nu \rangle. \end{aligned} \tag{CS3}$$

The pairing is the invariant real Lie-algebra pairing and its bilinear holomorphic continuation. It is not replaced by a Hermitian pairing in the identities.

126.2 Exact summation by parts for the same-direction terms

For two Lie-algebra matrices A, B and scalar weights q_-, q_+,

$$\langle A - B, q_- A - q_+ B \rangle = \frac{1}{2}(q_- - q_+)(\langle A, A \rangle - \langle B, B \rangle) + \frac{1}{2}(q_- + q_+)\langle A - B, A - B \rangle. \tag{CS4}$$

Apply this with $A = T_\mu^{-1} X_{\mu\nu}(x)$ and $B = X_{\mu\nu}(x)$. Conjugation preserves the bilinear pairing also for complex group elements, so the first difference is the literal scalar backward difference of $\langle X, X\rangle$. Moreover

$$q_{\nu;\mu,-} - q_{\nu;\mu,+} = a D_\mu^* F_{\mu\nu}^f. \tag{CS5}$$

On the original periodic lattice the adjoint of $D_\mu^* = (T_\mu^{-1} - 1)/a$ for scalar tests is $D_\mu = (T_\mu - 1)/a$. Exact summation by parts gives

$$\begin{aligned} a^4 \sum_{x,\nu,\mu\neq\nu} h_\nu \langle J_{\nu;\mu}, W_{\nu;\mu}^f \rangle &= \mathcal{D}_{f,h} + \mathcal{P}_{\omega,h}, \\ \mathcal{D}_{f,h} &= a^4 \sum_{x,\nu,\mu\neq\nu} \chi_{\nu;\mu}(x) |X_{\mu\nu}(x)|^2, \\ \chi_{\nu;\mu} &= \tfrac{1}{2} D_\mu \left(h_\nu D_\mu^* F_{\mu\nu}^f \right), \\ \mathcal{P}_{\omega,h} &= a^4 \sum_{x,\nu,\mu\neq\nu} h_\nu(x) \omega_{\nu;\mu}(x) |J_{\nu;\mu}(x)|^2. \end{aligned} \tag{CS6}$$

No field derivative has been estimated or added. Every derivative in χ acts on an explicit source weight. In particular, the forward shift of h_ν in $D_\mu(h_\nu D_\mu^* F^f)$ must be retained. On a local packet this operation keeps the complete union of the original and shifted source/field stencils.

126.3 Complete the total-current part and retain its scalar term

The exact link Casimir gives

$$\sum_A X_b^A (W_{b;\mu}^f)^A = C_F a^{-3} \sum_{\substack{p \ni b \\ p \text{ in the } (\mu\nu) \text{ plane}}} (F_p^f - 4 f_b) \operatorname{Re}\operatorname{tr} P_p. \tag{CS7}$$

Consequently the last line of CS3 is

$$-\frac{1}{3} \mathscr{E}[h W^{f,\mathrm{tot}}] - \frac{C_F}{3} \beta^{-1} a^{-2} \sum_p r_p^{f,h} \operatorname{Re}\operatorname{tr} P_p, \qquad r_p^{f,h} = F_p^f \sum_{b\in\partial p} h_b - 4 \sum_{b\in\partial p} f_b h_b. \tag{CS8}$$

The scalar is not discarded. Its exact source covariance form is

$$r_p^{f,h} = -\frac{1}{2} \sum_{b,c\in\partial p} (f_b - f_c)(h_b - h_c). \tag{CS9}$$

For common scalar smooth tests this is a product of two literal first source differences, of size $C a^2 \|D_a f\|_\infty \|D_a h\|_\infty$. For constant h it vanishes exactly. Its actual original anchors remain those in CS8–9.

Equations CS6 and CS8 already provide an exact nontrivial reduction. The next step isolates the physical part of the remaining directional quadratic rows.

126.4 Project the weighted directional square with its full Ward partner

Put $Q_{b;\mu} = J_{b;\mu} - J_b/3$, as in AI33, and define

$$\begin{aligned} \mathcal{H}_{\omega,h} &= a^4 \sum_{b=(x,\nu)} h_b \sum_{\mu\neq\nu} \omega_{\nu;\mu}(x) |Q_{b;\mu}|^2, \\ Y_b &= h_b \left\{ \frac{2}{3} \sum_{\mu\neq\nu} \omega_{\nu;\mu} J_{b;\mu} - \frac{1}{9} \left(\sum_{\mu\neq\nu} \omega_{\nu;\mu} \right) J_b \right\}. \end{aligned} \tag{CS10}$$

Expanding Q gives $\mathcal{P}_{\omega,h} = \mathcal{H}_{\omega,h} + a^4 \sum_b \langle J_b, Y_b \rangle$. Completing that full current and evaluating its exact divergence gives

$$\mathcal{P}_{\omega,h} = \mathcal{H}_{\omega,h} + \mathscr{E}[Y] + C_F \beta^{-1} a^{-2} \sum_p u_p^{\omega,h} \operatorname{Re} \operatorname{tr} P_p, \tag{CS11}$$

where

$$u_p^{\omega,h} = \sum_{b=(x,\nu)\in\partial p} h_b \left\{ \frac{2}{3}\omega_{b,p} - \frac{1}{9} \sum_{\rho\neq\nu} \omega_{\nu;\rho}(x) \right\}. \tag{CS12}$$

Here $\omega_{b,p}$ is the plane weight in CS1 for the plane of p at its actual edge b. It includes both incident plaquette source sums in that plane. CS12 must not replace it by a weight at the root of p.

Define

$$\mathcal{V}_{f,h} = Y - \tfrac{1}{3} h W^{f,\mathrm{tot}}, \qquad v_p^{f,h} = u_p^{\omega,h} - \tfrac{1}{3} r_p^{f,h}. \tag{CS13}$$

The complete identity is

$$\boxed{\begin{aligned} \mathcal{B}_{f,h} &= \mathscr{E}[\mathcal{V}_{f,h}] + \mathcal{D}_{f,h} + \mathcal{H}_{\omega,h} \\ &\quad + C_F \beta^{-1} a^{-2} \sum_p v_p^{f,h} \operatorname{Re} \operatorname{tr} P_p. \end{aligned}} \tag{CS14}$$

In particular, CQ42's term $-2C_F\beta^{-2}a^{-2}\mathcal{B}_{f,h}$ adds the complete current $-2C_F\beta^{-2}a^{-2}\mathcal{V}_{f,h}$, the two displayed source-weighted quadratic rows with that coefficient, and the explicit scalar trace row

$$-2C_F^2 \beta^{-3} a^{-4} \sum_p v_p^{f,h} \operatorname{Re} \operatorname{tr} P_p. \tag{CS15}$$

Splitting $\operatorname{Re} \operatorname{tr} P = N - s(P)$ preserves both the absolute scalar coefficient and the action coefficient.

There is an exact cancellation after the complete original periodic source sum. Put $\Omega_{\mathrm{sum}} = \sum_b h_b \sum_{\mu\neq\nu} \omega_{\nu;\mu}$. Each plane has two plaquettes incident at a bond, and each bond has six plaquettes. Hence

$$\sum_p u_p^{\omega,h} = \left(\frac{4}{3} - \frac{6}{9}\right) \Omega_{\mathrm{sum}} = \frac{2}{3}\Omega_{\mathrm{sum}}, \qquad \sum_p r_p^{f,h} = 2\Omega_{\mathrm{sum}}, \qquad \sum_p v_p^{f,h} = 0. \tag{CS15a}$$

The middle equality follows directly from $\sum_\mu \omega_{\nu;\mu} = \frac{1}{2} \sum_{p\ni b} F_p^f - 12 f_b$. Also, periodic summation gives

$$\sum_{x,\nu,\mu\neq\nu} \chi_{\nu;\mu}(x) = \sum_p \Xi_p = 0. \tag{CS15b}$$

Thus the newly introduced total constant in CS15 cancels after this complete paired source sum. Its individual local scalar atoms must remain until that exact pairing has been performed. A selected history, a partial carrier sum, or a later coefficient with different normalizers cannot use CS15a term by term. The original CQ vacuum contact coefficients are not removed by this cancellation.

126.5 The flux row is also an explicit action-plus-quartic row

With $R(P), A(P)$ from CQ6, the actual projected flux obeys

$$|X_{\mu\nu}(x)|^2 = a^{-4}\{2s(P_{\mu\nu}(x)) + Q_F(P_{\mu\nu}(x))\}, \qquad Q_F(P) = -\operatorname{tr} R(P)^2 + \frac{1}{N}(\operatorname{tr} A(P))^2. \tag{CS16}$$

Thus Q_F is an explicit fourth-order plaquette remainder, with the same real compact and common-small-complex bounds as CQ10–11. For the single counted plaquette $p = p_{\mu\nu}(x)$, $\mu < \nu$, set $\Xi_p = \chi_{\nu;\mu}(x) + \chi_{\mu;\nu}(x)$. The exact ordered-index sum in CS6 is

$$\mathcal{D}_{f,h} = \sum_p \Xi_p \{2s(P_p) + Q_F(P_p)\}. \tag{CS17}$$

This uses $X_{\nu\mu} = -X_{\mu\nu}$ at the common base point; it does not identify differently based fluxes. All differentiated source weights are explicit in Ξ_p.

126.7 Identify the already-defined pointed completion before using its symmetry

There is a stronger result for the specific pointed signed-coordinate average defined in AI4.1. This section identifies that source representative explicitly. It does not change a historical branch which used the unsymmetrized initial-point representative.

The per-bond scalar

$$e_b = |J_b|^2 - \kappa \sum_A X_b^A J_b^A = |J_b|^2 - C_F \beta^{-1} a^{-6} \sum_{p \ni b} \operatorname{Re} \operatorname{tr} P_p \tag{CS21}$$

and the per-bond scalar $H_b = \sum_{\mu \neq \nu} |Q_{b;\mu}|^2$ transform as scalar functions of the unoriented geometric bond. Under reversal of its own ν direction, the positive bond at x becomes the positive bond at the reflected point minus $a\hat{\nu}$. All of its partial forces acquire the same actual minus-adjoint transport. Their norms, their total-current subtraction, and the Casimir trace term are unchanged. Other signed coordinate permutations permute the partial plane actions. This establishes the transformation for the full two-plaquette force, rather than assuming it for a single forward based plaquette.

For either bond scalar $O_b = e_b$ or $O_b = H_b$, averaging the rooted outgoing-bond sum over all signed coordinate permutations fixing the root gives exactly

$$\operatorname{Av}\left(\sum_\nu O_{(x,\nu)}\right) = \frac{1}{2} \sum_\nu \{O_{(x,\nu)} + O_{(x-a\hat{\nu},\nu)}\}. \tag{CS22}$$

Each of the eight signed directions occurs equally often. After the local rooted stencil is oriented in this way, its words are read on the same lattice. This uses the local Wilson incidence pattern, not an assertion that a rectangular physical torus has a global coordinate-permutation symmetry. After smearing, the positive-bond source weights are therefore

$$f_\nu^c(x) = \frac{f(x) + f(x + a\hat{\nu})}{2}, \qquad h_\nu^c(x) = \frac{h(x) + h(x + a\hat{\nu})}{2}. \tag{CS23}$$

Thus this pointed EOM source is exactly $\mathscr{E}[f^c J]$, and this pointed physical quadratic source is exactly H_{h^c}. When this pointed completion is used, the identification implements WH18's instruction to retain its existing finite symmetrization.

The two operator labels must be averaged independently. Bilinearity of their exact contact then gives $C_{f^c,H_{h^c}}$. A single diagonal group average of C_{f,H_h} generally gives correlated weights and cannot be replaced by this product of averages. Every original finite label and every pair in the independent finite sums remains in the identified packet. No generic inherited history is assumed to have this symmetry.

For the endpoint weights the four-edge test sum is the literal four-vertex sum:

$$F_{\mu\nu}^{f^c}(x) = f(x) + f(x + a\hat{\mu}) + f(x + a\hat{\nu}) + f(x + a\hat{\mu} + a\hat{\nu}). \tag{CS24}$$

In particular, its exact two flux weights are

$$q_{\nu;\mu,\pm}^c(x) = f(x \pm a\hat{\mu}) + f(x \pm a\hat{\mu} + a\hat{\nu}) - f(x) - f(x + a\hat{\nu}). \tag{CS24a}$$

They obey $\|q_\pm^c\|_\infty \leq 2a \|D_a f\|_\infty$. This directly proves the corresponding W^f weak L^2 bound from CS2. It does not invoke CQ45's initial-point hypothesis $f_b = f(x)$, which the endpoint weights do not satisfy. Likewise CS24 and the endpoint formula for h^c directly prove the source-second-difference bound for χ. Substituting it and $4f_\nu^c = 2(f(x) + f(x + a\hat{\nu}))$ into CS1 proves

$$\begin{aligned} \omega_{\nu;\mu}^c(x) &= \frac{a^2}{2} \{\Delta_\mu^{\mathrm{cen}} f(x) + \Delta_\mu^{\mathrm{cen}} f(x + a\hat{\nu})\} \\ &= a^2 \Delta_\mu^{\mathrm{cen}} f_\nu^c(x), \qquad \Delta_\mu^{\mathrm{cen}} = \frac{T_\mu + T_\mu^{-1} - 2}{a^2}. \end{aligned} \tag{CS25}$$

This is an exact source identity. It is the reason an a^2 gain is valid for this identified completion, while CS19 remains correct for the raw initial-point packet.

There is a second source-anchor correction when the lower H row is compared with the same pointed primitive smeared by fh:

$$f_\nu^c(x)h_\nu^c(x) = (fh)_\nu^c(x) - \frac{a^2}{4}(D_\nu f)(x)(D_\nu h)(x). \tag{CS26}$$

The product of endpoint averages is not the endpoint average of the product. The difference in CS26 is retained as a separate tensor row with two explicit first source differences.

126.8 Complete bare projection and fourth-power gain for that identified packet

Here all quantities $t_p, \mathcal{Q}^H, \mathcal{V}, \Xi_p, v_p, \omega$ are the exact preceding expressions evaluated at the bond tests (f^c, h^c). Write

$$\alpha_H = \frac{C_A}{2} - 8C_F, \qquad \eta_\nu(x) = (D_\nu f)(x)(D_\nu h)(x), \qquad H_\eta = a^4 \sum_{x,\nu} \eta_\nu(x) \sum_{\mu\neq\nu} |Q_{\nu;\mu}|^2. \tag{CS27}$$

Combining CQ42 with CS14, CS17, and CS26 gives the exact pointwise identity

$$\begin{aligned} C_{f^c,H_{h^c}} &= \mathscr{E}[V^c] + k_0^c + \sum_p k_p^c s(P_p) + \alpha_H \beta^{-2} a^{-2} H_{(fh)^c} + \mathcal{R}_H^c, \\ V^c &= \kappa f^c X H_{h^c} - 2C_F \beta^{-2} a^{-2} \mathcal{V}_{f^c,h^c}, \\ k_0^c &= \frac{4}{3} d_G \beta^{-2} a^{-4} \sum_p t_p - 2NC_F^2 \beta^{-3} a^{-4} \sum_p v_p, \\ k_p^c &= \frac{4}{3} c_g \beta^{-2} a^{-4} t_p - 4C_F \beta^{-2} a^{-2} \Xi_p + 2C_F^2 \beta^{-3} a^{-4} v_p. \end{aligned} \tag{CS28}$$

The remainder is the following explicitly defined sum:

$$\begin{aligned} \mathcal{R}_H^c = {} & 2\beta^{-2} a^{-4} \mathcal{Q}_{f^c,h^c}^H - 2C_F \beta^{-2} a^{-2} \sum_p \Xi_p Q_F(P_p) \\ & - 2C_F \beta^{-2} a^{-2} \mathcal{H}_{\omega^c,h^c} - \frac{\alpha_H}{4} \beta^{-2} H_\eta. \end{aligned} \tag{CS29}$$

In particular, the two quadratic tensor terms have not been deleted; their literal source differences in CS25–26 make their scale gain quantitative. The complete current and both scalar coefficients in CS28 are equally necessary.

To state that gain, restore a protected physical readout length ℓ, with $a \le \ell$, on one declared complete carrier and collar of fixed geometry in those units. Its full read sets lie in a fixed bounded number of physical ℓ-boxes, of total volume at most $C\ell^4$, as in WH. A collection of such carriers retains its actual sum and component counts. The constants below may depend on the once-fixed weak chart, $p > 8$, collar geometry, and complex transport bounds, in addition to the group; they do not hide a growing component count. Use the dimensionless actual test norm $\|f\|_{C^2_{\ell,a}} = \max_{r\le 2} \ell^r \|D_a^r f\|_\infty$, including the required mixed and shifted differences. Let ϵ_ℓ bound the dimensionless weak flux strength, and let $\mathcal{M}_\ell$ be the full dimensionless WH current-reconstruction strength, including the reference multiplier and the current multiplier. WH's restored bounds are $\|a^{-2}(P-I)\|_{L^4}^4 \le C\ell^{-4}\epsilon_\ell^4$ and $\|\mathfrak{Q}\|_{L^2}^2 \le C\ell^{-2}\mathcal{M}_\ell^2$. The four summands of CS29 consequently satisfy

$$a^4 |\widehat{\mathcal{R}}_H^c| \le C_N \beta^{-2} (a/\ell)^4 \|f\|_{C^2_{\ell,a}} \|h\|_{C^2_{\ell,a}} \{\epsilon_\ell^4 + \mathcal{M}_\ell^2\}. \tag{CS30}$$

The first summand uses CQ24–28. The flux quartic correction has the stronger factor $(a/\ell)^6$: its weight Ξ_p has two physical source derivatives. For the third summand CS25 gives $\|\omega^c\|_\infty \le Ca^2\ell^{-2}\|f\|_{C^2_{\ell,a}}$, which multiplies the actual L^2 current-square bound. The last summand uses the two derivatives in η. No extra field derivative is invoked in any of these gains.

On the physical compatible KKT graph, the hat in CS30 equals the actual scalar remainder in CS29. Off that graph, the only source extensions are the already specified WH reconstructions in the directional-current terms; the flux words remain the actual weak-chart words. This gives a common weak analytic root, with the saved engineering factor intact. In canonical normalization for this same Wilson measure,

$$g_W^{-4}a^4|\widehat{\mathcal{R}}^c_H| \le C_N c_\beta^{-2}(a/\ell)^4\|f\|_{C^2_{\ell,a}}\|h\|_{C^2_{\ell,a}}\{\epsilon_\ell^4 + \mathcal{M}_\ell^2\}. \tag{CS31}$$

The lower physical row in CS28 is exactly the identified pointed primitive $H_{(fh)^c}$, with the canonical $O(g_W^2)$ coefficient computed in CQ39. The quadratic terms in the remainder have flat field valuation two, with their explicit two source differences; they are not mislabeled as fourth-order field words.

Thus the identified, independently pointed-symmetrized bare H-contact packet has a complete algebraic projection and the required fourth-power scalar remainder estimate on its actual admitted weak domain. The earlier unsymmetrized statement is not overwritten. The result still does not transport the full current V^c, its future contacts, its cutoff-boundary currents, or its normalization commutators through arbitrary mixed histories. It also does not admit an old field-dependent coefficient merely by replacing its test by an endpoint average. Those remain distinct all-history obligations.

127.2 The positive-order kernel jets retain the small factor

Set $q_\varepsilon = e^\varepsilon - 1 < 1$. Then

$$|m(z) - 1| \le q_\varepsilon, \qquad |m(z)| \ge 1 - q_\varepsilon, \tag{NC4}$$

and

$$\|K_z - K_0\|_{\mathrm{TV}} \le \frac{2q_\varepsilon}{1 - q_\varepsilon} \le C\varepsilon. \tag{NC5}$$

Indeed, $K_z - K_0 = m^{-1}(e^\sigma - m)K_0$, and $\int |e^\sigma - m|\, dK_0 \le \int |e^\sigma - 1|\, dK_0 + |m - 1|$. This proof involves no inverse power of Z_0.

Choose a strictly smaller outer chart with spare physical radius $c\rho$, in exactly the sense used in WCT22. For fixed order $r \ge 1$, apply the measure-valued Cauchy formula to $K_z - K_0$, rather than to K_z. A polydisc in the at most r displayed directions, of radius $c\rho/r$, gives

$$\|D_{u_r} \cdots D_{u_1} K_z\|_{\mathrm{TV}} \le C_r \varepsilon \rho^{-r} \prod_{i=1}^{r} \|u_i\|. \tag{NC6}$$

The formula can also be proved by writing the explicit density in NC2 relative to the one fixed measure K_0 and integrating its scalar Cauchy formula. Thus no assertion about an unspecified dual of the space of measures is required.

On bounded analytic invariant-coordinate frames and a radius at most one, the lower chart terms have the same upper bound. At the declared current canonical scale, $Y_i = g_k X_i$,

$$\|Y_{i_r} \cdots Y_{i_1} K_z\|_{\mathrm{TV}} \le C_r \varepsilon (g_k/\rho)^r, \qquad r \ge 1. \tag{NC7}$$

If different input directions carry different recorded couplings or radii, their literal product replaces $(g_k/\rho)^r$. An old birth coupling is not renamed a current coupling. An absolute sum over microscopic derivative coordinates is not a consequence of NC6; the admitted direction or tensor-kernel norm must already be supplied by the actual readout.

For the original radius $\rho_k \gtrsim g_k(\log g_k^{-2})^q$, NC3 and NC7 give a small g_k-times-logarithms bound for every fixed positive order. This is stronger than retaining only WCT's fixed $e^{3/4}$ total-variation ceiling. It is not by itself the uniform Cg_k row of FR: the logarithms and component multiplicities must still be paid in their actual norms.

The relative logarithm also retains the same factor. The branch $\mathcal{L}(z) = \log m(z)$, with $\mathcal{L}(z_0) = 0$, obeys

$$\sup |\mathcal{L}| \le -\log(1 - q_\varepsilon) \le C\varepsilon, \qquad \|D^r \mathcal{L}\| \le C_r \varepsilon \rho^{-r}. \tag{NC8}$$

Only $\log Z(z) - \log Z_0$ appears in NC8. The absolute vacuum constant $\log Z_0$ remains unestimated by this formula.

127.3 Exact ordered lift of a paired current measure

The following identity states precisely which normalization commutators have to be retained. Let x be admitted outer coordinates, and let τ, ν be ordinary finite signed or Banach-valued measures satisfying

$$\tau = \mathscr{D}_{Y_1 \cdots Y_r} \nu, \qquad \langle \mathscr{D}_{Y_1 \cdots Y_r} \nu, f \rangle = \int Y_1 \cdots Y_r f \, d\nu. \tag{NC9}$$

Their support may contain the original constraints. The vector fields and the kernel must be defined on the required common neighborhood. Write $(\nu \otimes K)(dx, d\xi) = \nu(dx) K_x(d\xi)$. This notation adjoins only the actual kernel coordinates. It does not duplicate an already retained physical variable.

For a subset $A \subseteq \{1, \dots, r\}$, denote by Y_A the subword in its original order and use the empty subword as the identity. Then

$$\tau \otimes K = \sum_{A \subseteq [r]} \mathscr{D}_{Y_A} \big(\nu \otimes (Y_{A^c} K) \big). \tag{NC10}$$

The output derivatives act on the outer variable with the new kernel coordinate fixed. To prove NC10, test its left side against $\varphi(x, \xi)$. By NC9 it equals $\int Y_1 \cdots Y_r [\int \varphi K_x] \, d\nu$. The exact ordered product rule distributes each derivative between φ and K, in its original order, and gives the right side. Noncommuting vector fields cause no omitted term: no subword is reordered. Terms arising from their coordinate coefficients are included in those ordered operators.

At order one this is

$$\tau \otimes K = \mathscr{D}_Y (\nu \otimes K) + \nu \otimes (YK). \tag{NC11}$$

For higher order, the $A = \varnothing$ term is an ordinary scalar measure. Thus adjoining the kernel does not preserve a pure positive-order current representation without a scalar correction. For a finite sum of paired currents, NC10 is applied term by term before any contact product is formed.

Every proper-subset correction has zero fiber mass:

$$(Y_{A^c} K_x)(1) = 0 \quad (A \neq [r]). \tag{NC12}$$

It vanishes against tests independent of ξ. It need not vanish against $\varphi(q(x, \xi))$. In that case the actual derivatives of q enter the finite current chain rule, and the correction is retained.

This identity applies to currents already expressed in the admitted outer variables of the original component. It is not a formula which turns a derivative in an internally integrated old variable into an outer derivative. Nor does it assert independence between a preexisting old variable and a new reference coordinate when the original kernel has a conditional dependence on that old variable. Such variables remain among the joint outer arguments and their actual derivatives must meet the chart hypothesis.

127.4 Quantitative current bound and preserved incoming weights

Use the represented current mass from WCT, with the original anchors and canonical normalization. The leading $A = [r]$ term of NC10 has coefficient variation at most $\sup_x \|K_x\|_{\mathrm{TV}} |\nu|$. Every proper term has at least one differentiated kernel factor. NC7 therefore gives, for $r \leq m$,

$$\mathcal{M}_m(\tau \otimes K - \mathscr{D}_{Y_1 \cdots Y_r}(\nu \otimes K)) \leq C_m \varepsilon \sum_{s=1}^{m} (g_k/\rho)^s \, |\nu|(X). \tag{NC13}$$

The same statement holds for a tensor packet by summing its existing absolute tensor variations. No ambient bond count is inserted. The scalar term is included on the left.

If an admitted incoming coefficient already has an envelope $A_{j,I}W_{e,j}$ in this measure norm, its right side in NC13 is multiplied by that same envelope. The operation is linear in the full source coefficient, so neither the old source degree nor its numerical saved factor is reset. This conclusion requires the incoming envelope on this actual outer support and all its required directions. It does not obtain that envelope from an extension identity.

For a bounded analytic coefficient $F_z(\xi)$ on the same original chart, the ordered product rule also yields

$$Y_{[r]}(K_zF_z) - K_z(Y_{[r]}F_z) = \sum_{\varnothing\neq A\subseteq[r]} (Y_AK_z)(Y_{A^c}F_z). \tag{NC14}$$

Here each pairing integrates the whole coefficient in the one original kernel. The formula does not replace a joint product of marks by a product of separately integrated marks. Each displayed kernel derivative has the small factor in NC7, while the coefficient derivatives retain their actual grades, anchors, and old prefactors.

127.5 Sharp selectors and genuine normalization factors

Suppose a selector is applied to the whole lifted packet. The complementary representation C7.29–34 applies to the ordinary scalar companion and all its current partners, including the scalar commutator in NC10. Its quantitative failure bound requires the variation envelope of every partner on the same realized failure set. NC13 supplies the kernel part of those envelopes on the admitted outer chart. It does not move the selector through a discarded variable or turn an internal moving cutoff into a frozen outer cutoff.

If a further source-zero factor $h(x)$ multiplies a current, it must likewise be included before using the adjoint identity. For an ordered word,

$$h\,\mathscr{D}_{Y_{[r]}}\nu = \sum_{A\subseteq[r]} \mathscr{D}_{Y_A}\big((Y_{A^c}h)\nu\big). \tag{NC15}$$

In particular $h\mathscr{D}_Y\nu = \mathscr{D}_Y(h\nu) + (Yh)\nu$, with the convention NC9. Relative derivatives of the original normalizing factor $h = Z^{-1}$ are controlled using NC8 and the ordered product rule for $e^{-\mathcal{L}}$. The base factor Z_0^{-1} must stay paired with its original unnormalized body; it is not bounded independently.

There is an exact cancellation when the original expression actually recombines $ZK = T$. Differentiating that identity retains every product-rule term and restores the derivative of the original unnormalized measure. This is the correct place to cancel a normalization commutator. One may not perform that cancellation after separately replacing the two factors by their absolute norms.

127.8 Sources and dependencies

The primary model input is T. Bałaban, *Large field renormalization II*, CMP **122** (1989), (1.71), the comparison preceding (1.73), (1.73)–(1.75), and the component treatment after (1.92): primary paper. WCT21–28 verifies the precise original-domain and fixed-coordinate interpretation used here. XP supplies the finite-source density bookkeeping; C7.29–34 supplies the constrained paired-measure and failure representation. All original weak-trajectory and admitted-domain assumptions remain in force.

128.4 An explicit all-history exponent-defect estimate

Assume two integrands on the literal common record satisfy

$$d\mu_{1,X,\omega} = e^{W_{X,\omega}} d\mu_{0,X,\omega}, \qquad |W_{X,\omega}(\xi)| \le \delta A_\omega(\xi). \tag{VC7}$$

The defect may be complex. It need not have a history-independent supremum. The inequalities are imposed on the original common constraint and selector support. In particular VC7 does not express two different sharp supports as an exponential tilt.

For bounded Banach-valued F, define the auxiliary unnormalized operator

$$T_{z,X}F = \sum_\omega \int F e^{zW_{X,\omega}} d\mu_{0,X,\omega}. \tag{VC8}$$

This is an analytic interpolation of integrands. It is not a claim that its intermediate values are original physical RG steps.

128.4.1 Proposition VC9: the defect retains the rare majorant

For $m \ge 0$,

$$\left\| \frac{1}{m!} \sum_\omega \int F W_{X,\omega}^m \, d\mu_{0,X,\omega} \right\| \le \|F\|_\infty B_X \left(\frac{\delta}{r}\right)^m. \tag{VC9}$$

Consequently, when $0 \le \delta < r$,

$$\begin{aligned} \|(T_{1,X} - T_{0,X})F\| &\le \frac{\delta}{r-\delta} B_X \|F\|_\infty, \\ \left\| (T_{1,X} - T_{0,X})F - \sum_\omega \int F W_{X,\omega} d\mu_{0,X,\omega} \right\| &\le \frac{\delta^2}{r(r-\delta)} B_X \|F\|_\infty. \end{aligned} \tag{VC10}$$

For $|z|\delta \le r$, one also has the direct bound

$$\|T_{z,X}F\| \le B_X \|F\|_\infty. \tag{VC11}$$

Proof. For $u \ge 0$, $u^m/m! \le r^{-m} e^{ru}$. Use this with $u = A_\omega$, then VC2, VC6 and VC7, before the sum over histories is taken. This proves VC9. Sum the resulting geometric series from $m = 1$ or $m = 2$ to prove VC10. Direct domination by e^{rA_ω} proves VC11. The same bound on smaller discs justifies the analytic series and all exchanges of integrals and history sums. No lower bound for $T_{0,X}1$ occurs. □

If the carrying function changes too, use the exact equality

$$T_{1,X}F_1 - T_{0,X}F_0 = T_{1,X}(F_1 - F_0) + (T_{1,X} - T_{0,X})F_0. \tag{VC12}$$

The estimate may also be applied with a nonconstant positive carrying majorant when VC6 is retained with that majorant inside its **same** integral. This is the form used when an original Mayer factor still depends on the internal history. It is not legitimate to integrate each Mayer factor against a separate copy of a component measure.

VC9–11 are the record-dependent completion of the exponential moment qualification after RC14. The elementary excess argument is the same positive argument as S15, but its use here is for an absolute two-integrand difference, including scalar normalization atoms and the original history sum.

128.6 Denominators: exact recombination and its limitation

For one already formed component let

$$Z_i = T_{i,X}1, \qquad K_i = Z_i^{-1}T_{i,X}. \tag{VC20}$$

Where this factorization is defined, the exact finite-difference identity is

$$\begin{aligned}(T_{1,X} - T_{0,X})F &= (Z_1 - Z_0)K_0F + Z_1(K_1 - K_0)F,\\ (K_1 - K_0)F &= Z_1^{-1}(T_{1,X} - T_{0,X})F - Z_1^{-1}(Z_1 - Z_0)K_0F.\end{aligned} \tag{VC21}$$

Substitution cancels the two displayed normalization terms exactly. This keeps the normalization commutator until the mass and the normalized kernel have been recombined. It does not estimate the two terms separately by reciprocals of a small mass.

For disjoint original components $X_1, \dots, X_q$ acting on a joint function $F(\xi_1, \dots, \xi_q)$, the relevant expression is

$$\left(\prod_{h=1}^{q} Z_{i,h}\right)(K_{i,1} \otimes \cdots \otimes K_{i,q})F = (T_{i,1} \otimes \cdots \otimes T_{i,q})F. \tag{VC22}$$

Each component is used once; F includes all Mayer factors touching it. The product is justified only for the disjoint component factorization of the original formula. If two live pieces share a variable, they first belong to one joint body, and VC22 is applied after that merger. The number q is the actual component count.

The component mass in VC20 is an optional factorization of a formed T. The preparation denominator D_b in VC18 is already inside T. VC21 does **not** cancel D_b, any old conditional denominator, or an external vacuum constant. This distinction is indispensable for the source-zero comparison.

128.6.1 Why the reserve does not bound every relative logarithm

For a one-point history with $N = m$, take

$$d\nu_m = e^{-2rm}\delta_m, \qquad W_m = \delta m. \tag{VC23}$$

Then $\int e^{rN} d\nu_m = e^{-rm}$, so the reserve-weighted absolute difference is bounded by $C_r\delta e^{-rm}$. Nevertheless

$$\log \frac{\int e^{W_m} d\nu_m}{\int d\nu_m} = \delta m. \tag{VC24}$$

It has no uniform bound of order δ. Thus the moment reserve alone cannot be inserted into RC3 as though it were a history-independent pointwise exponent bound. It also cannot control every normalized conditional moment. The original unnormalized activity formulation is what permits VC10 to be used without this false inference.

128.7 Passage through the completed scalar R formation

Fix the same current component geometry, complete records, and Mayer carrier conventions at the two endpoints. Let $V_1 - V_0 = \Delta V$ denote differences of the interactions outside the individual component exponents. Before its original connected resummation, a typical term is

$$\mathcal{A}_i(\mathcal{C}, \mathcal{D}) = \left(\bigotimes_{X \in \mathcal{C}} T_{i,X} \right) \prod_{Y \in \mathcal{D}} (e^{V_{i,Y}} - 1). \tag{VC25}$$

The sets and the localization of this formula are the ones in the original scalar formation. Any untouched external factors are retained as carrying factors. In particular the history sum inside $T_{i,X}$ still acts on every factor depending on that history.

Assume the following quantified input bounds:

1. Each original body satisfies VC6 and VC7, with a common δ and reserve r.
2. The external interaction differences have their actual rooted original formation norm at most δ.
3. The two endpoints lie within a common strict formation margin, on the same admitted argument set and carrier rules.

The original model supplies the source-zero reserve and formation margin. Equality of the two records and the small differences are the comparison assertions still to verify in a physical-torus application.

128.7.1 Proposition VC26: full formation difference, including its scalar part

Under these inputs, the completed R formation is Lipschitz in δ, with its weakened original carrier and rare majorants. Its absolute vacuum value has the same difference bound. No inverse component mass is used.

Proof. Use $T_{z,X}$ from VC8 and $V_{z,Y} = V_{0,Y} + z\Delta V_Y$. Choose a fixed $r_1 > 0$ smaller than the reserve r and the strict interaction margin. For $|z|\delta \le r_1$, VC11 bounds every body by its original weakened positive majorant. The interaction norm remains inside the original formation ball. Apply precisely the original positive Mayer and carrier sums with those bounds. Thus each formed activity, and the convergent connected output in its declared carrier norm, is analytic and uniformly bounded on

$$|z| < r_1/\delta. \tag{VC27}$$

Absolute convergence follows uniformly on each smaller disc; finite-cutoff sums can first be differentiated and then their positive majorants summed. If M_X denotes the resulting weakened output majorant, the Cauchy coefficient bound and its geometric sum give, for $\delta \le r_1/2$,

$$\|\mathcal{F}_X(1) - \mathcal{F}_X(0)\| \le \frac{\delta}{r_1 - \delta} M_X \le \frac{2\delta}{r_1} M_X. \tag{VC28}$$

This analytic argument is also a concise way to sum the exact pointing expansion: one differentiates either one body or one Mayer factor in VC25. It does not replace the actual q component sum by one component. The spare positive formation margin controls that differentiated count along with the ordinary Mayer count.

For an explicit bound on the connected scalar logarithm, retain the common Kotecky–Preiss majorant of RC15. If $z_i(P)$ are the completed scalar activities, then RC17 gives

$$|\log \Xi(z_1) - \log \Xi(z_0)| \le \sum_P |z_1(P) - z_0(P)| e^{a(P)}. \tag{VC29}$$

The $n = 1$ connected term is retained. Evaluation at the trivial external pair is a bounded evaluation in the same coefficient norm. Therefore VC28–29 include the scalar vacuum part of the completed R output. If the original convention subtracts that value from a displayed nonconstant activity, the equal scalar is placed in the original vacuum sum and compared there. It is not lost. □

The original empty-Mayer-family term already contains $T_X 1$. Consequently this proposition compares absolute R activity masses and the vacuum increment they generate; it is stronger than a statement only about $K_X F - K_X 1\, F(0)$. On the other hand, an independently displayed external scalar increment c_i contributes the separate difference $c_1 - c_0$. VC28 cannot erase it.

128.8 What this changes in the torus comparison

The new row can be inserted into the complete-state recursion of TD without a lower bound for a summed cold-component mass. It is sufficient to prove the matched input defects in VC7 and the exterior potential norm, and then use VC28 for the completed R output. In particular an old-scale multiplicity inside the same positive body need not be transported outside the integral and estimated by an uncontrolled sum $\sum_j L^{4(k-j)}$.

This does not settle either the clean-state input row or the reference comparison row. The quantity entering VC7 is the difference of the **actual** two exponents on the **same** complete measure support. Neither the norm of one exponent nor the locality of one physical Hessian proves that statement.

If a nonwrapping complete carrier has been literally matched between T_* and qT_*, its q^4 translates have the same multiplicities. Division by physical volume cancels that factor. If a complete carrier wraps, its tail must be paid by the actual carrier estimate. The complete carrier includes the retained constraints, determining prefixes, gauge paths, and normalizer read sets; a small visible support does not by itself certify that it is nonwrapping.

Once the complete differences at level j are bounded by Ae^{-cT_*/ℓ_j}, TD supplies the finite physical-volume sum

$$\sum_j \ell_j^{-4} A e^{-cT_*/\ell_j} < \infty, \qquad \ell_j = L^j a, \tag{VC30}$$

with constants independent of the fine cutoff and of q. The completed R formation estimate is now available for that argument. The hypothesis on the **complete** differences has not been proved by VC30.

The alternative pressure route remains unchanged. The original finite-torus stability bound compares the Wilson pressure with its own finite-ambient-torus vacuum. A common counterterm or a uniform finite-physical-size pressure estimate would close the remaining step, but neither follows from this local formation theorem. The elementary same-Haar seam estimate still has the ultraviolet surface cost described in V7N43–46 and cannot supply S94.

128.9 Exact remaining source-zero obligations

Obligation	What is now available	What remains unproved
History-dependent exponent differences in one original R body	VC6–11, including old-scale logarithmic scalar costs, the actual history sum, and one shared energy reserve	The bound VC7 for all actual cross-torus inputs
Formation normalization and absolute R vacuum increment	Exact VC21–22 recombination; VC25–29 compare raw activities and their scalar connected value	Differences of genuine internal preparation denominators and independent external scalar increments
Copied finite reference body	V7N38–40 gives equality of its minimum, finite raw denominator, and original count factors	Identification of every original prepared body with such a copied complete record
Original clean and localized seeds	HC and V7N compare physical current Hessians and nonlinear minima on their admitted charts; ME60–70 gives joint reference locality on its common weak chart	The corresponding bounds for all original typed factors, finite-domain prescriptions, background extensions, scalar normalizations, and inherited readouts
Sharp selector and constraint records	All are retained unchanged in VC1–29	A literal common record or a controlled original change of variables when a torus change moves a readout or selector boundary
Distance and scale counts	VC19 can pay old counts inside a reserve-bearing live body; RC13 pays birth-distance defects outside it	A suitable bound for every remaining mixed primitive and every uncanceled normalizer; adaptive locality alone is insufficient for those rows
Cover uniformity	Exact q^4 lifting once complete nonwrapping records are matched	Complete-carrier naturality and wrapped-tail bounds for the full original inherited construction
S94	A completed original R formation difference estimate, ready for a complete-state comparison	The complete source-zero comparison itself

The distinction between a quantitative formation map and its actual two-torus input hypotheses is maintained throughout. No claim here closes LM47, all-history MT36, the weak-trajectory issue, or

the interacting four-dimensional Yang–Mills existence proof.

131.6 A nonlinear Wilson example separates the two norm requirements

This example is not a counterexample to LM47. It proves that the unweighted AE8 ball is not necessary for the displayed scalar contact estimates, even on actual compact Wilson configurations. It proves one non-implication, not an ordering of the two combined flux/current domains.

Let $\ell = 1$. Choose a fixed smooth compactly supported real one-form B on the interior of a four-dimensional unit cube with $dB \neq 0$, and a fixed Cartan generator T of $SU(N)$. For

$$1 < \alpha < 2 - \frac{4}{p}, \qquad \sigma_n = L^{-n}, \qquad a_n = L^{-2n}, \tag{AE9.19}$$

put $A_\mu^{(n)}(x) = \sigma_n^{\alpha-1} B_\mu(x/\sigma_n)T$, extended by zero away from its small cube. Define actual links by line integration:

$$U_\mu^{(n)}(x) = \exp\left(T\sigma_n^{\alpha-1} \int_0^{a_n} B_\mu((x + t\hat{\mu})/\sigma_n)\, dt\right). \tag{AE9.20}$$

All links commute. The plaquette logarithm is therefore the exact surface integral of the ordinary curl. Since $a_n/\sigma_n \to 0$, the lattice Riemann sums give

$$\|F_{a_n}(U^{(n)})\|_{L^p} \asymp \sigma_n^{\alpha+4/p-2}, \qquad \epsilon_{\Delta_n}(U^{(n)}) \asymp \sigma_n^{\alpha}. \tag{AE9.21}$$

The first exponent is negative by AE9.19. Thus these configurations leave every fixed small unweighted weak curvature ball.

For completeness, the physical directional currents also have a direct lattice bound. In the commuting sector each is the adjacent-plaquette difference of the derivative of the same scalar Wilson potential, divided by a_n^3. The two plaquette angles are $O(a_n^2\sigma_n^{\alpha-2})$, and their difference is $O(a_n^3\sigma_n^{\alpha-3})$. The derivative of the scalar potential has uniformly bounded derivative on that small angle interval. Hence

$$|J_{\nu;\mu}(U^{(n)})| \le C\sigma_n^{\alpha-3}, \qquad \mathcal{M}_{\Delta_n}(U^{(n)})^2 \le C\sigma_n^{2\alpha}. \tag{AE9.22}$$

The same bound applies to $Q_{\nu;\mu}$. A fixed number of σ_n-cells contains the support and all its one-stencil extensions. The remaining cells have zero flux and current. Thus

$$\mathfrak{F}_4(U^{(n)};1) \le C\sigma_n^{4\alpha-4} \longrightarrow 0, \qquad \mathfrak{J}_2(U^{(n)};1) \le C\sigma_n^{2\alpha-2} \longrightarrow 0. \tag{AE9.23}$$

The common finite-prefix bounds are uniform because these are compact group elements. The exact CQ and CS scalar remainders therefore retain their fourth-power own-word-mesh bound, although the AE8 unweighted curvature norm diverges.

This example uses no formal continuum current in the contact identity. The continuum one-form only specifies a sequence of actual links; the stated asymptotics follow from exact commuting plaquette integrals and convergent lattice Riemann sums.

It does not show that the original prepared map outputs these configurations. A constrained minimization can smooth or remove a localized fluctuation, and the original selectors and target records have not been checked for this example. It must not be labeled a first-entry counterexample.

There is also a useful limitation of energy-only reasoning. If the same construction uses $0 < \alpha < 1$, then

$$\beta S_0(U^{(n)}) \asymp \beta\sigma_n^{2\alpha}, \qquad \mathfrak{F}_4(U^{(n)};1) \asymp \sigma_n^{4\alpha-4} \longrightarrow \infty. \tag{AE9.24}$$

At fixed β, the action tends to zero. The same holds along any sequence with $\beta_n\sigma_n^{2\alpha} \to 0$. Consequently an exponential energy reserve by itself cannot supply a uniform pointwise bound for the atlas moment. Actual preparation smoothing, selector restrictions, or a legitimate original record charge is necessary. This is again a statement about what an energy bound implies, not an assertion about the actual prepared image.

131.8 The endpoint identity and all-history grades remain separate

The M8 retraction agrees with the actual source on its identified physical critical graph. Neither AE9.8 nor AE9.15 identifies an arbitrary prepared wall field with that graph. They instead evaluate the explicit bare polynomials directly at the given field.

For a general previously generated marked coefficient, the following still have to be checked together:

1. Its actual argument is in its original complete paired core, including independent-current and nested-readout conditions.
2. Its chosen source extension has the correct physical endpoint with the original constraints, gauges, selectors and normalizers.
3. Its saved coefficient factors survive the required changes of domain, or any loss is paid from a specified unused fraction of the same original reserve.
4. Derivatives of field-dependent old coefficients, current partners, cutoff currents and subsequent contacts are included in that joint estimate.

The sharper mixed moments in AE9.6 address only the explicit bare scalar row where their sufficiency was proved. They do not remove any of these remaining obligations.

132.2 A small density tilt needs no internal score estimate

For a bounded measurable multiplier h, define the whole commutator directly by

$$\mathcal{C}_h(\mathcal{P}) = h\tau - \sum_w \mathscr{D}_w(h\nu_w). \tag{CT9.4}$$

Every multiplication in CT9.4 is a bounded function times an ordinary measure. The definition makes sense for a sharp selector. It never multiplies a discontinuous function by a derivative distribution.

Since CT9.2 says $\mathcal{C}_1(\mathcal{P}) = 0$, exactly

$$\mathcal{C}_h(\mathcal{P}) = (h-1)\tau - \sum_w \mathscr{D}_w((h-1)\nu_w). \tag{CT9.5}$$

Consequently

$$\boxed{\mathfrak{M}_{\mathcal{W}}(\mathcal{C}_h(\mathcal{P})) \le \int |h-1|\, dm_{\mathcal{P}} \le \|h-1\|_\infty \|\mathcal{P}\|_{\mathrm{pair}}.} \tag{CT9.6}$$

This proof is the displayed representation itself. It has no derivative of h, no inverse conditional mass, and no extra coordinate count. It preserves every numerical incoming prefactor that is already present in both partner measures.

For comparison, when h is smooth, the ordered product rule expands the same object as

$$\mathcal{C}_h(\mathcal{P}) = \sum_{w\in\mathcal{W}} \sum_{A\subsetneq[r_w]} \mathscr{D}_{w_A}\big((Y_{w_{A^c}}h)\nu_w\big). \tag{CT9.7}$$

The subwords have their original order; the empty word gives an ordinary scalar measure. Equation CT9.7 includes every inherited-score and lower-current term. CT9.6 estimates their complete sum in its paired representation, without separately estimating the possibly large derivatives in CT9.7.

For one direction Y, the usual effective-score identity is

$$\mathsf{D}^{\dagger}_{\rho_0 e^{-W},Y}F = \mathsf{D}^{\dagger}_{\rho_0,Y}F + (YW)F = e^{W}\mathsf{D}^{\dagger}_{\rho_0,Y}(e^{-W}F). \tag{CT9.8}$$

Thus the additional score is present. CT9.5 packages its complete differential completion when the relative multiplier is controlled. It does not say that an arbitrary full inherited W is small. In particular, a small complex perturbation of a real R reference body leaves the entire real inherited action in that reference body.

132.4 Later contacts with field-dependent source coefficients

Let G be smooth on the common neighborhood required by the original words. For a word w, denote its full action by Y_wG. Put

$$\begin{aligned}\tau_G &= G\tau - \sum_w (Y_wG)\nu_w,\\ \nu_{w,A}^G &= (Y_{w_{A^c}}G)\nu_w, \qquad \varnothing \neq A \subseteq [r_w].\end{aligned} \tag{CT9.13}$$

The exact ordered product rule proves

$$\tau_G = \sum_w \sum_{\varnothing \neq A \subseteq [r_w]} \mathscr{D}_{w_A} \nu^G_{w,A}. \tag{CT9.14}$$

This is the product-completed pair $\mathcal{P}_G$. Its ordinary scalar term includes the full last contact. Its positive-order coefficients include all intermediate contacts and field-dependent coefficient derivatives.

For bounded h, multiplication of a distribution by the smooth function G is defined. Applying the same product rule to CT9.4 gives

$$G\mathcal{C}_h(\mathcal{P}) = \mathcal{C}_h(\mathcal{P}_G). \tag{CT9.15}$$

No derivative of h is used to prove this equality. This matters when h includes a sharp selector.

If every word has length at most r, define

$$\|G\|_{\mathcal{Y},r} = \max_{\substack{w \in \mathcal{W} \\ A \subseteq [r_w]}} \|Y_{w_A} G\|_\infty.$$

Then, with the original word multiplicity retained,

$$\|\mathcal{P}_G\|_{\mathrm{pair}} \leq C_r \|G\|_{\mathcal{Y},r} \|\mathcal{P}\|_{\mathrm{pair}}, \qquad \mathfrak{M}(G\mathcal{C}_h(\mathcal{P})) \leq C_r \|h-1\|_\infty \|G\|_{\mathcal{Y},r} \|\mathcal{P}\|_{\mathrm{pair}}. \tag{CT9.16}$$

One can instead retain the integral of the pointwise derivative envelope against $m_{\mathcal{P}}$; this is the useful version when its polynomial energy cost is paid by the shared reserve.

For the finite nilpotent source polynomial $A(\eta) = \sum_{\varnothing \neq I} \eta_I A_I$, apply CT9.13–16 to each ordinary coefficient of $G = e^{A(\eta)}$. The exact multiplier h is source independent, as in the original frozen source-zero kernel; no sourced component denominator is introduced. The exact partition formula for that coefficient is XP4. An ordered derivative is distributed over those factors, with no reordering. Thus, at fixed finite source degree and current order,

$$\mathfrak{M}\Big([\eta_I] e^{A(\eta)} \mathcal{C}_h(\mathcal{P})\Big) \leq C_{I,r} \|h-1\|_\infty \|\mathcal{P}\|_{\mathrm{pair}} \sum_{\pi \in \Pi(I)} \prod_{B \in \pi} \|A_B\|_{\mathcal{Y},r}. \tag{CT9.17}$$

Empty coefficients and an already source-valued pair are treated by the literal coefficient convolution. This preserves the original finite labels, source anchors, mesh factors, and saved amplitudes. It does not assert the displayed derivative norm for an unadmitted inherited A_B. A relation valid only on a physical graph supplies only its physical tangent derivatives, not arbitrary ambient independent-current derivatives.

The proof never multiplies two preexisting derivative distributions. Every later physical insertion is first a smooth coefficient of the original finite-source integral; CT9.13 constructs its complete packet before another selected differential representation is used.

132.5 Application to the original R-II density comparison

In the actual fixed-coordinate component of R-II (1.71), NC1–4 gives

$$T_z = e^{\sigma(z,\xi)} T_0, \quad |\sigma| \leq \varepsilon < \tfrac{1}{4}, \quad m(z) = K_0(e^\sigma), \quad K_0 = T_0/T_0(1). \tag{CT9.18}$$

All original selectors, constraints, internal histories, scalar normalizers, and inherited real interactions are in T_0. The ratio concerns only the already admitted outer complex comparison. Its bound is pointwise in the actual integration coordinates.

For the unnormalized ratio $h = e^{\sigma}$, $\|h-1\|_{\infty} \le e^{\varepsilon}-1$. For the normalized ratio $h = e^{\sigma}/m$, exactly

$$|m-1| \le q_{\varepsilon}, \qquad \|h-1\|_{\infty} \le \frac{2q_{\varepsilon}}{1-q_{\varepsilon}} \le C\varepsilon, \quad q_{\varepsilon} = e^{\varepsilon}-1. \tag{CT9.19}$$

There is no inverse power of the possibly small $T_0(1)$. Consequently, for any complete pair satisfying CT9.2 on this same reference realization and the stated domination condition,

$$\mathfrak{M}(\mathcal{C}_h(\mathcal{P})) \le C\varepsilon\|\mathcal{P}\|_{\mathrm{pair}}. \tag{CT9.20}$$

Unlike NC7, CT9.20 requires no derivative of σ in the word's internal directions. Those directions need not be converted into outer Cauchy directions to prove this particular commutator bound. Their original coefficient and readout norms still occur in $\|\mathcal{P}\|_{\mathrm{pair}}$.

This closes an additional quantitative row: a bounded *relative* original density change acts on a previously controlled complete pair with its actual small factor. It does not prove that adjoining the reference kernel K_0 itself has a bounded internal derivative. That operation must already have produced the common pair in CT9.2. Nor does this calculation bound all internal derivatives of the inherited real density hidden in T_0.

For a genuine normalization, multiply its ordinary coefficients before taking the adjoint. Where $ZK = T$ is the actual original factorization, the corresponding products of ordinary coefficient measures agree exactly and give the same CT9.4. This retains every normalization commutator without bounding Z_0^{-1} in isolation. A genuine preparation denominator already present in T_0, or an independent external vacuum increment, is not canceled by this identity.

132.7 The bare CQ and CS complete pairs have explicit own-mesh energy envelopes

This subsection concerns the actual bare compact Wilson variables, before any action is canceled in a reference insertion. It supplies creation bounds, not all-history transport. The tests f, h are the original field-independent, color-scalar tests of CQ and CS.

Let μ be the original Wilson measure, $Y_i = g_W X_i$, $\beta = c_\beta g_W^{-2}$, and let V be either CQ20's complete vector $\widetilde{V}^{f,h}$ or CS28's complete vector V^c. The normalized paired identity is

$$\tau^{\mathrm{can}} = g_W^{-4} a^4 \mathscr{E}[V]\,\mu = \sum_i \mathscr{D}_{Y_i} \nu_i^{\mathrm{can}}, \qquad \nu_i^{\mathrm{can}} = g_W^{-5} \kappa a^8 V_i\,\mu. \tag{CT9.25}$$

Every displayed a is this contact word's original mesh.

For one fixed local source-root contribution and its complete plaquette star, the coefficient of ν_i^{can} is a sum of terms of the form

$$c_\beta^{-2} g_W^{-1} u_i(U) + c_\beta^{-3} g_W v_i(U), \tag{CT9.26}$$

where u_i, v_i are bounded-length compact Wilson words, vanish at least to first order in plaquette defects, and have bounded fixed finite Haar derivatives. Their coefficients retain the literal source weights.

For the first term, $P_h = a^{-2} \sum h(XS_0)^2$, or the analogous partial-force square for H, so $XP_h = a^{-2}$ times a first-order vanishing word. The leading vector is $\kappa X P_h = \beta^{-1} a^{-5} u$, yielding the first term of CT9.26 after CT9.25. The other vector terms are $\beta^{-2} a^{-5}$ times a first-order word. For CQ's weighted Casimir term, use the exact commutation of each link Casimir with all invariant derivatives:

$$\sum_b f_b \Delta_b (X_i S_0) = X_i \left(C_F \sum_p F_p^f \,\mathrm{Re}\,\mathrm{tr}\, P_p \right). \tag{CT9.27}$$

The derivative of the real trace vanishes at the trivial plaquette. For CS, the extra vector is a linear combination of the actual partial forces and weighted two-plaquette differences, so the same word bound holds. Source endpoint differences are retained, not estimated as field derivatives.

With the local original energy $\mathscr{E}_Y = \beta \sum_{p \subset Y} s(P_p)$, LM2 and LM8 now give

$$\sum_i \left| \frac{d\nu_i^{\mathrm{can}}}{d\mu} \right| \le C_{f,h} \mathscr{E}_Y^{1/2}, \qquad \left| \frac{d\tau^{\mathrm{can}}}{d\mu} \right| \le C_{f,h} (1 + \mathscr{E}_Y). \tag{CT9.28}$$

Indeed $g_W^{-1}|u| \le C \mathscr{E}_Y^{1/2}$, $Y(g_W^{-1} u) = Xu$ is bounded, and the canonical Wilson score is bounded by the square root of the same enlarged-star energy. The $g_W v$ term is smaller. The exact adjoint contains both the coefficient derivative and the score product, giving CT9.28. Finite later bare contacts have the corresponding higher but fixed polynomial degree by LM6–9 and CT9.13.

Thus a bare complete CQ/CS pair has a genuine same-measure, own-mesh creation envelope. For many original roots, retain their actual sum and enlarged energy union. A physical box containing order a^{-4} fine roots is not one fixed microscopic root. If the Wilson action for a compact slot has already been canceled, CT9.28 cannot borrow it as a reserve; only the appropriate compact supremum cost and its existing record bank are available.

132.8 The scalar fourth-power remainder is not a complete-pair bound

132.8.1 An exact nonlinear flat-field count

Take $f = h = 1$ on a periodic four-dimensional lattice with V_a fine sites and at least three sites in every periodic direction. There are $4V_a$ independent positive bonds and $6V_a$ plaquettes. CQ16–22 gives

$$A_0 = 216 d_G V_a, \qquad D_p = 2C_A - 32C_F. \tag{CT9.29}$$

At the trivial field, $J = 0$, the contact $C_{1,1} = 0$, the plaquette action vanishes, and the quartic remainder vanishes. CQ23 therefore gives the exact scalar companion value

$$g_W^{-4} a^4 \mathscr{E}[\widetilde{V}^{1,1}](I) = -\left[432 c_\beta^{-2} d_G + 6N C_F c_\beta^{-3} g_W^2 (2C_A - 32C_F)\right] V_a. \tag{CT9.30}$$

The leading coefficient is nonzero. On a fixed physical torus V_a is proportional to a^{-4}.

For the pointed H packet with the same constant tests, $\sum_p t_p = 72V_a$ and $v_p = 0$. CS28 gives, exactly,

$$g_W^{-4} a^4 \mathscr{E}[V^c](I) = -96 c_\beta^{-2} d_G V_a. \tag{CT9.31}$$

Consequently a pointwise norm of the whole scalar companion on the physical weak chart does not have the scalar remainder's fourth-power grade. It does not even have a uniform fixed-physical- volume ceiling before the quantum cancellation. The identity row and its opposite Ward trace must be retained in the completed quantum operation.

This does not contradict CT9.28: the latter is a bound per actual microscopic root, and CT9.30–31 retain all V_a roots.

132.8.2 A bounded test after the complete Gaussian cancellation

The obstruction is not confined to a pointwise trace before integration. Use CP33–39 on the fixed physical transverse slice. Let

$$\Delta_c = (QH^{-1}Q^{\mathsf{t}})^{-1}, \qquad R_a = QHQ^{\mathsf{t}}.$$

The complete coarse conditional Ward scalar, including its actual vacuum trace, is

$$W_c(B) = 2\beta^{-1}\left(\langle \Delta_c B, R_a \Delta_c B\rangle - \beta^{-1}\operatorname{Tr}(R_a\Delta_c)\right) = \mathsf{D}^\dagger_{\rho_c}(2\beta^{-2} R_a \Delta_c B). \tag{CT9.32}$$

The coarse Gaussian covariance is $\beta^{-1}\Delta_c^{-1}$. Fix a coarse vector u and $t > 0$, and use the bounded smooth test

$$\Psi(B) = 1 - \cos(t\sqrt{\beta}\langle u, B\rangle). \tag{CT9.33}$$

In canonical coordinates $B = g_W b$, this test and every fixed derivative have bounds independent of the fine mesh and g_W. Gaussian integration by parts, or differentiation of its characteristic function, gives exactly

$$\begin{aligned} g_W^{-4} a^4 \int W_c(B)\Psi(B)\, d\mu_c(B) = 2c_\beta^{-2} t^2 a^4 \langle u, R_a u\rangle \\ \times \exp\left(-\frac{t^2}{2}\langle u, \Delta_c^{-1} u\rangle\right). \end{aligned} \tag{CT9.34}$$

For completeness, $\nabla\Psi = t\sqrt{\beta}\,u\sin(t\sqrt{\beta}\langle u,B\rangle)$; the Gaussian identity

$$\mathbb{E}[\Delta_c B\sin(t\sqrt{\beta}\langle u,B\rangle)] = t\beta^{-1/2}ue^{-t^2\langle u,\Delta_c^{-1}u\rangle/2}$$

proves the displayed coefficient and sign.

Use exactly the CP28–39 configuration: a fixed even torus of side T, retained spacing one, $a = 1/n$ with n odd, and the normalized Cartan generator H_{C}. The coarse vector has $u_2(y) = \epsilon(-1)^{y_1}H_{\mathrm{C}}$ and all other components zero. Its actual adjoint image is the block step field $Q^{\mathrm{t}}u = j_a$, with the same values on the centered unit blocks. There are T jump planes with jump 2ϵ, so CP39 gives $\langle u, R_a u\rangle = 4\epsilon^2T^4/a$.

The same field gives the direct uniform covariance bound

$$\langle u,\Delta_c^{-1}u\rangle = \langle j_a, H^{-1}j_a\rangle \le CT^2\|j_a\|_{L^2_a}^2 = C\epsilon^2T^6.$$

Here constant/harmonic modes remain removed or fixed as in CP33, and the transverse periodic Laplacian has the fixed-volume spectral gap cT^{-2}. Thus the exponential factor in CT9.34 has a positive mesh-independent lower bound. It follows that

$$\left|g_W^{-4}a^4\int W_c\Psi\,d\mu_c\right| \ge c_{u,t,T,\epsilon,c_\beta}a^3. \tag{CT9.35}$$

Any output norm which controls pairing against this fixed bounded canonical smooth test therefore cannot give a uniform Ca^4 bound for this whole Ward distribution.

The test can also be put in the physical free test class. The chosen u is transverse, so its pairing is invariant under linearized local gauge shifts. To impose residual global color invariance, replace Ψ by

$$\overline{\Psi}(B) = \int_G \left[1-\cos\Big(t\sqrt{\beta}\langle \mathrm{Ad}_g\,u, B\rangle\Big)\right]\,dg,$$

with normalized Haar measure. The Gaussian covariance, R_a, and Δ_c are color diagonal. Their two quadratic forms on $\mathrm{Ad}_g\,u$ equal their values on u, so CT9.34–35 and the uniform canonical derivative bounds are unchanged by this average. This is a gauge-invariant test for the free linearized theory. It is not asserted to be a uniformly controlled nonlinear Wilson representative.

This is the actual flat constrained Gaussian sector already calculated in CP. It is not asserted to be a new nonlinear counterexample to all-history MT36. It identifies an unavoidable distinction between the projected scalar remainder and the current/contact sector. A satisfactory recursion must keep the latter's different grade or exhibit a further exact cancellation with the corresponding later contacts. Merely taking more generic regular-prefix jets does not produce the missing grade.

132.9 What a readout comparison would have to estimate

If $q : M \to N$ is an actual common smooth readout, then

$$\langle q_*\mathcal{C}_h(\mathcal{P}),\varphi\rangle = \int\varphi(q)(h-1)\,d\tau - \sum_w\int Y_w(\varphi\circ q)(h-1)\,d\nu_w. \tag{CT9.36}$$

The original full chain rule supplies all tensors in this formula. CT9.6 controls the input current coefficients, while the needed output norm uses the actual jets of q. Replacing a complete mixed readout by a regular-prefix map is not part of the identity.

For two actual maps q_0, q_1 in one common chart, subtract CT9.36. A uniform bound on their jets through order r, together with a C^r jet difference of size δ, gives

$$|\langle (q_{1*} - q_{0*})\mathcal{C}_h(\mathcal{P}), \varphi\rangle| \le C_{r,M}\delta \|h-1\|_\infty \|\varphi\|_{C^{r+1}} \|\mathcal{P}\|_{\mathrm{pair}}. \tag{CT9.37}$$

The extra derivative of the test comes from comparing its derivatives at two different output points. This is a weak test-functional estimate, not a total-variation estimate between nearby constraint graphs. Small coordinate displacement alone does not make two point masses close in total variation. The actual common chart and its mixed jets remain to be proved.

A genuine two-slot gradient contact behaves differently: if both slots are $F(q)$ and $G(q)$, its chain rule uses $Dq \otimes Dq$, not the higher jets of a one-slot differential distribution. DC9 proves the corresponding stronger regular-prefix contraction for the stated joint ordinary tensor, with its incoming mass and original couplings retained. For CQ, however, the displayed slots are the fine Wilson action and the fine current-square source. Neither is constant on a general fluctuating coarse fiber. Equality on a physical critical graph does not establish a two-slot pullback identity inside the original integral. An exact deterministic/ proper-difference splitting retains its mixed and proper terms.

133.1 The two-slot structure gives a different estimate

C7 proves that a fixed-order jet of a long regular averaging prefix has one strict first-column factor. It does not give one factor for each derivative of a single composite function. Consequently its generic second-current estimate does not close the D8 grade.

There is a narrower, useful class for which two factors really do occur. It consists of contact forms with two separately retained first-derivative slots. The same original observation must act on both slots, and their coefficient must be an ordinary joint tensor measure. No product of independently estimated distributions is formed. The first-jet estimate then applies twice, without invoking a second jet of the observation. This section proves that assertion, identifies an actual Wilson Ward contact having this structure, and states why it does not include the entire CQ or CS packet.

The identities below hold at finite cutoff. All source labels, old engineering factors, contact-word meshes, scalar normalizers and constraint records are retained in the coefficient measures. They are never reset when an observation is pushed forward.

133.2 Pushforward of an ordinary joint contact tensor

Let the input and output admit the actual fixed coordinate charts under discussion. Write their canonical directions as $Y_i = g_{\mathrm{in}} X_i$ and $Z_\alpha = g_{\mathrm{out}} X'_\alpha$. For the original smooth observation q, define its first response by the identity

$$Y_i(F \circ q) = \sum_\alpha A_i^\alpha[q](x)(Z_\alpha F) \circ q. \tag{DC9.1}$$

This defines the actual canonical matrix, including chart and frame factors. If ordinary dimensionless coordinates are used, its leading expression is $(g_{\mathrm{in}}/g_{\mathrm{out}})Dq$. At a relative-chart conversion all lower frame factors are those of the specified terminal and initial charts, not a new constant at each old step.

Let ν_{ij} be an ordinary finite tensor measure on the original joint configuration/record space. The form is

$$\mathcal{B}_\nu(F,G) = \sum_{i,j} \int (Y_i F)(Y_j G)\, d\nu_{ij}, \qquad \|\nu\|_{1,\mathrm{TV}} = \sum_{i,j} |\nu_{ij}|(\mathcal{X}). \tag{DC9.2}$$

The actual coefficient norm may be a stronger weighted version of this norm. Any saved weight which is constant on a literal record is carried inside its summand; the calculation is performed before summing histories. The measure may be signed, complex, or finite dimensional tensor valued. It need not be a product measure or a probability measure. Original delta constraints remain part of it.

Define

$$\eta_{\alpha\gamma} = q_* \left(\sum_{i,j} A_i^\alpha A_j^\gamma \, \nu_{ij} \right). \tag{DC9.3}$$

Applying DC9.1 once to each separate function proves exactly

$$\mathcal{B}_\nu(F\circ q, G\circ q) = \mathcal{B}_\eta(F,G). \tag{DC9.4}$$

No D^2q term occurs in this identity. By the triangle inequality and the pushforward variation bound,

$$\|\eta\|_{1,\mathrm{TV}} \le \left(\sup_{x,i}\sum_{\alpha}|A_i^\alpha(x)|\right)^2 \|\nu\|_{1,\mathrm{TV}}. \tag{DC9.5}$$

For a tensor carrying two different admitted observations, replace the square by the product of their two actual column bounds. This does not prove that two unrelated source preparations have a common admitted chart.

The same proof holds for an explicitly retained r-slot form

$$\mathcal{B}_\nu^{[r]}(F_1,\dots,F_r) = \sum_{i_1,\dots,i_r}\int \prod_{s=1}^{r} Y_{i_s}F_s\, d\nu_{i_1\cdots i_r}. \tag{DC9.6}$$

Its pushed tensor has one first-response factor for each distinct slot. This is a statement about the given joint tensor. It does not create a bounded cross-diagonal operation on two separately bounded weak source functionals; C7.23–24 still applies to that different operation.

133.3 Quantitative application to the actual regular prefix

Take the admitted compatible regular prefix Q_n of C7.13–22, with the original physical action and complete determining prefixes. Its dimensionless first-column bound is

$$\sup_i\sum_\alpha |(DQ_n)_i^\alpha| \le C\lambda^n, \qquad \lambda \le 2L^{-3} < 1. \tag{DC9.7}$$

At fixed initial and final canonical charts, DC9.5 gives

$$\|\eta_n\|_{1,\mathrm{TV}} \le C\lambda^{2n}(g_{\mathrm{in}}/g_{\mathrm{out}})^2\|\nu\|_{1,\mathrm{TV}}. \tag{DC9.8}$$

The fixed chart constants are included once. The actual weak trajectory, when invoked, bounds the displayed coupling ratio; its proof is not supplied by this estimate. Every original birth coupling already in ν remains there as well.

For a two-slot packet whose **already specified** engineering conversion is the D8 factor L^{4n}, the transported bound is

$$L^{4n}\|\eta_n\|_{1,\mathrm{TV}} \le C(4/L^2)^n(g_{\mathrm{in}}/g_{\mathrm{out}})^2\|\nu\|_{1,\mathrm{TV}}. \tag{DC9.9}$$

Thus this restricted contact form has a strict saved-age gain for the once-chosen large L. In contrast, the generic D8 whole current has only $L^{4n}\lambda^n$ from C7.19. No stronger higher-jet bound has been asserted.

More generally an r-slot tensor with declared dimension $d = 4+2r$ has the factor $L^{2rn}\lambda^{rn} \le (2/L)^{rn}$, with its literal canonical ratios. Dimension is a recorded property of the packet, not a consequence of how many slots one chooses to write. If the incoming tensor mass itself contains an inverse mesh power, that power remains on the right side of DC9.8–9. The estimate cannot hide an extensive identity coefficient or an unproved incoming norm.

For individually gauge-invariant terminal functions, CL15 and C7.21 identify their actual raw pullbacks with their compatible pullbacks. Their first derivatives therefore agree. DC9.4–9 then describes the

same raw contact on that test class. A color-linked function across different depths must retain the common endpoint/gauge data and prove its own identity; separate gauge restoration is not assumed for it.

For the literal compatible stack $(Q_0, \dots, Q_N)$, C7.22 bounds the first-column sum by $C/(1-\lambda)$. DC9.5 therefore gives a bounded, depth-independent two-slot stack transport. It gives no strict age gain for a slot which still reads the identity level. A general mixed R map is not identified with this stack.

133.4 The exact Wilson contact in the retained sector

Here use the real original Wilson measure and its finite source conventions CP1–3. For two smooth invariant functions F, G on one admitted retained space, let

$$V_i = X_i(F \circ q), \qquad H = G \circ q. \tag{DC9.10}$$

The exact pointwise product identity is

$$H\mathscr{E}[V] = \mathscr{E}[HV] + \kappa a^4 \sum_i X_i(F \circ q) X_i(G \circ q). \tag{DC9.11}$$

The final term is the Dirichlet contact just estimated. In canonical input directions its joint coefficient measure, when paired with the original body, is $\nu_{ij} = \kappa a^4 g_{\mathrm{in}}^{-2} \delta_{ij}\, d\mu$, with any original scalar carrying coefficient included once. Its actual mass and source normalization must still be used; they are not declared scale independent by writing this formula.

The whole current has an exact measure-level pushforward as well. For the convention $\langle \mathscr{D}_{X_i}\zeta, \varphi\rangle = \int X_i \varphi \, d\zeta$, the bare Ward identity is

$$\mathscr{E}[V]\, d\mu = \kappa a^4 \sum_i \mathscr{D}_{X_i}(V_i\, d\mu). \tag{DC9.12}$$

Testing against $\varphi \circ q$ and using the first chain rule gives the coarse flux measure

$$q_*\big(\mathscr{E}[V] d\mu\big) = \kappa a^4 \sum_\alpha \mathscr{D}_{X'_\alpha} q_* \left(\sum_i (X_i q)^\alpha V_i \, d\mu \right), \tag{DC9.13}$$

where $(X_i q)^\alpha$ denotes the actual invariant response, including the stated frame. No conditional density or division by its mass is needed to state this identity. For a constrained measure, DC9.13 remains true whenever its incoming paired-current identity is the original one, with all constraint and boundary partners retained; the unconstrained bare identity is not silently imposed on a selected fiber.

If $V = X(F \circ q)$, the coarse flux in DC9.13 uses the same two first-response factors as DC9.3. The equality is a divergence-form identity for a measure, not a claim that an individual fine gradient is the gradient of the coarse effective action.

133.5 Inherited densities and later retained insertions

For an actual smooth inherited density $d\mu_W = Z_W^{-1} e^{-\beta S_0 - W} \prod dU$, define

$$J_i^W = J_i + \kappa X_i W, \qquad \mathscr{E}_W[V] = a^4 \sum_i (J_i^W V_i - \kappa X_i V_i). \tag{DC9.14}$$

Haar integration by parts proves DC9.12 with $(\mathscr{E}_W, \mu_W)$. The product identity DC9.11 has the same contact term. Thus the retained Dirichlet sector and its full divergence-form current survive the **actual** inherited density; the score contribution $\kappa X W$ is included exactly.

For fixed ordinary coefficient tensors, multiplication by any bounded density factor h simply replaces ν by $h\nu$ in DC9.3. Consequently

$$\|\eta[h\nu]-\eta[\nu]\|_{1,\mathrm{TV}} \le \left(\sup_{x,i}\sum_{\alpha}|A_i^{\alpha}|\right)^2 \|h-1\|_{\infty}\|\nu\|_{1,\mathrm{TV}}. \tag{DC9.15}$$

The model-specific small factor of NC1–7 can be used here on its actual comparison chart and support. Its multiple-component payment uses NC19–23 and the same original VC reserve. No new reserve is allocated merely because two derivative slots occur.

For later smooth retained scalar insertions $G_1(q),\dots,G_m(q)$, the complete product rule is

$$\begin{aligned}\mathscr{E}_W[V]\prod_{r=1}^{m}G_r(q) = \mathscr{E}_W\left[V\prod_{r=1}^{m}G_r(q)\right]\\ +\kappa a^4\sum_{r=1}^{m}\sum_i V_iX_i(G_r\circ q)\prod_{s\neq r}G_s(q).\end{aligned} \tag{DC9.16}$$

If $V=X(F\circ q)$, each displayed contact has two separately retained first-derivative slots. The remaining scalar insertions are ordinary bounded carrying factors. This proves the exact transport of one such current and all of these later scalar contacts. It uses no derivative estimate for W in the contact row and does not differentiate a selector by a formal substitution. When selectors are exposed, their full paired commutator is the one in CT9.

This statement does not multiply two current distributions. Contacts between several current insertions must be formed in the original finite source algebra with all their scalar partners before a weak representation is estimated.

133.6 The actual CQ and CS contacts still contain other slots

For the CQ packet, $fJ=a^{-3}fXS_0$. With constant f, the formal first slot is the fine Wilson action S_0, and the second is P_h. Neither is generally a function of a retained observation on an entire fluctuating fiber. Agreement of their values on a physical minimum does not imply equality of their derivatives on that fiber. Nonconstant tests also retain the original weighted gradient corrections. Therefore DC9.9 is not a bound for the full CQ23 current.

An exact source splitting of the kind used in XP16 has

$$F=T_F\circ q+D_F,\qquad G=T_G\circ q+D_G. \tag{DC9.17}$$

The contact then has the literal four terms

$$\begin{aligned}\mathcal{B}_\nu(F,G)=\mathcal{B}_\nu(T_F\circ q,T_G\circ q)+\mathcal{B}_\nu(T_F\circ q,D_G)\\ +\mathcal{B}_\nu(D_F,T_G\circ q)+\mathcal{B}_\nu(D_F,D_G).\end{aligned} \tag{DC9.18}$$

Only the first is in the sector proved here without further work. All four share their actual original joint measure and source labels. The remainders need their own admitted derivative/tensor bounds and may not be replaced by their scalar suprema. A small scalar remainder alone does not bound its arbitrary fine-coordinate derivative.

The positive conclusion is a strict quantitative regular-prefix estimate for an identified retained Dirichlet sector, plus exact transport of its full current through inherited smooth densities and later retained scalar insertions. It supplies a valid way two first response gains can occur. It does not improve the generic higher-jet estimate, admit actual mixed fields, establish an incoming complete D8 pair grade, or close all-history MT36.

134.2 The whole conditional family, with the scalar trace

Let

$$\Delta_\ell = (Q_{a,\ell} H^{-1} Q_{a,\ell}^*)^{-1}$$

on its actual admitted physical row space. The conditional mean and covariance are the CP33 ones. For the original fine insertion

$$W_{8,a}^0(A) = 2\beta^{-1}\langle A, H^3 A\rangle_a - 2\beta^{-2}\operatorname{Tr} H^2, \tag{GG9.5}$$

the exact conditional output is

$$\mathcal{W}_{a,\ell}(B) = 2\beta^{-1}\left\{\langle \Delta_\ell B, R_{a,\ell}\Delta_\ell B\rangle_\ell - \beta^{-1}\operatorname{Tr}(R_{a,\ell}\Delta_\ell)\right\}. \tag{GG9.6}$$

The determinant factors are those of the original conditional Gaussian density. The fine vacuum trace in GG9.5 and the conditional covariance trace have already been combined in GG9.6.

In retained units put

$$\widehat{B} = \ell B, \quad b = \widehat{B}/g_e, \quad \widehat{\Delta} = \ell^2\Delta_\ell, \quad \gamma_{e,j} = g_e/g_j.$$

The original-word-normalized scalar and its full vector are exactly

$$\begin{aligned} \mathcal{S}_r(b) &:= g_j^{-4} a^4 \mathcal{W}_{a,\ell}(B) \\ &= 2r^4\left\{c_\beta^{-1}\gamma_{e,j}^2\langle \widehat{\Delta} b, \widehat{R}\widehat{\Delta} b\rangle - c_\beta^{-2}\operatorname{Tr}(\widehat{R}\widehat{\Delta})\right\}, \\ v_r(b) &= 2c_\beta^{-2} r^4 \widehat{R}\widehat{\Delta} b. \end{aligned} \tag{GG9.7}$$

The coarse Gaussian density in b is proportional to $\exp[-c_\beta\gamma_{e,j}^2\langle b, \widehat{\Delta} b\rangle/2]$. Therefore

$$\mathcal{S}_r\, d\mu_e = \mathscr{D}_b(v_r\, d\mu_e). \tag{GG9.8}$$

This proves the displayed coupling factors, including the absence of an extra running-coupling factor in v_r.

The original flat reconstruction bounds FW12–19 give $\|\widehat{\Delta}\|_{\mathrm{op}} \le C$ on the physical row space. Equivalently, its quadratic action is no larger than the uniformly bounded-energy admissible flat reconstruction. Let m_e be the actual number of independent coarse coordinates. GG9.4 yields

$$\begin{aligned} |\mathcal{S}_r(b)| &\le Cr^3\{\gamma_{e,j}^2\|b\|_2^2 + m_e\}, \\ \sum_\alpha |v_r^\alpha(b)| &\le Cr^3\sqrt{m_e}\,\|b\|_2. \end{aligned} \tag{GG9.9}$$

Every trace count is retained. In a fixed bounded number of retained-size boxes, m_e is bounded. On a larger torus it remains the actual coarse count. There is no additional fine-site count to insert after GG9.6: the fine count occurred in its two traces and was canceled by that exact covariance calculation.

For a canonical cutoff $\|b\|_\infty \le P_e$, the pair envelope is consequently

$$|\mathcal{S}_r| + \sum_\alpha |v_r^\alpha| \le Cr^3 m_e\{1 + P_e + \gamma_{e,j}^2 P_e^2\}. \tag{GG9.10}$$

It has a fixed polynomial degree and can be inserted into the same one-reserve formation estimate, provided that this is the actual Gaussian packet and its admitted formation domain.

The full Gaussian expectation gives a bound of the same type. The flat covariance has $\operatorname{Tr}\widehat{\Delta}^{-1} \le C m_e$ in four dimensions with the harmonic mode removed. This follows from FW's coarse symbol bound $C|k|^{-2}$ and the uniform four-dimensional sum of $|k|^{-2}$ per lattice volume. Thus

$$\|\mathcal{S}_r\,\mu_e\|_{\mathrm{TV}} + \sum_{\alpha} \|v_r^{\alpha}\mu_e\|_{\mathrm{TV}} \le C r^3 m_e (1 + g_j/g_e). \tag{GG9.11}$$

For a fixed torus the spectral-gap bound alone also gives this conclusion with its explicit fixed-volume constant.

134.3 Exact closure under subsequent regular Gaussian conditioning

Let the next retained observation be the original flat $Q_{\ell,\ell'}$. The direct actual averages satisfy

$$Q_{a,\ell'} = Q_{\ell,\ell'} Q_{a,\ell}. \tag{GG9.12}$$

This is the box-and-line filter composition, not a replacement macro-average. On the admitted rows,

$$\begin{aligned} \Delta_{\ell'}^{-1} &= Q_{\ell,\ell'} \Delta_{\ell}^{-1} Q_{\ell,\ell'}^{*}, \\ R_{a,\ell'} &= Q_{\ell,\ell'} R_{a,\ell} Q_{\ell,\ell'}^{*}. \end{aligned} \tag{GG9.13}$$

Condition GG9.6 once more. Its new mean term is the quadratic term with $(\Delta_{\ell'}, R_{a,\ell'})$. The covariance correction to the old quadratic term is

$$\beta^{-1}\left\{\operatorname{Tr}(R_{a,\ell}\Delta_{\ell}) - \operatorname{Tr}(R_{a,\ell'}\Delta_{\ell'})\right\}.$$

Its first trace cancels the old scalar trace in GG9.6 exactly. Consequently

$$\mathbb{E}[\mathcal{W}_{a,\ell} \mid Q_{\ell,\ell'} B = B'] = \mathcal{W}_{a,\ell'}(B'). \tag{GG9.14}$$

This closes the entire Gaussian family through every regular conditional step. Its original mesh a, initial coupling g_j, complete scalar trace, and source label are unchanged.

Later smooth retained sources satisfy

$$\int \mathcal{S}_r G \, d\mu_e = \int v_r \cdot \nabla_b G \, d\mu_e. \tag{GG9.15}$$

The whole current and the resulting contacts therefore keep the GG9.9–11 bound with the actual derivative envelope of G. Repeated finite-source products at this retained level use CT9.13–17. This gives the closed Gaussian Ward family and its exact contacts with functions of the current retained field. Insertions at different retained levels must keep their joint Gaussian record and prove its corresponding derivative and polynomial envelope; they are not all identified with one current retained function. This is not a closure for arbitrary mixed reference kernels.

135.1 A second-current identity with the correct ordered word

For $f = h = 1$, CP5 and CQ20 give

$$\widetilde{V}_i = \kappa X_i P_1 + \gamma \beta^{-2} a^{-2} J_i, \qquad \gamma = C_A/2 - 8C_F. \tag{NH9.1}$$

Use the original Wilson measure μ and $Y_i = g_j X_i$. With CT9.1's convention,

$$\mathscr{D}_{X_j}(F\mu) = \kappa^{-1} J_j F \mu - (X_j F)\mu. \tag{NH9.2}$$

Since $X_i P_1 = 2a^4 \sum_j J_j X_i J_j$, this gives

$$X_i P_1 \, \mu = 2\kappa a^4 \sum_j \Big\{ \mathscr{D}_{X_j}((X_i J_j)\mu) + (X_j X_i J_j)\mu \Big\}. \tag{NH9.3}$$

The full Haar Casimir $\Delta_H = \sum_j X_j^2$ is central. The adjoint Casimir has the compact sign $\sum_j [X_j, [X_j, X_i]] = -C_A X_i$ on the same bond. Consequently

$$\sum_j X_j X_i X_j = X_i \Delta_H + \frac{C_A}{2} X_i, \qquad \sum_j X_j X_i J_j = (-4C_F + C_A/2) J_i. \tag{NH9.4}$$

The second equality uses CP9, $\Delta_H S_0 = 4C_F \sum_p \operatorname{Re}\operatorname{tr} P_p$. Derivatives on other bonds commute, so they add no adjoint-Casimir term.

Put $\zeta = 3C_A/2 - 16C_F$. Substituting NH9.3–4 into the complete Ward expression proves the exact distribution identity

$$\boxed{\begin{aligned} g_j^{-4} a^4 \mathscr{E}[\widetilde{V}] \, \mu = {} & 2c_\beta^{-3} \sum_{i,j} \mathscr{D}_{(Y_j, Y_i)}((X_i X_j S_0)\mu) \\ & + \zeta c_\beta^{-3} \sum_i \mathscr{D}_{Y_i}(g_j X_i S_0 \, \mu). \end{aligned}} \tag{NH9.5}$$

The word is (j, i): applying $\mathscr{D}_{X_i}$ to $\mathscr{D}_{X_j}\nu$ tests ν against $X_j X_i \varphi$. No symmetrization or reordering is made in NH9.5. If one does symmetrize, the exact Lie-bracket contribution must be transferred to the first-order row.

This identity contains the whole nonlinear Wilson current with its original scalar companion. It does not estimate that scalar companion by the quartic scalar remainder.

There is an optional second-current representation for the explicit lower row alone. Since $\mathscr{D}_{Y_i}\mu = \beta g_j X_i S_0 \, \mu$,

$$\zeta c_\beta^{-3} \mathscr{D}_{Y_i}(g_j X_i S_0 \, \mu) = \zeta c_\beta^{-4} g_j^2 \mathscr{D}_{(Y_i, Y_i)}\mu. \tag{NH9.5a}$$

It retains its explicit g_j^2 and its lower engineering assignment. It is not absorbed into the Hessian tensor with an unrecorded constant.

135.2 The exact principal tensor and remaining drift after observation

Let $q_{\log}$ be the admitted actual compatible regular observation in one final analytic relative chart, and put $q_{\mathrm{can}} = q_{\log}/g_e$. In NH9.6–8 the symbol q denotes this canonical map q_{can}. The physical differential C_U used below is always the differential of $q_{\log}$, before division by g_e. Define the actual canonical response by

$$Y_i(\varphi \circ q) = A_i^{\alpha}(\partial_\alpha \varphi) \circ q. \tag{NH9.6}$$

The coefficient A_i^{α} is a function of the fine input, not necessarily of its coarse image. The ordinary chain rule gives

$$\begin{aligned} Y_j Y_i(\varphi \circ q) = {} & A_j^{\gamma} A_i^{\alpha}(\partial_\gamma \partial_\alpha \varphi) \circ q \\ & + (Y_j A_i^{\alpha})(\partial_\alpha \varphi) \circ q. \end{aligned} \tag{NH9.7}$$

All endpoint/frame factors are included in the declared chart and in A.

Writing $B_{ij} = X_i X_j S_0$, NH9.5 pushes forward to its second-order tensor and first-order drift measures

$$\begin{aligned} \eta^{\alpha\gamma} &= 2c_\beta^{-3} q_* \left(\sum_{i,j} B_{ij} A_i^{\alpha} A_j^{\gamma}\, \mu \right), \\ \xi^{\alpha} &= c_\beta^{-3} q_* \left(\left[2 \sum_{i,j} B_{ij} Y_j A_i^{\alpha} + \zeta \sum_i g_j X_i S_0\, A_i^{\alpha} \right] \mu \right). \end{aligned} \tag{NH9.8}$$

Thus the pushed whole current is $\sum_{\alpha,\gamma} \partial_\gamma \partial_\alpha \eta^{\alpha\gamma} - \sum_\alpha \partial_\alpha \xi^{\alpha}$ in the usual distributional derivative convention. The signs follow from CT9.1: its order-two word has the usual positive second derivative and its order-one word has a minus derivative.

The two response factors in η may use a compensated Hessian form. The contracted second response in ξ is not that form.

135.3 A local Wilson Hessian estimate on the weak regular chart

Work on the common regular chart with a small physical connection $A \in W^{1,p}$, $p > 8$, and its actual weak plaquette bound. Let

$$\mathcal{F}_p = a^{-2} \max(\|P_p - I\|, \|P_p^{-1} - I\|).$$

The established nonlinear weak plaquette estimate gives $\|\mathcal{F}\|_{L^p} \le CM$, and $\|A\|_\infty \le CM$. The current formula as a difference of two transported fluxes in each plane also gives

$$\|aJ\|_{L^p} \le C \|\mathcal{F}\|_{L^p}. \tag{NH9.9}$$

No derivative of curvature is bounded in this assertion.

For physical link tangents v, w, define the ordered Hessian form

$$\mathcal{H}_U(v, w) = a^2 \sum_{i,j} v_i w_j X_i X_j S_0(U). \tag{NH9.10}$$

A direct four-link plaquette differentiation gives

$$\begin{aligned} |\mathcal{H}_U(v,w)| \le {} & C(\|\nabla_a v\|_2 + M\|v\|_2)(\|\nabla_a w\|_2 + M\|w\|_2) \\ & + C\big(\|\mathcal{F}\|_{L^p} + \|aJ\|_{L^p}\big) \|v\|_{L^{2p/(p-1)}} \|w\|_{L^{2p/(p-1)}}. \end{aligned} \tag{NH9.11}$$

Here is the local derivation. Transport the four varied edge directions to the base of their actual plaquette. Its first logarithmic variation is the oriented sum $\Xi_p(v) = a\sum_{b\in\partial p} \pm \mathrm{Ad}(P_{p,b})v_b$. In the small link chart it obeys the actual discrete estimate

$$|\Xi_p(v)| \le Ca^2 \left(|\nabla_a v|_{\mathrm{star}(p)} + M|v|_{\mathrm{star}(p)} \right).$$

The symmetric second variation of the trace is a product of $\Xi_p(v), \Xi_p(w)$, plus commutators of two transported edge directions. A commutator has zero trace at $P_p = I$; its remaining coefficient is bounded by $C\|P_p - I\|a^2 \sum_{b,c\in\partial p} |v_b||w_c|$. Summing with the physical counting measure and bounded plaquette incidence gives the first line and the $\mathcal{F}$ term of NH9.11. The difference between the ordered and symmetric Hessians is $\frac{1}{2}[X_i, X_j]S_0$. It is on one bond and, in physical normalization, is bounded by $Ca\int |J||v||w|$, giving the last term. This also verifies that ordered derivatives have not been replaced by a symmetric Hessian without compensation. The bilinear holomorphic continuation has the same absolute estimate with its common bounded transport factors.

Choose

$$\frac{2}{p} < s < \frac{1}{2}. \tag{NH9.12}$$

The uniform discrete Sobolev embedding $H^s \subset L^{2p/(p-1)}$ then controls the second line. For the actual physical prefix differential C_U, CL18 and the once-only endpoint chart factors give

$$\|C_U^{\mathfrak{t}}\lambda\|_{H_a^s} \le C_s\|\lambda\|_2.$$

Here the base exponent in CL18's full-space fractional adjoint argument is two: its block-hold face estimate requires $2s < 1$. This use of a Hilbert Sobolev norm does not change the weak-field existence exponent $p > 8$. For the endpoint factors choose $s < 1/2 < \alpha < 1 - 4/p$. Discrete Morrey control of the admitted $W^{1,p}$ connection, followed by the bounded analytic endpoint chart maps, gives uniformly bounded C^α coefficient multipliers. Their multiplication norm on H^s is uniform: in the fractional difference quotient, write $m(x)v(x) - m(y)v(y) = m(x)(v(x) - v(y)) + (m(x) - m(y))v(y)$; the second term is bounded by $C\|m\|_{C^\alpha}^2\|v\|_2^2 \int_0^{O(1)} r^{2\alpha-2s-1}\,dr$, uniformly in the fine mesh. These endpoint factors occur once, as in CL18. Thus no iteration of their multiplier constants and no unsupported weak-field $p = 2$ construction enters this estimate. The discrete inverse-difference estimate yields $\|\nabla_a C_U^{\mathfrak{t}}\lambda\|_2 \le Ca^{s-1}\|\lambda\|_2$. Substitution into NH9.11 proves

$$|\mathcal{H}_U(C_U^{\mathfrak{t}}\lambda, C_U^{\mathfrak{t}}\lambda')| \le C_s a^{-2+2s}\|\lambda\|_2\|\lambda'\|_2. \tag{NH9.13}$$

This is weaker than the sharp flat a^{-1} face estimate GG9.3, but stronger than a bare a^{-2} Hessian-operator estimate. It holds on the specified actual weak regular chart. An unadmitted mixed readout does not acquire it.

135.4 Exact conversion to the canonical principal tensor

At retained spacing one, the physical transpose includes the volume ratio a^{-4}. Thus the response in NH9.6 is

$$A_i^\alpha = (g_j/g_e)a^3(C_U^{\mathfrak{t}})_i^\alpha. \tag{NH9.14}$$

This equality uses physical fine tangents v with dimensionless link variation av, followed by the declared final canonical chart. It includes the input and output spacings exactly. Using the factor a^2 in NH9.10 gives

$$\sum_{i,j} B_{ij}A_i^\alpha A_j^\gamma = (g_j/g_e)^2 a^4 \mathcal{H}_U(C_U^{\mathfrak{t}}e_\alpha, C_U^{\mathfrak{t}}e_\gamma). \tag{NH9.15}$$

Hence the principal tensor coefficient obeys

$$\left\| \sum_{i,j} B_{ij} A_i \otimes A_j \right\|_{\mathrm{op}} \le C_s (g_j/g_e)^2 a^{2+2s}. \tag{NH9.16}$$

The original prefix determining sets have diameter bounded in retained units. Two output coordinates with disjoint enlarged determining sets have zero contraction through the local plaquette Hessian. Thus only a fixed number of other output coordinates couple to any given one. Summing the entrywise variation of the principal tensor keeps one actual coarse coordinate count m_e, not an unrecorded fine count:

$$\sum_{\alpha,\gamma} |\eta^{\alpha\gamma}| \le C_s c_\beta^{-3} (g_j/g_e)^2 a^{2+2s} m_e \, q_* |\mu|. \tag{NH9.17}$$

A local packet uses its complete actual determining neighborhood; a global packet keeps its actual m_e. The real Wilson case has a positive measure. For a complex admitted comparison use its original absolute majorant, with the same one reserve and normalization.

The entrywise estimate NH9.17 uses the stated local relative coordinate realization on actual coarse bonds. It is not an entrywise estimate in an arbitrary transverse or gauge-reduced basis. After a nonlocal change of output coordinates, NH9.16 retains its operator-norm meaning under isometric changes, but the entrywise variation must carry the actual row overlaps of that change; the factor m_e cannot be retained by assertion. No output gauge or constraint record is changed in NH9.17.

The word's a^4 has not been reset. If an actual physical source lacks that factor, or carries a different recorded power, the conversion in GG9.16 still has to be made. In particular, this principal estimate alone is not the physical D8 MT36 row.

135.5 The drift remains a different quantitative problem

The unresolved coefficient in NH9.8 is

$$\mathfrak{d}_U^\alpha = 2 \sum_{i,j} B_{ij} Y_j A_i^\alpha + \zeta \sum_i g_j X_i S_0 \, A_i^\alpha. \tag{NH9.18}$$

It contains the contracted second derivative of the *actual* observation. The two first responses in NH9.15 cannot replace it.

The explicit lower first-current summand does have a direct quantitative bound on the same chart. Its physical form is

$$\sum_i g_j X_i S_0 \, A_i^\alpha = (g_j^2/g_e) a^2 \langle J, C_U^{\mathrm{t}} e_\alpha \rangle_a. \tag{NH9.18a}$$

CL17 bounds the last adjoint in $L^{p/(p-1)}$; its complete support has bounded retained volume. NH9.9 therefore gives

$$\left| \sum_i g_j X_i S_0 \, A_i^\alpha \right| \le C (g_j^2/g_e) a M. \tag{NH9.18b}$$

On a protected physical critical chart where the actual current itself has $\|J\|_{L^p} \le C M_J$, the sharper bound is $C(g_j^2/g_e) a^2 M_J$. Both keep the actual ratio and the original field strength; a physical-core strength is not silently replaced by a canonical one. Summing output coordinates retains their actual coarse count. Thus the remaining unbounded-by-these- results part of NH9.18 is the contracted Hessian/second-response term, rather than this explicit lower row.

There is an exact flat cancellation when both the variable and its fixed reference are the identity. The Wilson Hessian is then diagonal in color. Restrict all varied links to one fixed Lie-algebra generator. Their holonomies commute, the actual logarithmic means are arithmetic, and the compatible prefix is exactly its flat linear box-and-line map on this one-color subspace. Its second derivative along any two such directions is zero. The color-diagonal contraction therefore vanishes, and $X_i S_0(I) = 0$ gives

$$\mathfrak{d}_I^{\alpha} = 0. \tag{NH9.19}$$

This proves the flat value, not a uniform nonlinear age bound.

An absolute use of the available generic second-prefix-jet estimate still produces a fine coordinate count multiplying its one column factor. In the present setting it has the form

$$C(g_j^2/g_e) N_{\mathrm{fine}} \Gamma_2(n), \qquad N_{\mathrm{fine}} \asymp m_e a^{-4}, \quad \Gamma_2(n) \le C\lambda^n. \tag{NH9.20}$$

That expression does not give the needed positive power. A weak-chart zero at $U = I$ does not remove this count or create additional powers of a. A compensated nonlinear trace identity, together with its full scalar/effective-density completion, is still required.

The identity NH9.5 is for the bare Wilson density. Under an inherited density $e^{-\beta S_0 - W}$, solving NH9.2 for the bare $J_j F$ adds the term $-\kappa(X_j W)F$, and converting the bare EOM to the full effective EOM adds its corresponding score term. CT9 packages a bounded relative density perturbation of an already controlled whole pair. It does not delete these terms for an arbitrary inherited W.

136.1 The terminal coordinate contribution is already controlled

Use NH9's original contact mesh a, original coupling g_j, and final canonical coupling g_e. Let $W(V)$ be the actual retained group-valued observation, and keep its reference W_{ref} fixed while taking the incoming directions $Y_i = g_j X_i$. In the final relative chart put

$$b = g_e^{-1}\log(WW_{\mathrm{ref}}^{-1}), \qquad Z_\alpha = g_e X'_\alpha, \qquad M^a{}_\alpha(b) = Z_\alpha b^a. \tag{CD9.1}$$

The primed fields are the actual left-variation fields on output bonds. The matrix M is block-local on those bonds. On the admitted endpoint chart its norm and inverse norm are bounded, and $\|ZM\| \le Cg_e$. These are finite-dimensional estimates for the logarithm; the endpoint is used once.

Define the intrinsic response $\widehat{A}$ by $Y_i(F\circ W) = \widehat{A}_i^\alpha (Z_\alpha F)\circ W$. The coordinate response in NH9.6 is $A_i^a = M^a{}_\alpha \widehat{A}_i^\alpha$. For the actual ordered Wilson Hessian $B_{ij} = X_iX_jS_0$, the ordinary product rule gives the exact split

$$\begin{aligned}\sum_{i,j} B_{ij}Y_jA_i^a &= \Theta^{\alpha\gamma}(Z_\gamma M^a{}_\alpha)(W)\\ &\quad + M^a{}_\alpha(W)\sum_{i,j} B_{ij}Y_j\widehat{A}_i^\alpha,\\ \Theta^{\alpha\gamma} &= \sum_{i,j} B_{ij}\widehat{A}_i^\alpha\widehat{A}_j^\gamma.\end{aligned} \tag{CD9.2}$$

There is no change of integration map in this identity. The first line is the terminal-frame drift; the second is the remaining intrinsic response contraction.

In the local output coordinate realization of NH9.17, conjugating the principal tensor by the bounded block-local M^{-1} preserves its actual overlap count. Hence NH9.16–17 prove

$$\sum_a \left|2c_\beta^{-3}q_*[\Theta^{\alpha\gamma}Z_\gamma M^a{}_\alpha\,\mu]\right| \le C_s c_\beta^{-3}\frac{g_j^2}{g_e}a^{2+2s}m_e\,q_*|\mu|, \qquad 2/p < s < 1/2. \tag{CD9.3}$$

This is a positive quantitative subrow on NH9's admitted regular chart. It keeps the original word's a^4 convention. The fractional adjoint proof uses Hilbert base exponent two, while the weak connection still has $p > 8$; the separate endpoint multiplier justification is now given after NH9.12.

If the reference is a live inherited variable, the chain rule also contains $(D_{W_{\mathrm{ref}}}M)Y_jW_{\mathrm{ref}}$. That term belongs to the actual joint-readout record. CD9.3 does not supply its coefficient or derivative bound. Similarly, an entrywise m_e bound cannot be transported to a nonlocal transverse basis without its actual row-overlap factors.

136.3 The centered real regular prefix

Let $a = a_0 < a_1 < \cdots < a_n = 1$, $a_t = L^t a$, with fixed odd L. Start from a real connection U_0 on a protected, nonwrapping, buffered coordinate-rectangular region, and use the actual raw reference chain

$$U_{t+1} = \mathcal{M}(U_t). \tag{CD9.7}$$

Assume the physical plaquette curvature on the complete original neighborhood has supremum α, with $C_L\alpha$ sufficiently small. On the protected real physical chart this is provided by CL28; PG19–20 then give the same bound, up to one fixed factor, on all genuine intermediate references. All neighborhoods

below include the original paths, the Wilson Hessian plaquettes, and the readsets of the indicated maps. They also contain all coordinate-tree ribbon fillings used in the proof and the curvature-propagation stencils. More precisely, a prefix ending at scale a_t has a coordinate-path-convex buffered box of radius $K_L a_t$ with this closure and the same actual curvature bound. A mere nonwrapping graph of small diameter, possibly with holes or missing controlled plaquettes, does not satisfy this hypothesis.

Hold the chain CD9.7 fixed while varying V_0 near U_0. Set $V_{t+1} = \mathcal{B}_{U_t}(V_t)$, as in CL14, and use dimensionless relative logarithms

$$x_t = \log(V_t U_t^{-1}), \qquad x_{t+1} = \Phi_t(x_t), \qquad \Phi_t(0) = 0. \tag{CD9.8}$$

These are the actual compatible maps, not collapsed all-depth means. In taking jets, the reference is fixed and all evaluations are at the centered point $V_0 = U_0$.

The result proved below is

$$\sum_{\alpha \in E} \left| \sum_{i,j} (X_i X_j S_0)(U_0) X_j X_i x_n^{\alpha}(U_0) \right| \le C_L \, m_E \, a\alpha. \tag{CD9.9}$$

Here E is an actual finite set of local final bond coordinates; m_E includes its fixed determining enlargement. The estimate is a zeroth-order coefficient estimate. It does not assert the same bound after arbitrarily many incoming marked derivatives.

136.3.1 One common gauge for the whole prefix comparison

A separate gauge at every depth would not justify the needed tensor comparison. Instead, fix a prefix ending at scale $\ell = a_t$ and choose one ordinary fine coordinate-tree gauge on its entire enlarged neighborhood of diameter $C_L \ell$. Use the buffered coordinate-path-convex box specified above, so the ribbons in the next sentence lie inside the controlled plaquette region. First take the actual jets with their fixed reference, then choose this gauge to estimate their values. Equivalently one may hold that chosen gauge constant during all variations, using joint covariance. It need not satisfy the original gauge pins: it is not used to replace the integration variables, constraints, selectors, or history record. Fine plaquette telescoping gives

$$\|U_0^g(b) - I\| \le C_L a \ell \alpha. \tag{CD9.10}$$

For adjacent endpoints the two tree paths enclose a ribbon of width a, length at most $C_L \ell$, and thus at most $C_L \ell / a$ plaquettes. Multiplying real unitary transports does not amplify the sum of these plaquette errors.

The same gauge controls all the actual intermediate reference links. PG9's loop comparison gives, at each genuine one-step average, the difference between its output and the straight lower-scale path with the same endpoints as $C_L a_s^2 \alpha$. Let e_s be the difference between an actual U_s bond and the original fine straight product with those endpoints. Product telescoping yields

$$e_s \le L e_{s-1} + C_L a_s^2 \alpha, \qquad e_s \le C_L a_s \alpha \sum_{r=1}^{s} a_r \le C_L a_s^2 \alpha. \tag{CD9.11}$$

No associativity of the nonlinear mean is used. The comparison is of its actual value with a straight path at each step. The fine straight product has at most a_s/a factors in CD9.10. By actual gauge covariance of the raw means, the restrictions of this one fine gauge therefore satisfy

$$\|U_s^{N^s g} - I\| \le C_L a_s \ell \alpha, \qquad 0 \le s \le t. \tag{CD9.12}$$

The same reasoning applies with a fixed extra collar to include the next one-step map. All gauges in this proof are real compact transformations. No claim about an arbitrary complex wall chart follows from CD9.10–12.

136.3.2 First responses and the transported tensor defect

Use norms of color blocks induced by the fixed Ad-invariant Euclidean norm. Their column sum is submultiplicative, and an adjoint rotation is isometric. This avoids a color-dimension factor at every depth. Let $A_s = D\Phi_s(0)$ and $S_t = A_{t-1}\cdots A_0$; a superscript zero denotes the same actual flat hierarchy and the same coordinate/readset records. At the diagonal the dimensionless flat first map is L times the physical box-line average, with column sum exactly L^{-3}. CL17's finite-dimensional analytic estimate in the common gauge CD9.12 gives

$$\begin{aligned} \|A_s\|_{\mathrm{col}} &\le L^{-3}(1 + C_L a_{s+1}\ell\alpha), \\ \|A_s - A_s^0\|_{\mathrm{col}} &\le C_L L^{-3} a_{s+1}\ell\alpha. \end{aligned} \tag{CD9.13}$$

The centered relative endpoints have no independent repeated chart factor. Multiplying these inequalities and telescoping the difference from the flat product proves

$$\begin{aligned} \|S_t\|_{\mathrm{col}} &\le C_L(a/a_t)^3, \\ \|S_t - S_t^0\|_{\mathrm{col}} &\le C_L(a/a_t)^3 a_t^2\alpha. \end{aligned} \tag{CD9.14}$$

Indeed $\sum_{s<t} a_{s+1}a_t \le C_L a_t^2$, so the product of the perturbations is bounded by one fixed exponential. There is no replacement of every leading factor by $2L^{-3}$.

Let $H_{ij} = (X_iX_jS_0)(U_0)$, with precisely this ordered word. In the gauge CD9.10 it has fixed finite plaquette incidence, bounded row mass, and

$$\|H - H^0\|_{\mathrm{row}} \le C_L a a_t \alpha. \tag{CD9.15}$$

This follows by differentiating the finite four-link Wilson plaquette twice: each local Hessian coefficient is an analytic function of those link variables, all within CD9.10 of identity. It is not an assertion that the transported Hessian is a new Wilson Hessian.

The actual tensor entering the next map is

$$R_t = S_t H S_t^{\mathsf{T}}, \qquad R_t^0 = S_t^0 H^0 (S_t^0)^{\mathsf{T}}. \tag{CD9.16}$$

For a fixed scale-a_t output patch, the complete original fine readset has at most $C_L(a_t/a)^4$ bonds. With entry mass defined by summing the norms of the color blocks, CD9.14–15 give

$$\begin{aligned} \|R_t^0\|_{\mathrm{ent}} + \|R_t\|_{\mathrm{ent}} &\le C_L(a/a_t)^2, \\ \|R_t - R_t^0\|_{\mathrm{ent}} &\le C_L a^2\alpha. \end{aligned} \tag{CD9.17}$$

For example the two response errors contribute $(a_t/a)^4(a/a_t)^6 a_t^2\alpha = C_L a^2\alpha$. The fine-Hessian error contributes $(a_t/a)^4(a/a_t)^6 a a_t\alpha = a^3\alpha/a_t \le a^2\alpha$. These are the actual fine counts. No second root count is added. The ordered antisymmetric part is included in these estimates.

136.3.3 Flat color cancellation and all intermediate forcing terms

Write $K_t = D^2\Phi_t(0)$. In a common gauge on the slightly larger prefix patch through scale a_{t+1}, analyticity and CD9.12 give

$$\|K_t\| \le C_L, \qquad \|K_t - K_t^0\| \le C_L a_{t+1}^2\alpha. \tag{CD9.18}$$

Use this same enlarged stage-$(t+1)$ analysis gauge for both R_t and K_t. In CD9.13–17 its radius is larger by only the fixed factor L, so their bounds change only by C_L. One does not contract tensors written in unrelated comparison frames. The flat tensor R_t^0 is scalar in color: its spatial matrix can have arbitrary actual coefficients, but every color block is a scalar multiple of the identity. For a fixed generator T_A, all inputs in its one-generator Abelian subgroup commute. The actual symmetric means become arithmetic means of logarithms, and the compatible endpoint gauge gives exactly the linear box-line map. Thus the mixed second derivatives in any two input bonds with this same color vanish. Consequently

$$R_t^0 : K_t^0 = 0. \tag{CD9.19}$$

This is why the transported tensor defect in CD9.17 had to be estimated; arbitrary transported tensors do not satisfy CD9.19. Combining the complete difference $(R_t - R_t^0) : K_t + R_t^0 : (K_t - K_t^0)$ with CD9.17–18 gives, per actual next-scale output coordinate,

$$|R_t : K_t| \le C_L a^2 \alpha. \tag{CD9.20}$$

Define the full ordered contracted response

$$B_t^\alpha = \sum_{i,j} H_{ij} X_j X_i x_t^\alpha(U_0).$$

The chain rule, with its original fine derivative order, is

$$B_{t+1} = A_t B_t + R_t : K_t. \tag{CD9.21}$$

The initial value is not silently set to zero. Since $x_0(V) = \log(V U_0^{-1})$, its second derivative at the center is a same-bond Lie bracket. Its contraction uses the antisymmetric part of H, which is exactly the same-bond bracket applied to S_0. Wilson's single-plaquette derivative and the actual curvature bound give

$$|B_0(b)| \le C|XS_0(b)| \le C_L a^2 \alpha. \tag{CD9.22}$$

This retains the initial frame/score row. Neither an ordered Hessian nor an ordered current word has been replaced by its symmetric part.

For final coordinates in E, the determining enlargement at mesh a_s contains at most $C_L m_E a_s^{-4}$ actual coordinates. The suffix column norm from mesh a_s to one is at most $C_L a_s^3$, by the same product estimate as CD9.14. Propagating CD9.22 and every forcing in CD9.20 proves

$$\begin{aligned} \sum_{\alpha \in E} |B_n^\alpha| &\le C_L m_E a^2 \alpha \left(a^{-1} + \sum_{s=1}^{n} a_s^{-1} \right) \\ &\le C_L m_E a \alpha. \end{aligned} \tag{CD9.23}$$

This proves CD9.9. The gain comes from curvature, two prefix responses, the complete transported-tensor comparison, and the actual remaining column/count factors. It does not come from assigning extra decay to an ordinary higher jet.

136.4 Exact use on invariant terminal tests and its limitations

Let φ be invariant under the terminal lattice gauge group. For each evaluation field U, hold the reference chain $\mathcal{M}^t(U)$ fixed and differentiate in a separate variable V. CL15 is an identity on a neighborhood:

$$\varphi(\mathcal{B}_{U_{n-1}} \circ \cdots \circ \mathcal{B}_{U_0}(V)) = \varphi(\mathcal{M}^n(V)). \tag{CD9.24}$$

The left output is a gauge transform of the right output. Thus its first and second jets at $V = U$ compute exactly the jets of the same physical scalar function on the right. There is no derivative of a chosen moving reference: the reference is held fixed for this neighborhood identity before its jet is evaluated. Define the terminal test in the centered chart by $\varphi_U(x) = \varphi(e^x \mathcal{M}^n(U))$. Its first derivative at zero is the actual invariant-frame derivative, and its second ordinary derivative is the symmetrized invariant second derivative. The ordered fine-frame contribution has already been retained in CD9.21–22. Accordingly the invariant-test second-current coefficient may be written as $\operatorname{sym} R_n$; antisymmetric output coefficients have zero pairing with this ordinary second derivative. This is a consequence of the displayed chain rule, not a deletion of the initial ordered-bracket term in B_n.

Multiplying the contraction CD9.23 by NH9's exact canonical factor gives the represented first-current row

$$C_L c_\beta^{-3} \frac{g_j^2}{g_e} a\alpha \, m_E \, |\mu| \tag{CD9.25}$$

before its pushforward by the actual raw physical observation $\pi(U) = \mathcal{M}^n(U)$, tested in the invariant class of CD9.24; the displayed constant includes the factor two in NH9.8. Pushing a coefficient measure cannot increase its total variation. This bounds coefficient mass carried by the stated real body. It does not assert that the original Wilson measure is supported there. Restricting a Ward identity by an original selector, or inserting a new curvature cutoff, creates its full boundary partners. Those partners are neither discarded nor estimated by CD9.25, and no new cutoff is inserted here.

There is also a principal estimate on this same stronger real curvature geometry, without inferring a $W^{1,p}$ bound from the axial gauge. At final scale one the sharp flat face estimate GG9.2–3, with its physical-adjoint conversion, gives $\|R_n^0\|_{\mathrm{op}} \le Ca^3$. Explicitly, the dimensionless response is the physical average multiplied by a^{-1}, the flat Wilson Hessian is a^2 times its physical operator, and converting its final ordinary transpose to the physical fine adjoint contributes a^4; thus the dimensionless tensor is $a^4 Q H_{\mathrm{phys}} Q^*$, bounded by Ca^3. Its actual ambient local coordinate matrix has fixed final range. CD9.17 therefore proves

$$\sum_{\alpha,\gamma \in E} |R_n^{\alpha\gamma}| \le C_L m_E (a^3 + a^2 \alpha). \tag{CD9.26}$$

The canonical principal measure has the further exact factor $2c_\beta^{-3}(g_j/g_e)^2$. This estimate uses the original local coordinate basis, not an arbitrary projected transverse basis. Alternatively NH9.17 gives its a^{2+2s} principal bound when that actual weak $W^{1,p}$ chart is also available; the two sets of hypotheses are not identified.

At the centered endpoint $b = 0$, the coordinate frame satisfies $M(0) = I$. Its first derivative is a Lie-bracket tensor, whose contraction with the flat color-diagonal R_n^0 vanishes. Thus CD9.17 bounds this centered terminal-frame contribution by $Ca^2\alpha$, before the common canonical factor. Subtracting it in the exact split CD9.2 shows that the centered intrinsic contracted response also obeys

$$\sum_{\alpha \in E} \left| \sum_{i,j} H_{ij} Y_j \widehat{A}_i^\alpha \right| \le C_L \frac{g_j^2}{g_e} m_E a\alpha. \tag{CD9.28}$$

This conclusion is still diagonal in the frozen-reference family; it is not an off-diagonal or live-reference assertion.

The explicit lower score coefficient can likewise be bounded on the same strong geometry. Each fine bond obeys $|XS_0| \le Ca^2\alpha$, and the complete fine count and column estimate give

$$\sum_{\alpha \in E} \left| \sum_i (X_i S_0) X_i x_n^\alpha \right| \le C_L m_E a\alpha. \tag{CD9.27}$$

Its actual factor is $\zeta c_\beta^{-3} g_j^2/g_e$. If the separate protected current bound $\|J\| \le M_J$ in the required norm is available, NH9.18a improves that score row to $Ca^2 M_J$ with the same coupling factor.

All original couplings and engineering assignments are retained. With a smooth physical source inserted later, its actual derivative factors and the resulting contact partners must still be kept; CD9.25–27 are not bounds for all of them.

The re-centering in CD9.24 is available for a terminal invariant test on the actual same-field regular hierarchy. A joint test that color-links several retained histories, a fixed-reference off-diagonal chart, or a test with live reference/constraint partners need not be invariant under the terminal gauge alone. Those tests cannot be replaced by CD9.24. Changing the reference inside an already differentiated old marked coefficient also requires its derivative and gauge partners. No such change is made here.

Two further distinctions are essential. First, CD9.25 bounds a represented current on the specified invariant test class; it does not by itself bound the pushed ordinary scalar Ward companion in the joint total-variation envelope required by CT9. Second, the source convention remains the one in NH9: the word originally contains $g_j^{-4} a^4$. Removing an a^4 absent from an actual physical source turns the displayed $a\alpha$ into $a^{-3}\alpha$, as dictated by GG9.16. One may not insert that factor anew to obtain a marked grade.

The all-history obligations therefore remain: source-graded derivatives of these coefficients on actual mixed domains; complete live-reference, constraint, later-contact and cutoff partners; quantitative control through inherited effective density and genuine normalizers; and the ordinary scalar part of the whole pair. The pointwise gauge in this proof changes none of those records, introduces no new reserve, and supplies no permission to spend an existing reserve twice.

137.4 A nonlinear Wilson oscillation bound for a protected physical reference sector

The following bound uses an actual nonlinear Wilson action, not a quadratic substitute. Its domain is the clean physical target sector already established in V7N2–37. It is not an admission theorem for all original mixed histories.

Use its common fixed physical reference and compatible raw target chart on two tori T_0, T_1. Let θ_i lie in the same strict small target ball. Suppose the two arrays agree on a common injectively lifted collar U. Let Y be a set of actual target coordinates at distance D from the outside of that collar. Its complete prefixes, endpoint gauges, and scalar rows are included. Let n_Y be its actual current coordinate count.

Choose a relative output C supported on Y. Since the original target values there are common, left multiplication by that same C changes the target logarithms by the same finite local function $\zeta(C)$, supported on Y. For instance its literal one-link expression in the fixed logarithmic chart is

$$\zeta(C)_y = \log(C_y e^{\theta_y}) - \theta_y, \qquad \zeta(1) = 0. \tag{VM9.14}$$

The corresponding full mean/pivot expression is used when the original output variable has first been straightened. Its actual complete finite paths and Jacobian are unchanged. Assume the entire original relative cutoff lies in $\|\zeta(C)\|_\infty \le t$ and that the paths $\theta_i + s\zeta(C)$, $0 \le s \le 1$, stay inside the established smaller target domain. No original selector is shrunk in making this check.

Let F_i be the full constrained Wilson critical action in V7N. Write

$$\begin{aligned} u_i(C) &= -x_k\{F_i(\theta_i + \zeta(C)) - F_i(\theta_i)\}, \\ c &= -x_k\{F_1(\theta_1) - F_0(\theta_0)\}. \end{aligned} \tag{VM9.15}$$

For the classical reference density, the exact logarithmic ratio is $c + u_1 - u_0$. The baseline actions may be extensive; they have not been bounded by an exponentially small local error.

V7N30 applies along each pair of paths, because their targets still agree throughout U. If η bounds both path arrays, then the actual multipliers satisfy

$$\sup_{y \in Y} |\mu_1(\theta_1 + s\zeta; y) - \mu_0(\theta_0 + s\zeta; y)| \le C\eta e^{-c_0 D}. \tag{VM9.16}$$

The physical action derivative is V7N14, with its original current pairing. The common $\kappa\theta$ and reference-multiplier contributions cancel on Y. Integrating the difference of these derivatives in s proves

$$\sup_C |u_1(C) - u_0(C)| \le C x_k n_Y t\eta e^{-c_0 D}. \tag{VM9.17}$$

This is a full nonlinear difference, uniform in the hidden fine cutoff on the specified domain. It preserves the original coupling, target count, logarithmic-coordinate map, and physical action. The complex version uses the same common complex target chart and its separate endpoint curvature bounds; it does not infer those bounds from a weak potential norm alone.

For actual canonical output cutoffs of size $g_k P_k$ and background targets of size at most $g_k Q_k$, the explicit bound is

$$\varepsilon_k \le C n_Y P_k (P_k + Q_k) e^{-c_0 D}, \qquad x_k = g_k^{-2}. \tag{VM9.18}$$

The fixed target-chart comparison factors are included in C. If the background is only bounded by a fixed small ceiling, use the literal factor $x_k t\eta$ in VM9.17 instead; no extra coupling smallness is claimed.

Combining VM9.3–5 with VM9.17 proves a centered reference-kernel comparison for this actual nonlinear classical sector. It does not require equality of the extensive baseline actions or a lower bound for the raw reference denominator. Its unused scalar remainder is exactly $\log K_0(e^{u_1-u_0})$, with magnitude at most $C\varepsilon_k$, on the stated small comparison disc.

An original R denominator is covered by this estimate only when its own finite constrained action and target map have been identified with this physical problem on the stated domain. The occurrence of a reference minimum in its notation does not establish that identification. The exact conditional-fiber identities in Sections 1–3 apply separately to the original insertion already identified by S58b.

An actually admitted mixed version follows from V7N24–26 with the adapted distance and the physical row weights. Every row retains the factors $r_c^4, r_c^{-3}, r_c^{-1}$; their product is one. This extension still requires the actual MC domain and complete common record. It is not a proof that every prepared R input satisfies it.

137.4.1 How inherited terms enter the exact comparison

For a full reference density with an inherited term E_i, write

$$r_i(C) = \exp\{-x_k F_i(\theta_i + \zeta(C)) + E_i(C) + c_i^{\mathrm{ext}}\}\, j_i(C). \tag{VM9.19}$$

Here j_i retains the actual Haar/pivot/gauge factor and c_i^{ext} denotes a scalar which actually belongs inside this reference numerator. Its original placement is part of the notation. The centered logarithmic defect is

$$\begin{aligned} v(C) = u_1(C) - u_0(C) \\ &+ [E_1(C) - E_1(1)] - [E_0(C) - E_0(1)] \\ &+ \log \frac{j_1(C)}{j_0(C)} - \log \frac{j_1(1)}{j_0(1)}. \end{aligned} \tag{VM9.20}$$

VM9.17 bounds the first line. Literal common finite-step pivot and Haar factors make the last line zero in the sector where V7N18a identifies them. The middle line is an inherited *oscillation* comparison. It is a weaker requirement than comparing arbitrary absolute constants in E_i, but it is not proved merely by a bound on either E_i. Its actual old radii and physical readouts remain necessary. A factor not admitting the stated logarithm must be compared in its original nonlogarithmic representation instead.

Scalars genuinely independent of C cancel from the kernel in VM9.20. Their original multiplicities and any compensation outside the reference fraction remain in VM9.8 and the full vacuum record. An external scalar not in this reference numerator is not included in c_i^{ext} and does not cancel.

137.6 Consequences for complete source-zero matching

The preceding results sharpen two of the missing input rows:

1. A genuine reference fraction needs a comparison of its reference-coordinate oscillation. Its own constant numerator and denominator factors cancel exactly, with their action-completion partners still recorded. The nonlinear classical part of that oscillation is bounded by VM9.17 on the verified physical sector.
2. Different output selector labels can be compared by one common refinement. Smallness of the selector difference is unnecessary if the physical carrying function and the actual conditional space are common. Otherwise the precise residual is VM9.12.

The following are still unproved for the complete original two-torus construction:

- identification of a common actual conditional fiber and all nonlinear constraint records at every mixed preparation;
- the inherited oscillation in VM9.20, including actual paired-domain admission and original shrinking radii;
- the value and current-jet difference VM9.12 for every surviving history-dependent readout;
- all original typed local factors and zero-order scalar seeds under the intertorus identification;
- independent absolute vacuum increments, completion atoms outside a canceled reference fraction, and terminal terms;
- complete wrapped-carrier bounds with the original scale counts.

The old direct seam argument remains insufficient. Cutting or rewiring the original fine Wilson action across a physical face changes $O(a^{-3})$ plaquettes, each with exponent of size $O(\beta_a)$. Its unrenormalized bound is $O(\beta_a a^{-3})$, not a cutoff-independent physical-volume comparison. No lower bound obtained by selecting a thin seam tube removes its own ultraviolet entropy cost without another estimate.

After complete input matching, VC28 and TD provide the desired formation and ultraviolet summation. The present results do not establish that full matching, and therefore do not prove

$$\frac{V_{a,T_*}}{\mathrm{Vol}(T_*)} - \frac{V_{a,qT_*}}{\mathrm{Vol}(qT_*)} \le C \tag{VM9.22}$$

uniformly in a and q. S94 and the full interacting four-dimensional existence proof remain incomplete.

138.1 The primitive in the original R formula

The denominator in R-II (1.1), expanded in (1.2) and factored in (1.10), is a cutoff-weighted **pure Wilson reference integral**. It is not an arbitrary inherited effective reference density. Retain its original relative variable $C = V'$, action cutoff ζ_0, relative field cutoff χ, and axial gauge forest G_0:

$$D_{\mathfrak{R}}(b) = \int dC\, \delta_{G_0}(C)\, \chi(C) \exp\{-x_k A_{\mathfrak{R}}(\zeta_0, U_{k,Z}(CV_\Lambda; b))\}, \qquad x_k = g_k^{-2}. \tag{RN9.1}$$

All terms in this formula have their original meaning. The prescribed retained data b include old variables that will be integrated later. The support and factors in χ are unchanged. The gauge delta denotes the original normalized Haar/tree fiber, not a new Lebesgue delta of a constraint already imposed in the coordinates.

There are two different minima in RN9.1.

- For each admitted C, $U_{k,Z}(CV_\Lambda; b)$ is the original localized Wilson minimum with the **entire prescribed target array fixed**, including the target induced by C in Λ.
- The reference U_0 in R-II (1.3) minimizes the unweighted Wilson action while the targets in Λ are **free**; the original retained constraints outside Λ stay fixed. Its current target supplies V_Λ in the original gauge.

In particular, U_0 is not obtained by imposing an arbitrarily chosen fixed interior target. Neither minimum is defined here by minimizing the weighted action $A(\zeta_0, \cdot)$. The latter is evaluated on the original unweighted minimum in RN9.1. The weighted first variation at U_0 therefore need not vanish.

R-II printed pp.357–359 verifies the relaxed variational condition, its current identity, and uniqueness of its small axial-gauge branch through (1.12)–(1.13). This does **not** establish uniqueness for every fixed-target inner field in RN9.1. The separate fixed-target theorem and its exact application requirement are stated in Section 3 below. No uniqueness among unrelated rough or topological critical branches is asserted.

138.3 Equality of the two original variational problems

Let $Q_{\mathfrak{R}}$ denote the complete original raw determining map, with all its prescribed rows. The Wilson action is a finite sum of literal plaquette words, and each raw mean is built from its literal finite ordered paths. Therefore, for every field on the common original domain,

$$\begin{aligned} A_{\mathfrak{R}'}(\iota U) &= A_{\mathfrak{R}}(U), \\ Q_{\mathfrak{R}'}(\iota U) &= \iota Q_{\mathfrak{R}}(U), \\ A_{\mathfrak{R}'}(\zeta_0', \iota U) &= A_{\mathfrak{R}}(\zeta_0, U). \end{aligned} \tag{RN9.3}$$

For the means, prove the second equality recursively through the original one-step averaging formula. Each product and its endpoint conjugation has the same ordered factors after copying. Its allowed logarithmic branch is the copied branch. No new averaging map or linear approximation is introduced.

First apply RN9.3 to the relaxed problem: retain only its original constraints outside Λ, with the same fine exterior data and admissible regularity/gauge restrictions. Its admissible fiber is carried bijectively onto the relaxed fiber on $\mathfrak{R}'$. The action values agree. The copied field is therefore a solution in the same small branch, and the original uniqueness gives

$$U_0' = \iota U_0, \qquad V_\Lambda' = \iota V_\Lambda. \tag{RN9.4}$$

The output equality uses the actual recorded coarse target and gauge, not an independently reselected axial representative. If one has kept the original finite gauge-choice average, apply this argument to each choice and retain that average with its original factor.

Next fix an admitted relative output C. Its original full target array CV_Λ, with the unchanged retained rows, is carried to the full target array $(\iota C)V_\Lambda'$. RN9.3 now identifies the two **fixed-target** variational fibers. This proves an unconditional set statement for those identified finite variational problems. If $\mathcal{F}_{\mathfrak{R}}(C, b)$ is the complete admitted fiber, then

$$\iota\left(\operatorname*{argmin}_{U\in\mathcal{F}_{\mathfrak{R}}(C,b)} A_{\mathfrak{R}}(U)\right) = \operatorname*{argmin}_{U'\in\mathcal{F}_{\mathfrak{R}'}(\iota C,\iota b)} A_{\mathfrak{R}'}(U'). \tag{RN9.3a}$$

The analogous equality holds for critical sets and for their allowed gauge-orbit sets. Equality of fibers and action values alone does not select one member of a multivalued minimum set. Moreover, equal *unweighted* minimum values would not by themselves identify a *weighted* action value on two different minimum orbits.

138.3.1 The separately inspected fixed-target theorem

The primary variational paper, CMP **102** (1985), printed pp.277–279, specifies the fixed-target problem before Theorem 1. Its data are the nested separated hierarchy $\{\Omega_j\}$ of (1), the layer-dependent plaquette and current regularity class of (2), the complete fixed mixed averages in (3), and the pinned gauge group in (4). The Wilson action and admitted space are (5) and (6). The target condition (7) includes the stated lower-scale completion of plaquettes crossing mixed-layer interfaces. Theorem 1 gives a minimal orbit and makes it the unique critical orbit in (6) when

$$\epsilon_1 < a_1, \qquad B_3\epsilon_1 \le \epsilon_0 \le a_0. \tag{RN9.5a}$$

Thus the fixed-target uniqueness source has actually been inspected; it is not inferred from R-II's relaxed U_0 argument or from a publisher abstract. Its conclusion is gauge orbit uniqueness, which is exactly enough for RN9.1.

To apply that theorem to a literal localized $U_{k,Z}$, the following identification remains part of the application:

- The localized hierarchy and its separation are the hierarchy in (1), with the original action plaquettes and boundary rule in (5). No box action or alternative exterior prescription is substituted.
- The whole recorded determining set $B_k(Z)$, its ordered averages and pins give precisely the fiber (3) and gauge group (4). Any fixed exterior data must be incorporated in the theorem's actual admissible space, rather than imposed by an extra unverified restriction.
- Every real target CV_Λ admitted by the original χ satisfies (7), including its interface convention, and the original selected inner field lies in (6) with RN9.5a. An output cutoff being small in a different norm is not a substitute for these clauses.

These are the **fixed-target identification clauses**. Whenever they have been checked for a primitive, Theorem 1 and RN9.3a give

$$\big[U_{k,Z'}((\iota C)V'_\Lambda;\iota b)\big] = \big[\iota U_{k,Z}(CV_\Lambda;b)\big], \tag{RN9.5}$$

where brackets denote the allowed gauge orbit. Weighted Wilson action is invariant under that gauge group, so RN9.5 and RN9.3 give equality of the actual weighted integrands. Equality of the chosen representatives as matrices is unnecessary; it requires, in addition, the copied gauge-selection rule.

At $C = 1$, this also gives the reference orbit of U_0: it is already stationary and minimizing on the relaxed fiber, and fixing its own interior target restricts that fiber to one containing U_0. The separately justified orbit uniqueness identifies the two readouts modulo the allowed gauge. This does not identify their variational problems for arbitrary C.

The complete finite boundary prescription is essential in this argument. The original localized map is not replaced by a periodic minimum, by a box inverse with newly chosen boundary conditions, or by a newly defined source-only extension. The inspected theorem does cover admissible mixed hierarchies, but this supplement has not checked its clauses for every $U_{k,Z}$ and every selector range in an arbitrary original mixed history. On those histories, the fixed-target identification remains an explicit premise. If the localized object is instead a weakened reconstruction, its equality with the minimum requires a separate argument.

No numerical formula for the CMP99 typed propagator expansion is needed for the finite-record implication RN9.3a plus the stated uniqueness theorem. This observation does not supply the missing domain identification. Without uniqueness, a separately verified equivariant selection rule would suffice; no such substitute is being assumed here.

138.4 Exact equality of the genuine denominator and its scalar factors

The relative forest fiber is a product of the original free group coordinates. Relabeling them by ι preserves normalized Haar measure exactly. It also preserves the forest delta, every pinned value, and the original cutoff:

$$\iota_*\{dC\,\delta_{G_0}(C)\chi(C)\} = dC'\,\delta_{G'_0}(C')\chi'(C'). \tag{RN9.6}$$

There is no determinant from relabeling. The original tree disintegration has Jacobian one as in S54, with precisely the same number of free links and gauge rows. Any allowed residual compact gauge integration has its original normalized volume one on both sides.

For a copied primitive satisfying the fixed-target identification clauses on its full original cutoff support, substitution of RN9.3, RN9.5 and RN9.6 into the actual integral RN9.1 proves

$$\boxed{D_{\mathfrak{R}'}(\iota b) = D_{\mathfrak{R}}(b).} \tag{RN9.7}$$

The equality includes the full cutoff-weighted nonlinear action, not merely its quadratic expansion. Positivity fixes the real logarithm, so the corresponding logarithms agree. Section G, particularly Proposition 9 on printed p.309 of the variational paper, supplies the analytic fixed-target branch in its specified domains. Where both actual continuations are identified with that branch on a common chart, their gauge-invariant weighted actions are the same analytic germ under ι. The equality of their integrals then continues on their corresponding common domain. Equality of logarithms continues only on the common nonvanishing branch; no global complex zero-free claim is added.

Equation (1.10) extracts a classical action and explicit coupling, Haar-density and coordinate-count constants. Write their unchanged signed combination as $c^{\mathrm{coord}}_{k,\mathfrak{R}}$, so that the remaining logarithmic factor is defined by

$$\log D_{\mathfrak{R}} = -x_k A_{\mathfrak{R}}(\zeta_0, U_0) + c^{\mathrm{coord}}_{k,\mathfrak{R}} + I_{\mathfrak{R}}. \tag{RN9.8}$$

The original sign convention for each extracted factor is retained in c^{coord}. Its actual counts are copied, so RN9.3, RN9.4 and RN9.7 imply equality of c^{coord} and $I_{\mathfrak{R}}$, separately. No extensive scalar has been discarded. The reference probability kernels likewise satisfy $K_{\mathfrak{R}'} = \iota_* K_{\mathfrak{R}}$ on this original fiber.

For a connected complete nonwrapping record the deck group gives exactly q^4 lifts, one for each lift of its root. They are distinct copies with identical counts. If these are the actual disjoint reference components in the original product, their log denominator contribution and the physical volume both multiply by q^4. Factorization is used only in that situation. This does not factor an independently prepared wrapping component or a pair of overlapping live components.

Thus one genuine S94 matching row is closed for an actual copied primitive whose fixed-target identification clauses have been checked: its denominator, extracted logarithmic factor, original count constants, and normalized reference kernel agree exactly. The remaining model-specific assertions are the fixed-target identification where not yet verified and the claim that native complete histories to be compared supply these copied records, including all their read sets and boundary data.

138.5 The cutoff-weighted action tail on the verified physical sector

VM9.17 compared the full nonlinear critical action. To use that estimate for the original weighted action, the spatial cutoff must also be accounted for. Here the fields are in the established physical target sector of V7N, with its common reference chart, complete constraints, endpoint curvature bound, and local inverse. This paragraph does not prove admission of an arbitrary original mixed history to that sector.

Use current units, with fine spacing a_f. Let W_s be the physical solution with target $\theta + s\zeta(C)$, where the full changed-target carrier is Y, its actual coordinate count is n_Y, and $\|\zeta(C)\|_\infty \le t$. Let η bound the common physical background/path strength. Suppose the original fixed spatial cutoff ζ_0 equals

one on a complete D-neighborhood of Y, and is bounded by a fixed constant everywhere. It remains fixed along s.

For the holomorphic Wilson plaquette function $s(P) = N - \frac{1}{2}\operatorname{tr}(P + P^{-1})$, the exact derivative is

$$ds(P)[\dot{P}] = -\frac{1}{2}\operatorname{tr}[(I - P^{-2})\dot{P}]. \tag{RN9.9}$$

On the common small plaquette chart this is bounded by $C\|P - I\|\|\dot{P}\|$. Set $F_{s,p} = a_f^{-2}(P_p(W_s) - I)$. Therefore

$$|ds(P_p(W_s))/ds| \le Ca_f^4|F_{s,p}|\,|\dot{F}_{s,p}|. \tag{RN9.10}$$

The factor a_f^4 is retained before summing fine plaquettes. On each current unit cell, the endpoint physical curvature bound gives an L^2 norm at most $C\eta$. Differentiating the actual five-equation system along this path gives a forcing only in its original target rows on Y. The locally weighted inverse of its full linearization, as in V7N27–30, bounds the solution response by the sum of its target-source kernels. The subtracted current and plaquette identities in V7N33–37 then give

$$\|\dot{F}_s\|_{L^2(Q)} \le Ct \sum_{y \in Y} e^{-c_0 \operatorname{dist}(Q,y)}. \tag{RN9.11}$$

All five rows and their gauge/prefix factors remain in this response. The estimate follows either by a difference quotient in their common strict domain or by differentiating their already established analytic solution. It uses the local curvature comparison and no unproved higher spatial jet.

Sum RN9.10 with $|1 - \zeta_0|$, use Cauchy–Schwarz on each current cell, and then RN9.11. Only cells at distance at least $D - O(1)$ from the complete changed-target carrier contribute. The four-dimensional exponential kernel has a summable tail, so, after a fixed reduction of its exponent,

$$\begin{aligned} &|[A(\zeta_0, W_1) - A(\zeta_0, W_0)] - [A(W_1) - A(W_0)]| \\ &\qquad \le Cn_Y t\eta e^{-cD}. \end{aligned} \tag{RN9.12}$$

The current-cell volume sum and the fine plaquette volume factor have both been retained. The constant is independent of hidden fine depth and of the ambient covering degree. In restored physical units the corresponding local norms and current volume factors are rescaled together. There is no naked count of fine plaquettes.

The argument applies to complex solutions on the established common chart with its actual bounded transports and separate curvature control. If the spatial cutoff moves with the varied field, there is an additional $\sum_p \dot{\zeta}_{0,p} s(P_p)$ term. RN9.12 does not erase that term. Likewise its mixed-domain version requires the actual adapted locality and distance, with each cell's original physical scale retained.

For two physical target problems satisfying VM9.14–17 and the preceding cutoff geometry at both endpoints, apply RN9.12 to each side. The centered weighted classical exponent then obeys

$$\sup_C |v_{\zeta,\mathrm{cl}}(C)| \le Cx_k n_Y t\eta e^{-cD}. \tag{RN9.13}$$

Thus VM9.3–5a applies to the original *weighted* classical factor when its actual primitive has been identified with this physical problem. The original text's (1.5)–(1.6) uses the same separation:

unweighted stationarity removes the full linear current term, and the remaining weighted term is an exponentially small remote tail. RN9.12 gives the displayed nonlinear path version on the already verified physical sector.

138.6 What this closes and what it leaves

For a copied complete finite variational problem, RN9.3a identifies the actual minimizer sets without a uniqueness assumption. For a copied primitive meeting the separately stated fixed-target identification clauses, RN9.7 is exact: there is no denominator defect to pay, no small-mass inverse, and no additional reserve consumption. Its original Haar/tree constants and every scalar factor extracted in (1.10) are also copied. The implication is proved, but the identification clauses have not been verified for every original mixed-history primitive.

For nearby verified physical inputs, RN9.13 supplies the weighted classical oscillation needed by the centered reference comparison. If that oscillation is bounded by δ times an actual charged count, VM9.5a and the one previously allocated VC reserve handle its full genuine reference fraction. Neither argument modifies a source word's original mesh or saved scale factors.

These results do not identify every native two-torus history with a copied record. They also do not compare the inherited potential $V(X^{\sim})$, the scalar $E_k(Z)$ retained in the later R operator, independent vacuum increments, uninspected typed seed factors, or complete wrapped tails. Those are not part of the pure primitive denominator RN9.1 and do not cancel with it. The action completion partners in VM9.8 remain in the complete comparison.

Complete S94 therefore remains unproved. The new conclusion is the original primitive-denominator implication with its exact graph, measure, cutoff, scalar bookkeeping and explicit fixed-target admissibility requirement, together with the weighted-action tail estimate on the stated physical sector. The unconditional minimizer-set identity is kept distinct from the selected-kernel identity wherever that requirement has not been checked.

141.2 Primary input and original geometry

The primary input here is T. Balaban, *The variational problem and background fields in renormalization group method for lattice gauge theories*, Commun. Math. Phys. **102** (1985), 277–309, printed pp. 277–279, equations (1)–(10) and Theorem 1. The local primary PDF has been inspected, including the printed geometry on p. 277.

In the notation of that paper, the nested domains satisfy

$$\Omega_0 \supset \Omega_1 \supset \cdots \supset \Omega_k, \qquad (L^j a)^{-1}\,\mathrm{dist}(\Omega_j^c, \Omega_{j+1}) > RM_1, \qquad a = L^{-k}. \tag{AE10.1}$$

The domain Ω_j is a union of the prescribed big blocks of side $M_1 L^j a$. Its mixed determining lattice is $\mathcal{B}_k = \bigcup_j \Lambda_j$, with $B^j(\Lambda_j) = \Omega_j \setminus \Omega_{j+1}$. The actual constraint (3) is

$$\overline{U}^{\,j} = V \quad \text{on } \Lambda_j. \tag{AE10.2}$$

At level zero this is the literal identity constraint

$$U_b = V_b \quad (b \in \Lambda_0). \tag{AE10.3}$$

No tangent linearization, replacement average, or deletion of a constraint row is used below.

The target condition (7) bounds the mixed plaquettes of V, including plaquettes at interfaces. For those interfaces, the missing bonds are filled by the actual finer-level averages prescribed in the paper. It is not enough to check only plaquettes wholly inside a layer. Theorem 1 gives a minimal orbit for every such target of sufficiently small size ϵ_1. The orbit is contained in the original own-scale regularity class of size $B_3\epsilon_1$, and is the unique critical orbit in the indicated regularity range. Admissible gauge transformations obey the paper's condition (4), fixing the retained nodes.

Here is an admissible choice of geometry for the construction. Take a fixed sufficiently large physical torus and, inside it, nested aligned cubes. Starting from a fixed current-scale innermost cube, enlarge backwards by $HM_1L^j a$ at interface j, where the fixed integer H exceeds both the separation constant in AE10.1 and the constants for the actual complete averaging prefixes. These are the original prefixes used in M8; they are not replaced by one-step or endpoint-only read sets. The total added physical width is bounded uniformly because

$$\sum_{j=0}^{k-1} L^j a < (L-1)^{-1}. \tag{AE10.4}$$

Rounding to the original big-block alignment only changes the fixed constant. There is room in the interior of the outer fine collar $\Omega_0 \setminus \Omega_1$ for a fine bond b_*, its plaquette star, and the adjacent stars needed to evaluate the currents below, all at a fixed number of fine blocks from either boundary.

Choose this fixed collar width large enough that the support of the defect does not meet the complete read set of any coarser determining row or of any mixed-interface target plaquette. This is an explicit separation of the original read sets. A level-one interface reads a fixed number of fine blocks. Higher-level interfaces are farther inward by the corresponding collars. All other target links will be the identity. Thus every mixed-interface plaquette, including its prescribed finer-level reconstruction, is exactly the identity.

Let D be a fixed current-size box containing this retained fine star. This last requirement is essential: this example is about a norm measured on such a D.

141.4 The directional-current obstruction is also on the minimal graph

The complete fine star can be chosen large enough that every plaquette needed in the following current calculation consists of retained rows. Consequently all of those links are fixed by AE10.6 also in the actual minimum.

Let b_* point in direction ν, and let b' be the parallel bond displaced by one lattice spacing in a transverse direction μ. Choose the side of the displacement so that b' is the opposite ν-edge of an incident plaquette. Exactly one of the two (μ,ν)-plaquettes incident to b' contains b_*. Every other plaquette entering a ν-component force at b' is the identity.

Use the actual physical Wilson partial current of CQ/CS,

$$J_{b';\rho} = a^{-3} X_{b'} S_0^{(\rho\nu)}, \qquad \rho \neq \nu. \tag{AE10.11}$$

For a nontrivial plaquette $e^{\pm\delta T}$, its Lie-algebra gradient is the traceless anti-Hermitian part of that matrix, with the fixed orientation sign. Its component along T is a fixed nonzero constant times $\sin\delta$. It follows exactly that

$$J_{b';\mu} = j_a(\delta), \qquad J_{b';\rho} = 0 \quad (\rho \neq \mu,\nu), \qquad \|j_a(\delta)\| \geq c\delta a^{-3}. \tag{AE10.12}$$

The three directional projections in four dimensions therefore are

$$(Q_{b';\mu}, Q_{b';\rho_1}, Q_{b';\rho_2}) = (2j_a/3, -j_a/3, -j_a/3). \tag{AE10.13}$$

In particular,

$$\sum_{\nu,\rho\neq\nu} \|Q_{\nu;\rho}(U_a(\delta))\|^2_{L^2_a(D)} \geq c\delta^2 a^{-2}. \tag{AE10.14}$$

There is no contradiction with stationarity. The minimum is critical under the original mixed constraints. The level-zero identity rows force every admissible tangent on this star to vanish. The physical Wilson force there is a constraint reaction and need not vanish. Replacing constrained criticality by the unconstrained equation $J = 0$ would remove an actual row of the determining problem.

For test/readout length $\ell = 1$, AE9.6 and AE9.6a now imply

$$\mathfrak{F}_4(U_a(\delta); 1) \geq c\delta^4 a^{-4}, \qquad \mathfrak{J}_2(U_a(\delta); 1) \geq c\delta^2 a^{-2}. \tag{AE10.15}$$

For the flux statement, use either its original fine-cell assignment or $\mathfrak{F}_4 \geq c\|F_a\|_4^4$. For the current statement, use its literal disjoint assignment identity. No extra component or cell count is suppressed; one retained plaquette and one adjacent bond already give the lower bounds.

141.5 The explicit minimum and its exact small action

For the separated geometry in Section 2, define the fine field $\widetilde{U}$ to equal AE10.6 on the single bond and to be the identity everywhere else. Its complete original coarser averages are the identity, since none of their read sets meets the perturbed bond. Thus it satisfies the same original mixed constraints as $U_a(\delta)$, not merely linearized constraints.

This field satisfies the own-scale regularity conditions (2) of the primary paper with size $C\delta$: its plaquette defects and dimensionless Wilson-current defects are supported in a fixed fine star, and every coarser region is flat. Choose the fixed K in AE10.7 large enough that $C\delta \leq B_3 K\delta$. For sufficiently small δ, this field is in the same regularity space in which Theorem 1 supplies the minimum.

Every plaquette incident to the perturbed bond has all its links fixed by level-zero constraints. Its action contribution is therefore the same in every admissible compact configuration. Every other Wilson plaquette term is nonnegative. The configuration $\widetilde{U}$ makes all of those other plaquettes exactly the identity, so it attains this absolute lower bound. It is an explicit global constrained minimum, even before restricting to the small regularity class. Theorem 1 identifies its orbit with the native minimal orbit. We may consequently choose $U_a(\delta) = \widetilde{U}$ itself.

In the CQ normalization $s(P) = n_c - \operatorname{Re}\operatorname{tr} P$, there are exactly $2(d-1) = 6$ unoriented plaquettes incident to the interior perturbed bond, and each contributes $2(1-\cos\delta)$. Hence

$$S_0(U_a(\delta)) = S_0(\widetilde{U}) = 12(1-\cos\delta), \qquad c\delta^2 \leq S_0(U_a(\delta)) \leq C\delta^2. \tag{AE10.16}$$

The original paper's fixed trace normalization rescales the action and coupling together; it does not change the minimum or these scale powers. Multiplication by the same Wilson coupling β gives the corresponding equality and inequalities for βS_0. At fixed β, choose

$$\delta = a^\alpha, \qquad 0 < \alpha < 1. \tag{AE10.17}$$

Then the original target size, the native own-scale regularity size, and the Wilson action all tend to zero, while for $p > 8$

$$\begin{aligned} \|F_a(U_a(a^\alpha))\|_{L^p_a(D)} &\geq c a^{\alpha+4/p-2} \longrightarrow \infty, \\ \mathfrak{F}_4(U_a(a^\alpha);1) &\geq c a^{4\alpha-4} \longrightarrow \infty, \\ \mathfrak{J}_2(U_a(a^\alpha);1) &\geq c a^{2\alpha-2} \longrightarrow \infty. \end{aligned} \tag{AE10.18}$$

The action assertion remains true along any coupling sequence with $\beta(a)a^{2\alpha} \to 0$, but no running-coupling theorem is required or asserted here.

This improves the earlier raw-field counterexamples in one precise respect: the field on the left side of every lower bound is an actual minimal configuration supplied by the original nonlinear mixed Wilson variational theorem. It still does not identify a realizable R-I selector history for that field.

142.1 Statement and the identified domain

This chapter extends the centered real-prefix estimate CD9.9 to a specified common complex regular chart. It retains the original raw means, their analytic branches, every intermediate compatible map, the ordered Wilson Hessian, and all intermediate forcing counts. The new estimate also permits

parameter Cauchy bounds on the same complex domain. It does not admit an original mixed R field to that domain or identify an independently varying reference with a centered reference family.

Let $a_s = L^s a$, $s = 0, \dots, n$, with $a_n = 1$ and fixed odd L. Fix the complete, nonwrapping, coordinate-path-convex protected neighborhoods used in CD9, including every original prefix, Wilson-Hessian star, ribbon filling, and curvature-propagation stencil. Their sizes and overlap constants are fixed in current units.

Let z range over one declared complex parameter polydisc $\mathcal{D}$. On the complete neighborhood suppose the actual holomorphic connection has a potential in one controlled analysis chart satisfying

$$U_0(z) = e^{aA(z)}, \qquad \sup_{\mathcal{D}} \|A(z)\|_{W_a^{1,p}} \le M, \qquad \sup_{\mathcal{D}} \|a^{-2}(P(U_0(z)) - I)\|_\infty \le \alpha, \quad p > 8. \tag{CD10.1}$$

Choose $M \le M_*$ and then $\alpha \le \alpha_*(M_*)$ sufficiently small for the original analytic means and logarithms. These are separate bounds; the curvature bound is not inferred from the weak potential norm. The parameter domain and its radii are not enlarged in the proof.

There are already constructed branches to which CD10.1 applies. The CC branch on a small complex $\mathcal{Y}_a$ input ball has both bounds by CC3–5. CL25–28 give them on the protected regions where every complete equation carrier keeps weight one, uniformly over their allowed exterior walls. M8.9–10 give them for the physical source reconstruction with reconstruction walls all equal to one and with complex weak input varying on its proved ball. For these local applications the entire neighborhood required above must lie inside the region where those estimates hold. Arbitrary nontrivial M8 reconstruction walls are not included. This is a restriction to identified strong branches, not an assertion about original R-selector support.

Use exactly the actual reference chain and centered compatible maps

$$U_{s+1} = \mathcal{M}(U_s), \qquad V_{s+1} = \mathcal{B}_{U_s}(V_s), \qquad x_s = \log(V_s U_s^{-1}), \qquad x_{s+1} = \Phi_s(x_s), \quad \Phi_s(0) = 0. \tag{CD10.2}$$

For each value of z, the reference chain is held fixed while the derivatives in V are taken. The means are the original reference-logarithm outer averages and their original inner endpoint means, as specified in PG15a and CL1–4. A different group mean is not substituted.

Write $H_{ij} = (X_i X_j S_0)(U_0)$, with that ordered word and the holomorphic Wilson action $s(P) = N - \frac{1}{2}\operatorname{tr}(P + P^{-1})$. Define

$$\mathscr{B}_n^\gamma(z) = \sum_{i,j} H_{ij}(z) X_j X_i x_n^\gamma(U_0(z)). \tag{CD10.3}$$

For an actual finite set E of final local bond coordinates, let m_E include its original determining enlargement. The result is

$$\boxed{\sup_{z \in \mathcal{D}} \sum_{\gamma \in E} |\mathscr{B}_n^\gamma(z)| \le C_{L,p,M_*}\, m_E\, a\alpha.} \tag{CD10.4}$$

There is no loss exponential in the number n of blocking steps.

142.2 Fine path control and a coupled quantity for the actual means

Uniform discrete Morrey on the fixed protected box gives $\|A(z)\|_\infty \le C_{p,D} M =: m$. Thus every original fine path γ of physical length r satisfies

$$\|U_0(\gamma)\| + \|U_0(\gamma)^{-1}\| \le 2e^{Cmr}. \tag{CD10.5}$$

No intermediate potential bound is assumed. It will follow from the actual mean-versus-straight comparison and curvature together.

Let $R_s(c)$ be the original fine straight-path product joining the endpoints of a genuine scale-a_s bond. Let e_s be the maximum of the differences between $U_s(c)^{\pm1}$ and $R_s(c)^{\pm1}$, and put $d_s = e_s/a_s^2$. Initially $e_0 = d_0 = 0$. The fine-path estimate implies, without choosing a logarithm of U_s,

$$\|U_s(c)^{\pm1}\| \le e^{Ca_s(m+a_sd_s)}, \qquad \|U_s(\gamma)^{\pm1}\| \le e^{Cr(m+a_sd_s)} \tag{CD10.6}$$

for a scale-a_s path of physical length r. The inverse is included in the definition of e_s; it is not estimated by an assumption of unitarity. A one-step tree gauge and its inverse adjoint consequently cost $e^{C_La_{s+1}(m+a_sd_s)}$.

142.3 A coupled curvature and path-error induction in the original frame

Let f_s be the supremum of the physical logarithmic plaquette curvature of U_s, measured throughout in the same original frame on the required neighborhoods. The small matrix/logarithmic conversion gives $f_0 \le 2\alpha$.

At one step put $\ell = a_{s+1}$ and $b_s = m + a_sd_s$. A local complex tree gauge is used temporarily to prove the finite-stencil estimates, then their values are returned to the original frame. Its forward and inverse endpoint factors cost $e^{C_L\ell b_s}$ by CD10.6. The local loop expansion puts the dimensionless links in that tree frame within $C_Le^{C_L\ell b_s}\ell^2 f_s$ of the identity.

The actual flat curl identity PG19 is complex linear and its two-form map is a normalized positive scalar average. Hence its operator norm for the absolute color norm is exactly at most one. Taylor expansion of the actual analytic mean retains that leading coefficient one. The nonlinear remainder, including the fine logarithmic-versus-linear curl conversion, is quadratic in local dimensionless link size and is bounded after physical rescaling by $C_L\ell^2 f_s^2$, with the displayed endpoint factors.

The same contractible-loop comparison bounds the difference between the actual output mean and the straight product of its L input links by $C_Le^{C_L\ell b_s}\ell^2 f_s$, for the output and its inverse. The difference between that straight product and R_{s+1} is the exact product-difference sum of L terms. Its partial products obey CD10.6. After increasing one fixed constant, these facts give the coupled recurrences

$$\boxed{\begin{aligned} &f_{s+1} \le E_s(f_s + C_La_{s+1}^2 f_s^2),\\ &d_{s+1} \le L^{-1}E_sd_s + C_LE_sf_s, \qquad E_s = \exp\!\big(C_La_{s+1}(m + a_sd_s)\big). \end{aligned}} \tag{CD10.7}$$

The second equation is the first-power product comparison $e_{s+1} \le LE_se_s + C_LE_sa_{s+1}^2 f_s$, divided by a_{s+1}^2. It is not an associativity assertion for the nonlinear mean.

Here is an explicit uniform bootstrap. Choose a fixed $q \in (L^{-1}, 1)$. Choose the ceiling M_* first, small enough for $L^{-1}e^{C_Lm_*} < q$ with room to spare, where $m_* = C_{p,D}M_*$. Choose fixed constants K, D with

$$K > 4\exp\!\left(\frac{C_Lm_*}{1 - L^{-1}}\right), \qquad D > \frac{2C_Le^{C_Lm_*}K}{1-q}. \tag{CD10.8}$$

Finally choose α_* small enough that under $d_s \le D\alpha$, $f_s \le K\alpha$, one has $L^{-1}E_s \le q$, $C_LE_sK \le (1-q)D$, and

$$\exp\!\left(C_LD\alpha_*\sum_s a_{s+1}a_s + C_LK\alpha_*\sum_s a_{s+1}^2\right) < 2. \tag{CD10.9}$$

All these choices are independent of the number of steps because both squared-scale sums are bounded geometric sums. The second recurrence closes $d_s \le D\alpha$. For the first use $1 + C_La_{s+1}^2 f_s \le e^{C_La_{s+1}^2 f_s}$

and multiply the actual E_s's. Their logarithm sums to $C_L m \sum a_{s+1} + C_L \sum a_{s+1} a_s d_s$. Together with $f_0 \le 2\alpha$, CD10.8–9 close $f_s \le K\alpha$, with strict room in the ceiling. If $\alpha = 0$, both recurrences give $f_s = d_s = 0$ directly. In all cases,

$$f_s \le K\alpha, \qquad e_s \le D a_s^2 \alpha, \qquad \|U_s(c) - I\| \le C a_s (m + D a_s \alpha). \tag{CD10.10}$$

The last estimate follows from the original fine straight product and e_s. It proves the needed intermediate branch smallness. Choosing the same ceilings within the finite one-step analytic radius ensures inductively that the mean just used is always the original identity-connected branch. There is no independent selection of a different mean or logarithm.

Thus the leading curvature multiplier is one plus a summable $O(m a_{s+1} + \alpha a_{s+1} a_s)$ error. The auxiliary normalized path error has the genuine contraction L^{-1}. Neither is replaced by a fixed per-level norm loss, and no complex positivity of the Wilson density is used.

142.4 One fine analysis gauge for each complete prefix

Fix a prefix ending at scale $\ell = a_t$, with its complete buffered coordinate box of diameter $C_L \ell$. Choose one fine coordinate-tree gauge g on that entire box. Its original fine path lengths and CD10.5 give

$$\sup_x \{\| \operatorname{Ad} g(x)\|, \| \operatorname{Ad} g(x)^{-1}\|\} \le e^{C_L m \ell}. \tag{CD10.11}$$

In the neighboring-path comparison each ribbon has width a and length $O_L(\ell)$. A nonabelian product-difference expansion has $O_L(\ell/a)$ plaquette errors. The conjugating paths and every partial ribbon boundary have physical length $O_L(\ell)$, so their norms are bounded by CD10.11, rather than by a separate factor for every plaquette. It follows that

$$\|U_0^g(b) - I\| \le C_{L,M_*} a \ell \alpha. \tag{CD10.12}$$

All ribbon plaquettes are in the declared controlled neighborhood.

The actual intermediate fields are now compared in this same one gauge. The original-frame recursion already proved in CD10.7 is

$$e_s \le L e^{C_L a_s (m + a_{s-1} d_{s-1})} e_{s-1} + C_{L,M_*} a_s^2 \alpha, \qquad e_0 = 0. \tag{CD10.13}$$

The complete coupled induction, without assuming intermediate potential bounds, has established

$$e_s \le C_{L,M_*} a_s^2 \alpha. \tag{CD10.14}$$

The transformation of this error costs the two endpoints of the one gauge g only. The fine straight product after that gauge has a_s/a factors satisfying CD10.12. Gauge covariance of the original means consequently yields

$$\|U_s^{N^s g}(c) - I\| \le C_{L,M_*} a_s \ell \alpha, \qquad 0 \le s \le t. \tag{CD10.15}$$

The same argument on the next fixed enlargement includes a one-step map leaving scale a_t.

Every gauge in this proof is an analysis gauge. It is fixed while the incoming V-jets are taken. It changes none of the original gauge pins, determining rows, selectors, or reference records. Smallness of M_* and α_* keeps both the original and these transformed finite one-step arguments on the same analytic mean and logarithm branch. The complex covariance is the analytic continuation of CL5–6 on that branch.

142.5 Exact frame telescoping and the leading response products

Let G_s be the block-diagonal matrix of $\mathrm{Ad}(N^s g)$ at the starting sites of the scale-a_s bonds. Put $A_s = D\Phi_s(0)$, $K_s = D^2\Phi_s(0)$, and $S_t = A_{t-1}\cdots A_0$. Covariance and the centered logarithm give the exact identities

$$A_s^g = G_{s+1} A_s G_s^{-1}, \qquad S_t^g = G_t S_t G_0^{-1}. \tag{CD10.16}$$

The intermediate matrices cancel before norms are taken. Only the two endpoint matrices in the second identity are estimated. For the complex bilinear invariant color form, one also has $G_s^{\mathsf{T}} G_s = I$. This is bilinear orthogonality and does not imply Hermitian norm isometry.

The flat dimensionless one-step response is exactly L times the physical box-line average. Its color-block column sum is exactly L^{-3}. CL's finite-dimensional analytic estimate, applied to CD10.15, therefore gives in the one common gauge

$$\begin{aligned} \|A_s^g\|_{\mathrm{col}} &\le L^{-3}(1 + C_{L,M_*} a_{s+1}\ell\alpha), \\ \|A_s^g - A_s^0\|_{\mathrm{col}} &\le C_{L,M_*} L^{-3} a_{s+1}\ell\alpha. \end{aligned} \tag{CD10.17}$$

The block norm is induced by the fixed absolute complex color norm, and the column sum is submultiplicative. There is no per-step color dimension equivalence factor. Multiplication and exact difference telescoping give

$$\begin{aligned} \|S_t^g\|_{\mathrm{col}} &\le C_{L,M_*}(a/a_t)^3, \\ \|S_t^g - S_t^0\|_{\mathrm{col}} &\le C_{L,M_*}(a/a_t)^3 a_t\ell\alpha. \end{aligned} \tag{CD10.18}$$

Here $\sum_{s<t} a_{s+1}\ell \le C_L a_t \ell$. For a prefix-sized box $\ell \asymp a_t$, the relative defect is $O(a_t^2\alpha)$, exactly as in CD9. Returning the complete product to the original frame costs CD10.11 once at each endpoint. It cannot create e^{CnM}.

142.6 The transported Hessian and the flat color cancellation

Take one common gauge on the prefix patch through scale a_{t+1} for both S_t and K_t. The ordered fine Wilson Hessian in this gauge has fixed plaquette incidence and

$$\|H^g\|_{\mathrm{row}} \le C_L, \qquad \|H^g - H^0\|_{\mathrm{row}} \le C_{L,M_*} a a_{t+1}\alpha. \tag{CD10.19}$$

This is a finite four-link analytic estimate, with the inverses in the holomorphic Wilson action retained. The bilinear invariance noted after CD10.16 makes the transported contraction covariant:

$$R_t^g = S_t^g H^g (S_t^g)^{\mathsf{T}} = G_t R_t G_t^{\mathsf{T}}, \qquad R_t = S_t H S_t^{\mathsf{T}}. \tag{CD10.20}$$

The actual fine read set of a fixed scale-a_t patch contains at most $C_L(a_t/a)^4$ bonds. Hence CD10.18–19 give

$$\begin{aligned} \|R_t^g\|_{\mathrm{ent}} + \|R_t^0\|_{\mathrm{ent}} &\le C_{L,M_*}(a/a_t)^2, \\ \|R_t^g - R_t^0\|_{\mathrm{ent}} &\le C_{L,M_*} a^2\alpha. \end{aligned} \tag{CD10.21}$$

The response defects cost $(a_t/a)^4(a/a_t)^6 a_t^2\alpha$, and the Hessian defect costs $(a_t/a)^4(a/a_t)^6 a a_t\alpha$. These are the same literal fine counts as in CD9. The ordered antisymmetric part is included.

In this same comparison frame, finite-dimensional analyticity gives

$$\|K_t^g\| \le C_L, \qquad \|K_t^g - K_t^0\| \le C_{L,M_*} a_{t+1}^2\alpha. \tag{CD10.22}$$

The flat tensor R_t^0 is scalar in color. If every input is in one fixed generator direction, the actual means and compatible endpoint correction reduce exactly to their arithmetic logarithmic map. Thus every same-generator mixed second derivative is zero, over the complex field as well as over the reals. Consequently

$$R_t^0 : K_t^0 = 0. \tag{CD10.23}$$

Apply this to the full difference, not to an arbitrary transported tensor. Equations CD10.21–23 prove, per actual next-scale output coordinate,

$$|R_t^g : K_t^g| \le |(R_t^g - R_t^0) : K_t^g| + |R_t^0 : (K_t^g - K_t^0)| \le C_{L,M_*} a^2 \alpha. \tag{CD10.24}$$

Its return to the original frame is a single output adjoint factor, bounded by CD10.11. Tensors in unrelated comparison gauges are never contracted.

142.7 Initial bracket, every forcing term, and the strict gain

The actual ordered chain rule is unchanged:

$$\mathscr{B}_{t+1} = A_t \mathscr{B}_t + R_t : K_t. \tag{CD10.25}$$

The initial centered logarithm has a same-bond bracket as its ordered second derivative. Contracting with the antisymmetric part of the ordered Hessian gives its original Wilson-score term. The single-plaquette force bound in the original weak frame is $C_{M_*} a^2 \alpha$, including its bounded basepoint transport. Therefore

$$|\mathscr{B}_0(b)| \le C_{L,M_*} a^2 \alpha. \tag{CD10.26}$$

It is not set to zero by replacing the Hessian with its symmetric part.

A complete suffix from scale a_s to scale one has column norm at most $C_{L,M_*} a_s^3$, by CD10.16–18 on its full prefix-closed comparison box. The determining enlargement of E contains at most $C_L m_E a_s^{-4}$ actual coordinates at that scale. Propagating CD10.26 and every forcing term CD10.24 in the original frame now gives

$$\begin{aligned} \sum_{\gamma \in E} |\mathscr{B}_n^\gamma| &\le C_{L,M_*} m_E a^2 \alpha \left(a^{-1} + \sum_{s=1}^{n} a_s^{-1} \right) \\ &\le C_{L,M_*} m_E a \alpha. \end{aligned} \tag{CD10.27}$$

Each complete contraction and suffix has only its bounded endpoint frame costs. They are not multiplied once for every preceding depth. This proves CD10.4.

The same proof gives the CD9 principal and score bounds on this complex domain:

$$\begin{aligned} \sum_{\gamma,\delta \in E} |R_n^{\gamma\delta}| &\le C_{L,M_*} m_E (a^3 + a^2 \alpha), \\ \sum_{\gamma \in E} \left| \sum_i (X_i S_0) X_i x_n^\gamma \right| &\le C_{L,M_*} m_E a \alpha. \end{aligned} \tag{CD10.28}$$

For the principal row, use the flat face bound GG9 at the terminal scale and CD10.21. Its bounded endpoint adjoints are included once. These are coefficient bounds in the original local coordinate basis, not in an arbitrary nonlocal transverse basis.

142.8 Parameter derivatives on the literal common domain

The coefficient CD10.3 is a holomorphic function of the original parameters: its maps, reference chain, and centered evaluation are all holomorphic on their fixed branches. The analysis gauge used to prove the pointwise bound is not differentiated in defining this coefficient. Pointwise estimates followed by Cauchy's formula for the original coefficient avoid introducing derivatives of a chosen comparison gauge.

For definiteness suppose

$$\mathcal{D} = \{z : |z_r - z_r^*| < R_r,\ 1 \le r \le m\}, \qquad |z_r - z_r^*| \le \rho_r < R_r, \qquad d_r = R_r - \rho_r. \tag{CD10.29}$$

For every multi-index ν, Banach-valued Cauchy estimates applied to the actual output ℓ^1 norm give

$$\boxed{\sum_{\gamma \in E} |\partial_z^\nu \mathscr{B}_n^\gamma(z)| \le C_{L,p,M_*} m_E a \alpha\, \nu! \prod_{r=1}^{m} d_r^{-\nu_r}.} \tag{CD10.30}$$

The principal and score rows in CD10.28 have the corresponding Cauchy bounds with their displayed base factors. For normalized Taylor coefficients the $\nu!$ is divided out. For distinct nilpotent finite-source labels the mixed derivative is taken once in each of the actual label parameters; no factorial or extra independent label is added.

These margins are literal. For a linear input family $A(z) = A_* + \sum_r (z_r - z_r^*) h_r$, a sufficient declared polydisc in a strong input ball of radius t obeys $\|A_*\| + \sum_r R_r \|h_r\| \le t$, in that ball's actual norm. For a reconstruction wall polydisc $|s_C| < R_C$, derivatives at physical wall values $s_C = 1$ use margins at most $R_C - 1$. The proof does not replace these radii with a larger source radius, nor assign a new scale gain to a derivative.

If the parameters also vary the centered reference, their actual responses occur automatically when differentiating the holomorphic family CD10.3. Such derivatives are different from an off-diagonal derivative with a fixed independent reference. Equation CD10.30 does not identify those two derivatives and does not delete any live-reference term from an inherited joint current.

142.9 Application and limits in the marked construction

For the periodic CC construction, the small complex strong input ball CC5 bounds both the input weak potential and its curvature. CC3–4 then give CD10.1 for its actual same-target critical branch, uniformly in the cutoff. On its protected nonwrapping subregions, CD10.4 and CD10.30 therefore apply directly. This uses the identified CC solution and the actual raw chain of that solution.

For CL25–28, take a protected region with enough fixed current collars that the entire CD10 prefix, ribbon, and curvature closure lies where every complete protected equation carrier has weight one. CL28 is uniform over the permitted exterior walls on that common holomorphic domain. Its weak potential and physical curvature bounds transfer to the controlled original analysis chart. This supplies CD10.1 with $M \le C\epsilon$, $\alpha \le C\epsilon$, after the original smallness ceiling is chosen.

For M8.9–10, apply the theorem at the physical reconstruction-wall values all equal to one, while the complex weak input b varies on the original proved ball. The same full protected closure is required. Any further assertion involving nontrivial exterior reconstruction walls needs the identified protected MC/CL theorem to keep all those complete equation carriers unchanged. It does not follow merely from analyticity of the M8 map. Thus the centered complex bounds and actual-margin derivatives

follow on the stated input domain; arbitrary M8 wall derivatives have not been added. If the protected closure is missing, the corresponding local application is unavailable.

The exact invariant-terminal-test identity CD9.24 still holds on these complex branches by CL15 and analytic continuation. Thus these are bounds for the same physical scalar tests on the actual regular raw hierarchy. A test coupling several live histories or independent reference/constraint variables need not have this terminal invariance and is not covered by that identification.

Finally, the physical normalization is unchanged. CD9/NH9's first current row carries its exact factor $2c_{\beta}^{-3}g_{j}^{2}/g_{e}$, and that convention includes the word's original a^{4}. Removing a factor absent from a specified physical source changes $a\alpha$ to $a^{-3}\alpha$. Cauchy differentiation cannot supply an absent original-mesh birth factor. It also does not bound an ordinary scalar companion in total variation, selector-boundary currents, later contacts, or the response of genuine denominators.

The positive closure here is the centered complex regular-prefix coefficient estimate and its parameter derivatives on an already constructed strong branch. AE8/LM47 on original mixed histories, the complete all-history MT36 pair, and S94 remain separate unproved obligations. The argument changes no constraint record, normalization, component count, or reserve assignment.

143.2 The complete quadratic generating function on one actual Gaussian

Work on the positive flat physical row space of CP33 and GG9, with harmonic modes fixed or separately retained. All matrices are written in orthonormal physical bases, so their traces are literal traces over the actual independent coordinates. The physical volume weights are already in the inner products and adjoints.

Let

$$\Sigma = \beta^{-1} H^{-1}, \qquad d\mu_0 = N(0, \Sigma), \tag{PC10.9}$$

where $H = d_a^* d_a$ is the original flat Maxwell Hessian on this space. For any symmetric matrix $T(\eta)$ and scalar $s(\eta)$ with zero source-free part, put

$$A_\eta(A) = \langle A, T(\eta) A \rangle + s(\eta), \qquad S = \Sigma^{1/2} T \Sigma^{1/2}. \tag{PC10.10}$$

The scalar s is the actual scalar companion. It is not silently replaced by a centered value.

Take the complete joint observation

$$QA = (Q_1 A, \ldots, Q_m A). \tag{PC10.11}$$

The Q_e may be the actual boxed averages at different levels, with their actual offsets and any specified canonical factors. The joint coarse covariance can be singular because some retained observations are determined by others. No inverse of that joint covariance is needed below.

Define

$$\begin{aligned} K_\eta &= (\Sigma^{-1} - 2T)^{-1} = \Sigma^{1/2}(I - 2S)^{-1}\Sigma^{1/2}, \\ C_0 &= Q\Sigma Q^*, \qquad C_\eta = QK_\eta Q^*, \qquad \delta C = C_\eta - C_0, \\ v(\eta) &= s(\eta) - \tfrac{1}{2}\operatorname{Tr}\log(I - 2S). \end{aligned} \tag{PC10.12}$$

All inverse and logarithm series terminate in the source algebra. For every retained linear probe z, Gaussian completion of the square gives the coefficientwise exact formula

$$\boxed{\mathbb{E}_0 \exp\{A_\eta(A) + \langle z, QA\rangle\} = \exp\{v(\eta) + \tfrac{1}{2}\langle z, C_\eta z\rangle\}.} \tag{PC10.13}$$

One proof first proves the ordinary Gaussian formula for real matrices in a neighborhood of zero and then takes the finitely many Taylor coefficients. Equivalently, expand the nilpotent exponential; every coefficient is an integrable Gaussian polynomial times a linear exponential. Neither proof invokes a source radius uniform in the fine mesh.

For individually centered quadratic currents, $s = -\operatorname{Tr}(\Sigma T) = -\operatorname{Tr} S$, this becomes

$$\begin{aligned} v(\eta) &= \sum_{r\geq 2} \frac{2^{r-1}}{r} \operatorname{Tr}(S^r), \\ \delta C(\eta) &= \sum_{r\geq 1} 2^r Q\Sigma^{1/2} S^r \Sigma^{1/2} Q^*. \end{aligned} \tag{PC10.14}$$

In particular, centering the one-current insertion cancels only the linear trace. It does not remove higher absolute scalar contacts.

If $T = \sum_I \eta_I T_I$, the powers in PC10.14 contain all ordered matrix words in the original T_I. No commutativity of these matrices, common source anchor, or replacement of a fine test value by a coarse central value has been used. Cross-level contacts are the off-diagonal blocks of δC for the single joint Q.

143.2.1 The actual weighted CQ quadratic coefficient

On the same flat reference, retain the actual diagonal source multipliers $f_i = f_b$, $h_j = h_c$. The quadratic part of the centered CQ current, with its physical factor g_j^{-4}, is

$$\begin{aligned} t_{f,h}(A) &= 2g_j^{-4}\beta^{-1}\langle A, HfHhHA\rangle - 2g_j^{-4}\beta^{-2}\operatorname{Tr}(fHhH), \\ T_{f,h} &= g_j^{-4}\beta^{-1}H(fHh + hHf)H. \end{aligned} \tag{PC10.15}$$

The two forms agree because a scalar quadratic form uses the symmetric part of its matrix. The trace equality is cyclic. This is the flat limit of the actual first contact in CP4, with its complete constant companion; it is not an evaluation of the nonlinear contact on a non-Gaussian inherited density.

Thus PC10.13 applies without removing the original $f_b h_c$ offsets. It also makes precise the covariance cancellation due to the two outer Hessians: for $T = HKH$, with the original finite-source matrix K,

$$\begin{aligned} [\delta C]_r &= 2^r\beta^{-(r+1)}QK(HK)^{r-1}Q^*, \\ [v]_r &= \frac{2^{r-1}}{r}\beta^{-r}\operatorname{Tr}((KH)^r) \quad (r \geq 2) \end{aligned} \tag{PC10.16}$$

when the scalar is centered and $[\cdot]_r$ denotes the homogeneous degree in K. These are identities on the admitted physical rows. If an ambient coordinate representation is used, the corresponding physical projections at the ends remain. For nonconstant multipliers, those projections cannot be erased merely by calling K local.

143.3 An exact compensator, with its precise meaning

For a smooth retained test G, with the finite number of derivatives needed for the source degree and with integrable Gaussian growth, PC10.13 implies

$$\mathbb{E}_0[e^{A_\eta(A)}G(QA)] = e^{v(\eta)}\,\mathbb{E}_0\left[e^{\frac{1}{2}\delta C(\eta):D^2}G(QA)\right]. \tag{PC10.17}$$

Here the constant-coefficient differential exponential is a finite polynomial in the source algebra. A direct proof differentiates the Gaussian covariance in a finite-dimensional interpolation. Each covariance derivative is one half of contraction with the test's second derivative. The formula remains valid for a rank-deficient joint observation by doing this differentiation in the nonsingular fine Gaussian before applying Q.

Consequently the explicitly completed test

$$G_\eta^{\mathrm{comp}} = e^{-v(\eta)}e^{-\frac{1}{2}\delta C(\eta):D^2}G \tag{PC10.18}$$

satisfies the exact all-source identity

$$\boxed{\mathbb{E}_0[e^{A_\eta(A)}G_\eta^{\mathrm{comp}}(QA)] = \mathbb{E}_0[G(QA)].} \tag{PC10.19}$$

For $G(B) = \exp\langle z, B\rangle$, the compensator is simply

$$\exp\{-v(\eta) - \tfrac{1}{2}\langle z, \delta C(\eta)z\rangle\}. \tag{PC10.20}$$

This closes the redundant Gaussian current/contact module if these specific contact counterterms are chosen. Its proof does not give a mesh gain to an uncompensated packet. The original source exponential is preserved in PC10.17; the right side displays what it actually produces. PC10.18 is an additional, explicitly defined source-test completion. Its equality with a chosen physical renormalization prescription must be proved before it is used for that prescription.

For a field-dependent later source $A(B)$, even the first covariance term differentiates both its first and second field jets:

$$R : D^2e^{A(B)} = e^{A(B)}\{R : D^2A + (DA)^*R(DA)\}. \tag{PC10.21}$$

Thus the compensator includes lower currents and the last scalar contact. It cannot be replaced by a current coefficient with those partners omitted.

143.3.1 The corresponding coordinate transformation

Set

$$F_\eta = \Sigma^{1/2}(I - 2S)^{-1/2}\Sigma^{-1/2}, \qquad Q_\eta = QF_\eta^{-1}. \tag{PC10.22}$$

The square root is a finite binomial series. The weighted Gaussian measure in PC10.13 is exactly $e^{v(\eta)}$ times the formal pushforward of μ_0 by F_η. Therefore replacing the observation by Q_η cancels its covariance defect, while the absolute scalar mass e^v remains. Both the map change and that mass must be included to obtain PC10.19.

In an original RG constraint $\delta(QA-B)$, replacing Q by Q_η is a response of the actual constraint. It is not an identity with the unchanged constraint. PC10.8 is its nonlinear counterpart. Keeping a sharp selector fixed during this operation likewise leaves its boundary response. Nothing in the Gaussian matrix computation supplies those nonlinear bounds.

143.4 The physical D8 current and every repeated-current coefficient

Now specialize to exactly the constant-test packet of GG9.5, with no additional word-mesh factor:

$$\begin{aligned} t_8(A) &= g_j^{-4} W_{8,a}^0(A) \\ &= 2g_j^{-4}\beta^{-1}\langle A, H^3 A\rangle - 2g_j^{-4}\beta^{-2}\operatorname{Tr} H^2, \\ T_0 &= 2g_j^{-4}\beta^{-1}H^3, \qquad M = \Sigma T_0 = 2g_j^{-4}\beta^{-2}H^2. \end{aligned} \tag{PC10.23}$$

The vector field $A \mapsto MA$ has weighted Gaussian divergence exactly t_8:

$$D_{MA}\mu_0 = t_8\mu_0. \tag{PC10.24}$$

The equality follows from $\langle A, \Sigma^{-1}MA\rangle - \operatorname{Tr} M$.

Take $\tau = \varepsilon_1 + \cdots + \varepsilon_n$, with disjoint original source words ε_i. Applying PC10.14 to the original exponential $e^{\tau t_8}$ gives

$$\begin{aligned} v(\tau) &= \sum_{r\ge 2} \frac{2^{r-1}}{r}\tau^r \operatorname{Tr}(M^r), \\ C_\tau &= Q\Sigma(I-2\tau M)^{-1}Q^*, \\ [\tau^r]\delta C &= 4^r g_j^{-4r}\beta^{-(2r+1)} Q H^{2r-1} Q^*, \qquad r \ge 1. \end{aligned} \tag{PC10.25}$$

If $\beta = c_\beta g_j^{-2}$, the last prefactor is $4^r c_\beta^{-(2r+1)} g_j^2$. If the two external coordinates are normalized at couplings g_e, g_f, their corresponding block gains the recorded factor $g_j^2/(g_e g_f)$. No running coupling is silently identified with another.

For one current word $\varepsilon^2 = 0$, PC10.17 reduces to

$$\mathbb{E}_0[t_8 G(QA)] = 2g_j^{-4}\beta^{-3}\mathbb{E}_0[(QHQ^*) : D^2 G(QA)]. \tag{PC10.26}$$

This is the same complete current as GG9.6. In particular, the scalar trace and the quadratic part of that conditional current have already been combined. The first compensating test term is $-2\varepsilon g_j^{-4}\beta^{-3}(QHQ^*) : D^2 G$, not an extra a^2 or a^4 factor.

For two later linear source labels $z = \xi_1 u + \xi_2 v$, PC10.26 gives the scalar contact

$$4g_j^{-4}\beta^{-3}\langle u, QHQ^* v\rangle \quad \text{at degree } \varepsilon\xi_1\xi_2. \tag{PC10.27}$$

This is nonzero for suitable actual coarse tests. A source-independent scalar subtraction cannot cancel the current for all later tests: the constant test fixes that scalar subtraction to zero, whereas PC10.27 supplies a nonzero contact.

143.6 The higher absolute scalar masses

The trace in PC10.25 is a fine physical trace. On a periodic four-torus of fixed physical volume, the flat transverse Maxwell spectrum has $O(T^4a^{-4})$ nonzero coordinates, eigenvalues at most Ca^{-2}, and a fixed positive fraction of eigenvalues at least ca^{-2}. For example, restrict one Fourier coordinate to the middle half of its Brillouin interval. Removing the finitely many harmonic modes does not affect these bounds. Therefore, for each fixed $r \geq 1$,

$$c_r T^4 a^{-4r-4} \leq \operatorname{Tr}(H^{2r}) \leq C_r T^4 a^{-4r-4}. \tag{PC10.34}$$

The constants include the actual physical polarizations and color multiplicity; they do not remove those coordinates from the trace.

For $r \geq 2$, PC10.25 gives

$$[\tau^r]v = \frac{2^{2r-1}}{r} g_j^{-4r} \beta^{-2r} \operatorname{Tr}(H^{2r}). \tag{PC10.35}$$

At $\beta = c_\beta g_j^{-2}$, the coupling factor is c_β^{-2r}. In particular, these positive scalar coefficients do not acquire a small coupling merely by canonical normalization. For r distinct equal current words,

$$[\varepsilon_1 \cdots \varepsilon_r]v = 2^{r-1}(r-1)! \operatorname{Tr}(M^r). \tag{PC10.36}$$

At $r = 2$ this is $\mathbb{E}_0[t_8^2] = 2\operatorname{Tr}(M^2)$, of order a^{-12} in fixed volume. First centering gives $\mathbb{E}_0[t_8] = 0$; it plainly does not center this second source coefficient.

A genuine factor a^p on each current changes the scalar power to

$$a^{pr-4r-4}. \tag{PC10.37}$$

Even the genuine own-word factor $p = 4$ leaves a fine-volume mass of order a^{-4} at every fixed $r \geq 2$. Its component count is an extensive scalar contact, not a new fine-site factor to insert into the coarse bound PC10.32. The two quantities describe different rows of the exact same finite-source generating function.

Dividing by the original source-zero denominator does not cancel $v(\eta)$, because $v(0) = 0$. Dividing by the entire sourceful partition function would remove this scalar mass, but its logarithm would still have to be retained when reconstructing the original physical source correlations. The actual construction cannot make it disappear by an undocumented normalization change.

143.7 The original exponential and the Ward-flow completion differ

For a symmetric centered quadratic insertion with matrix T, the linear vector field $V(A) = \Sigma T A$ has that insertion as its weighted divergence. The Ward flow in PC10.5 has coordinate map $F_W = e^{\Sigma T}$ and covariance

$$\Sigma_W = \Sigma^{1/2} e^{2S} \Sigma^{1/2}. \tag{PC10.38}$$

Its logarithmic density ratio is

$$A_W(A) = -\operatorname{Tr} S - \tfrac{1}{2}\left\langle A, \Sigma^{-1/2}(e^{-2S} - I)\Sigma^{-1/2} A \right\rangle. \tag{PC10.39}$$

Its partition mass is exactly one. Its first source coefficient is the original centered current, but its higher source coefficients are fixed nonzero quadratic contacts.

For the repeated D8 current,

source family	fine covariance	log scalar mass
$e^{\tau t_8}$	$\Sigma(I - 2\tau M)^{-1}$	$\displaystyle\sum_{r\geq 2} \frac{2^{r-1}}{r} \tau^r \operatorname{Tr} M^r$
$e^{\tau D_{MA}} \mu_0/\mu_0$	$e^{2\tau M}\Sigma$	0

(PC10.40)

They agree at first order. Their covariance coefficients differ already at second order: $4\Sigma M^2$ in the original exponential, and $2\Sigma M^2$ in the Ward flow. Calling both expressions a completed current would conceal this distinction.

143.7.1 The D8 coefficient actually generated by a D6 Ward flow

There is a concrete second-order contact cancellation with no added mesh factor. Consider the Gaussian pure total-current insertion

$$q_6(A) = g_j^{-2}\langle A, H^2 A\rangle - g_j^{-2}\beta^{-1} \operatorname{Tr} H, \qquad M_6 = g_j^{-2}\beta^{-1} H. \tag{PC10.41}$$

It is the weighted divergence of $A \mapsto M_6 A$. Its exact Ward-flow source logarithm expands as

$$\begin{aligned} A_{W,6}(\tau, A) &= \tau q_6(A) - \beta\tau^2 \langle A, H M_6^2 A\rangle + O_{\mathcal{A}}(\tau^3) \\ &= \tau q_6(A) - \tfrac{1}{2}\tau^2 t_8(A) - \tau^2 \operatorname{Tr}(M_6^2) + O_{\mathcal{A}}(\tau^3). \end{aligned} \tag{PC10.42}$$

The equality uses exactly the physical t_8 of PC10.23. For $\tau = \eta_1 + \eta_2$, the second logarithmic coefficient is $-t_8 - 2\operatorname{Tr}(M_6^2)$. It is precisely the negative of the later contact created by the first current:

$$(M_6 A) \cdot D q_6 = t_8 + 2\operatorname{Tr}(M_6^2). \tag{PC10.43}$$

Thus a legitimate exact Ward completion subtracts precisely this D8 product contact together with its scalar companion. The coefficient is derived, not selected to improve an estimate. It also identifies exactly why simply deleting the D8 current is incorrect.

The remaining second-order density coefficient is itself a higher Ward current, rather than zero against every fixed later test. In particular, with the observation held fixed, the completed D6 flow still has

$$[\tau^2](C_{W,6} - C_0) = 2Q\Sigma M_6^2 Q^*, \tag{PC10.43a}$$

whereas the original exponential has $4Q\Sigma M_6^2 Q^*$. This remaining matrix still contains the sharp QHQ^* boundary form. Full cancellation against arbitrary later observables requires the simultaneous test and

kernel transport in PC10.5–8, or the explicit compensator PC10.18. The source-only Ward completion is not by itself a bound for the original fixed-constraint construction.

However, PC10.42 is not the original exponential $e^{\tau q_6}$. That exponential has its own nonzero second connected coefficient. Nor is the physical MT1 dimension-six family entirely the pure total-current direction PC10.41. Its nonredundant complementary directional-current and cubic sectors, and their contacts with this direction, remain present. An identification of the chosen physical source projection with PC10.42, including every later source and kernel response, would be needed to use this cancellation in MT36.

CP42 already provides a finite bare-word elimination of undifferentiated currents, and XP22–24 already provides its ordered product contacts. PC10.42 supplies the explicit Gaussian coefficient of that cancellation; it does not claim that the earlier formal algebra was absent. The new all-source determinant and covariance formulas expose the higher scalar and boundary rows that the quantitative induction must then control or explicitly renormalize.

143.7.2 The exact nonlinear Wilson D6 contact coefficient

PC10.6b also computes a corresponding contact on the original nonlinear Wilson measure, without a Gaussian approximation. Use the original definitions $J_i = a^{-3}X_iS_0$, $\kappa = (\beta a^3)^{-1}$, and field-independent source values $f_i = f_b$, $h_i = h_b$, independent of the color index. Set

$$\begin{aligned} P_f &= a^4 \sum_i f_i J_i^2, \\ B_f &= \kappa a^4 \sum_i f_i X_i J_i = \beta^{-1} a^{-2} \sum_i f_i X_i^2 S_0, \\ t_f &= g_j^{-2} \mathscr{E}[fJ] = g_j^{-2}(P_f - B_f), \\ V_f &= g_j^{-2} \kappa a^4 \sum_i f_i J_i X_i. \end{aligned} \tag{PC10.44}$$

Then $D_{V_f}\rho = t_f\rho$ exactly. The complete scalar companion B_f, including its constant vacuum part, remains in t_f. For the contact $C_{f,h}$ already defined in CP4, direct differentiation gives

$$\begin{aligned} V_f t_h &= g_j^{-4}\{C_{f,h} - L_{f,h}\}, \\ L_{f,h} &= \kappa a^4 \sum_i f_i J_i X_i B_h \\ &= \beta^{-2} a^{-4} \sum_{i,j} f_i h_j (X_i S_0)(X_i X_j^2 S_0). \end{aligned} \tag{PC10.45}$$

The third derivative here is ordered, and the source values remain at the original bonds. In particular, no test is replaced by its value at a common coarse center.

For the two-source derivation $\mathcal{V} = \varepsilon_f V_f + \varepsilon_h V_h$, PC10.6c gives the exact nonlinear mixed logarithmic coefficient

$$\boxed{[\varepsilon_f\varepsilon_h]A_W = -\frac{g_j^{-4}}{2}\{C_{f,h} + C_{h,f} - L_{f,h} - L_{h,f}\}.} \tag{PC10.46}$$

This is a derived same-Wilson contact completion. It subtracts the actual D8 contacts and retains the derivative of each original scalar companion. The associated mixed density coefficient is $t_f t_h - \frac{1}{2}(V_f t_h + V_h t_f)$, a higher complete Ward current. Its response to a fixed later kernel remains the one in PC10.8. The original exponential $e^{\varepsilon_f t_f + \varepsilon_h t_h}$ does not already contain the logarithmic term PC10.46.

The scalar-companion response admits a further exact reduction and a real-field bound. Write $s_p = N - \operatorname{Re}\operatorname{tr} P_p$ and $h_p = \sum_{b\in\partial p} h_b$, using the original four incidences of each plaquette. The color-summed link Casimir gives

$$\begin{aligned}\sum_j h_j X_j^2 S_0 &= C_F \sum_p h_p \operatorname{Re}\operatorname{tr} P_p,\\ B_h &= NC_F\beta^{-1}a^{-2}\sum_p h_p - C_F\beta^{-1}a^{-2}\sum_p h_p s_p,\\ X_i B_h &= -C_F\beta^{-1}a^{-2}\sum_p h_p X_i s_p.\end{aligned} \tag{PC10.47}$$

Thus the constant term disappears only after differentiating B_h; it is not dropped from the original insertion. Substituting PC10.47 into PC10.45 gives the actual finite-incidence expression

$$L_{f,h} = -C_F\beta^{-2}a^{-4}\sum_i f_i(X_i S_0)\sum_{p\ni b_i} h_p X_i s_p. \tag{PC10.48}$$

For $h = 1$, this yields the useful exact check

$$L_{f,1} = -4C_F\beta^{-2}a^{-2}P_f. \tag{PC10.48a}$$

It is generally nonzero. The flat Gaussian approximation made the corresponding scalar companion constant and therefore lost this nonlinear response; the full Wilson formula retains it.

For real compact link fields, the generator normalization and $\|P_p - I\|_{\mathrm{HS}}^2 = 2s_p$ imply

$$\sum_A |X_b^A s_p|^2 \le C s_p \quad (b \in \partial p). \tag{PC10.49}$$

Indeed the derivative is a trace pairing of a unitary conjugate of the generator with $P_p - I$. Cauchy–Schwarz and the finite color sum give the stated bound. This argument uses the first derivative of a plaquette action. No unsupported bound on a generic third derivative is being invoked in PC10.48.

Let Y^+ be the literal union of plaquette stars used by the nonzero terms of PC10.48, and retain its original energy $E_Y = \beta\sum_{p\in Y^+} s_p$. Since $X_i S_0 = \sum_{q\ni b_i} X_i s_q$, each bond has its original finite plaquette incidence, and $|h_p| \le 4\|h\|_\infty$, PC10.49 gives

$$\begin{aligned}|L_{f,h}| &\le C\beta^{-3}a^{-4}\|f\|_\infty\|h\|_\infty E_Y,\\ g_j^{-4}a^4|L_{f,h}| &\le Cc_\beta^{-3}g_j^2\|f\|_\infty\|h\|_\infty E_Y, \qquad \beta = c_\beta g_j^{-2},\\ g_j^{-4}|L_{f,h}| &\le Cc_\beta^{-3}g_j^2 a^{-4}\|f\|_\infty\|h\|_\infty E_Y.\end{aligned} \tag{PC10.50}$$

All plaquette and color multiplicities are retained in the finite incidence bound; the same original energy occurs once. The last line is the actual physical estimate if the source has no additional a^4. The middle line is only its explicitly own-word-normalized form. Neither line assigns a fresh reserve or restores energy that an earlier reference insertion has canceled.

This proves a bare nonlinear same-measure contact formula and a specific coupling-small scalar-companion response on its original real carrier. It is not a complex-wall estimate: the unitarity and positive plaquette energy used in PC10.49 require real compact fields. Previously generated effective coefficients also have additional score and source derivatives. Their inherited-density, selector, constraint and live-reference responses are still required before PC10.46 can be used in an all-history marked induction.

144.1 Conditioning a complete current pair retains a covariance packet

Fix one actual common realization with a positive finite reference measure μ. All original constraints, prescribed data and scalar normalizations are included in this measure. Write q for one identified observation, including any joint live-reference variables which the current directions actually vary. Suppose the full chain rule for the current words has already been performed. Its ordinary fine scalar and coarse-word coefficient functions are t and c_w, and

$$\overline{\tau} = q_*(t\mu), \qquad \overline{\nu}_w = q_*(c_w\mu), \qquad \overline{\tau} = \sum_w \mathscr{D}_w \overline{\nu}_w. \tag{PF10.1}$$

The coefficient list includes every lower word generated by the actual nonlinear observation. It does not discard derivatives of constraints or of live reference variables. For a signed coefficient measure, the notation means its Radon–Nikodym density with respect to the stated positive majorant; unrelated singular measures cannot be inserted without first constructing a common disintegration.

Disintegrate μ as $\rho(dy)\mu_y(dx)$, where $\rho = q_*\mu$ and μ_y is a conditional probability. The total mass of μ remains in ρ; it has not been normalized away. Define

$$\bar{t}(y) = \mu_y(t), \quad \bar{c}_w(y) = \mu_y(c_w), \quad \bar{h}(y) = \mu_y(h) \tag{PF10.2}$$

for a bounded multiplier h on the original fine realization. Let

$$\begin{aligned} k_t(y) &= \mu_y[(h-\bar{h})(t-\bar{t})], \\ k_w(y) &= \mu_y[(h-\bar{h})(c_w-\bar{c}_w)]. \end{aligned} \tag{PF10.3}$$

The same scalar or tensor pairings used by the actual coefficients are retained here; no Hermitian conjugation is inserted into a holomorphic coefficient.

For the fixed observation and fixed differential realization, the pushed density commutator satisfies the exact identity

$$\boxed{q_*\mathcal{C}_h(\mathcal{P}) = \mathcal{C}_{\bar{h}}(\overline{\mathcal{P}}) + k_t\rho - \sum_w \mathscr{D}_w(k_w\rho).} \tag{PF10.4}$$

Indeed $q_*(ht\mu) = (\bar{h}\bar{t} + k_t)\rho$, and the corresponding identity holds for every c_w. Substitution into the same coarse-word chain rule proves PF10.4. This argument takes place before a division by a conditional mass and before any estimate.

If $h = \widetilde{h} \circ q$, every covariance in PF10.3 vanishes. Only in that case, or after an actual estimate of these covariances, may the fine density change be replaced by its retained density change for this purpose. In general the scalar covariance remains an ordinary marked scalar atom. It is not a normalization current with zero integrated mass.

The represented norm of the additional packet has the direct bound

$$\mathfrak{M}\left(k_t\rho - \sum_w \mathscr{D}_w(k_w\rho)\right) \le \int \left(|k_t| + \sum_w |k_w|\right) d\rho. \tag{PF10.5}$$

For $\|h-1\|_\infty \le \varepsilon$, one may bound its right side by

$$2\varepsilon \int \left(|t - \bar{t} \circ q| + \sum_w |c_w - \bar{c}_w \circ q| \right) d\mu. \tag{PF10.6}$$

Thus a small multiplier preserves any already proved fine conditional oscillation bound. It does not replace that bound by the smaller mass of the conditioned coefficient. The original fine fluctuations are still visible in PF10.6.

The identity does not compare two different constraint fibers or two different observation maps. If the density change also changes the prepared readout, the actual readout and reference-response defects must first be retained. Likewise, dividing a family by a mass that depends on a differentiated retained variable produces the original normalization commutators. PF10.4 was deliberately stated before that division.

144.2 The fine Gaussian scalar mass diverges despite coarse cancellation

Use exactly the flat Wilson reference of GG9 on a fixed physical four-torus. Remove or separately fix its harmonic modes and use its positive physical transverse row space. Write its Maxwell Hessian as H_a, with eigenvalues $\lambda_k > 0$, and

$$\mu_a = N(0, \beta^{-1} H_a^{-1}), \qquad \beta = c_\beta g_j^{-2}. \tag{PF10.7}$$

The ordinary scalar companion of the bare Gaussian D8 current, with its genuine own-word factor, is

$$T_a(A) = g_j^{-4} a^4 \left[2\beta^{-1} \langle A, H_a^3 A \rangle_a - 2\beta^{-2} \operatorname{Tr} H_a^2 \right]. \tag{PF10.8}$$

In a physical orthonormal mode basis, $A = \beta^{-1/2} H_a^{-1/2} Z$, with independent standard real Gaussian coordinates. Consequently

$$T_a = 2c_\beta^{-2} a^4 \sum_k \lambda_k^2 (Z_k^2 - 1), \qquad \mathbb{E} T_a = 0, \tag{PF10.9}$$

and

$$\sigma_a^2 := \mathbb{E} T_a^2 = 8 c_\beta^{-4} a^8 \operatorname{Tr} H_a^4. \tag{PF10.10}$$

On a fixed four-torus the number of modes is $O(a^{-4})$, every eigenvalue is at most Ca^{-2}, and a fixed positive fraction of transverse Fourier modes has eigenvalue at least ca^{-2}. This follows directly from the lattice symbol $4a^{-2} \sum_\nu \sin^2(k_\nu a/2)$ on the transverse polarizations. Hence

$$\operatorname{Tr} H_a^4 \asymp a^{-12}, \qquad \sigma_a \asymp a^{-2}. \tag{PF10.11}$$

Constants retain the fixed physical volume and color count.

For any real weights w_k, the centered Gaussian quadratic $X = \sum_k w_k (Z_k^2 - 1)$ obeys

$$\mathbb{E} X^2 = 2 \sum_k w_k^2, \qquad \mathbb{E} X^4 = 12 \left(\sum_k w_k^2 \right)^2 + 48 \sum_k w_k^4 \le 15 (\mathbb{E} X^2)^2. \tag{PF10.12}$$

Hölder gives $\mathbb{E}|X| \ge (\mathbb{E} X^2)^{3/2} / (\mathbb{E} X^4)^{1/2}$. Together with Cauchy–Schwarz and PF10.10–12, this proves

$$\boxed{ca^{-2} \le \|T_a \mu_a\|_{\mathrm{TV}} \le Ca^{-2}.} \tag{PF10.13}$$

This is an ordinary scalar total-variation statement on the original fine Gaussian realization. It is compatible with GG9's much smaller Ca^3 own-word scalar and vector after conditioning onto a fixed retained lattice: that conditional computation uses exact signed quadratic and vacuum-trace cancellations. Total variation before conditioning cannot be replaced by total variation afterwards.

Without the word's a^4, the same calculation gives a scalar mass of order a^{-6}, retaining the corresponding coupling normalization. No missing source factor is added here.

144.3 An outer-small analytic density family can expose the fine scalar mass

The following family shows why the original R small-density comparison, by itself, cannot prove the needed current-pair estimate. It does not claim that this is an original R interaction.

For $0 < \varepsilon < 1/2$, let

$$h_z(A) = 1 + z\varepsilon \tanh(T_a(A)/\sigma_a), \qquad |z| \le 1. \tag{PF10.14}$$

The family is entire in the external parameter z, $h_0 = 1$, and on the real integrated Gaussian variables

$$\sup_{|z|\le 1} |h_z - 1| \le \varepsilon, \qquad |\log h_z| \le 2\varepsilon. \tag{PF10.15}$$

The real endpoint h_1 is positive. There is no assertion of a uniformly large holomorphic domain in every internal Gaussian coordinate; such a hypothesis would be additional information.

Since $\mathbb{E}T_a = 0$,

$$\mathbb{E}(h_1 T_a) = \varepsilon \mathbb{E}[T_a \tanh(T_a/\sigma_a)]. \tag{PF10.16}$$

Apply Paley–Zygmund to T_a^2. PF10.12 gives $\mathbb{P}(|T_a| \ge \sigma_a/\sqrt{2}) \ge 1/60$. The integrand in PF10.16 is nonnegative, so

$$\mathbb{E}(h_1 T_a) \ge \frac{\varepsilon \sigma_a}{60\sqrt{2}} \tanh(1/\sqrt{2}) \ge c\varepsilon a^{-2}. \tag{PF10.17}$$

The mass $\mathbb{E}h_1$ lies between $1-\varepsilon$ and $1+\varepsilon$. Thus the normalized perturbed expectation has the same lower order. Take q to be the constant observation: the original scalar pushforward is zero, while the perturbed scalar pushforward is PF10.17. This is precisely the conditional covariance term in PF10.4.

The example retains the complete scalar contribution. It does not contradict CT9: the right side of CT9.6 contains the fine scalar mass PF10.13. It rules out a stronger conclusion that would use only the small outer density factor and the already canceled coarse pair. Actual R interactions may have additional cancellations or derivative structure. Those properties have to be proved for their literal coefficients and supports.

144.4 What a scalar-recovery argument must actually estimate

There is a direct way to recover scalar mass from current coefficients on a smooth retained body, but its hypotheses are substantive. In a fixed flat coordinate realization let $\rho(db) = Ze^{-S(b)}db$, retaining its absolute scalar Z, and suppose the current coefficient measures are $v_w(b)\rho(db)$. There is no boundary contribution in this paragraph: use a full integration space with the required decay, coefficients compactly supported in the chart, or an explicit vanishing condition for every intermediate adjoint flux through order $r-1$. Vanishing of the initial coefficient on the boundary alone does not remove the later boundary terms of a higher current word.

For $L_i v = (\partial_i S)v - \partial_i v$, integration by parts gives

$$\mathscr{D}_{(i_1,\dots,i_r)}(v\rho) = (L_{i_r} \cdots L_{i_1} v)\rho. \tag{PF10.18}$$

The reversed operator order follows from the declared ordered-word convention. In particular, for a first-current representation,

$$\left\|\sum_i \mathscr{D}_i(v_i\rho)\right\|_{\mathrm{TV}} \le \int \sum_i (|\partial_i v_i| + |v_i \partial_i S|)\, d\rho. \tag{PF10.19}$$

The analogous second-order bound retains the coefficient derivatives, two score factors and score derivatives produced by PF10.18. A fixed-order analytic Cauchy estimate for the same coefficient on a common complex retained domain controls these derivatives with its literal radius. A physical radius R and canonical direction $g_e\partial$ give g_e/R, not a new arbitrary inverse coupling. Each actual coordinate incidence is retained.

This proves a recovery implication, not the required model-specific derivative bounds. CD9 gives a real centered coefficient estimate; it does not by itself give the complex conditional coefficient derivatives in PF10.19. The full inherited score is the score of the same retained density, including genuine denominators and any coordinate Jacobian. It is not replaced by the bare Wilson score.

If an original selector is present, its full boundary packet must be kept before using PF10.18. A sharp selector is not handled by taking its logarithm or by inventing a smooth replacement. The original failure-sector estimate and its one allocated reserve remain the required quantitative input.

144.5 The exact Gaussian derivative cancellation in the covariance defect

The covariance packet admits a more useful expression when the conditional law is the actual Gaussian reference and the multiplier has controlled fine derivatives. This identifies a possible additional estimate; it is stronger information than an outer density bound.

Keep the positive physical Gaussian $\mu = N(0,\Sigma)$ and the complete linear observation Q, which can be a joint stack of retained rows. For a compatible value y, its conditional mean and covariance are

$$\begin{aligned} C_Q &= Q\Sigma Q^*, \\ m_y &= \Sigma Q^* C_Q^\dagger y, \\ C &= \Sigma - \Sigma Q^* C_Q^\dagger Q\Sigma. \end{aligned} \tag{PF10.20}$$

The dagger is the Moore–Penrose inverse on the actual range. It does not delete a constraint row or declare dependent observations independent. The law of y remains supported on its original compatibility subspace, $QC = 0$, and $Qm_y = y$ there. No uniform bound on this inverse is asserted or required for the finite identity below.

Let

$$t(A) = \langle A, TA\rangle - \mathrm{Tr}(T\Sigma), \qquad T = T^*. \tag{PF10.21}$$

The matrix T contains its literal coupling, source-offset and mesh coefficients. Suppose h has two integrable Gaussian derivatives in the directions of the conditional fiber, with enough decay to justify the following integrations by parts. Writing $A = m_y + \xi$, $\xi \sim N(0,C)$, gives exactly

$$\boxed{k_t(y) = 2\langle Tm_y, C\,\mathbb{E}_y \nabla h\rangle + \mathrm{Tr}\left(CTC\,\mathbb{E}_y \nabla^2 h\right).} \tag{PF10.22}$$

To prove it, subtract the conditional mean from PF10.21:

$$t - \bar{t} = 2\langle Tm_y, \xi\rangle + \langle \xi, T\xi\rangle - \mathrm{Tr}(TC).$$

For a possibly singular conditional Gaussian the exact identities are

$$\mathbb{E}_y[\xi_i h] = \sum_k C_{ik}\mathbb{E}_y[\partial_k h], \qquad \mathbb{E}_y[(\xi_i\xi_j - C_{ij})h] = \sum_{k,l} C_{ik}C_{jl}\mathbb{E}_y[\partial_k\partial_l h]. \tag{PF10.23}$$

They follow by writing $\xi = C^{1/2}Z$ on its positive row space and integrating standard Gaussian coordinates. Contraction with the linear and quadratic coefficients proves PF10.22. Only conditional tangent derivatives occur, so the result is independent of how h is smoothly extended off that affine fiber. For complex coefficients, the contractions are extended bilinearly; no source conjugation is introduced.

For example, with Euclidean and Hilbert–Schmidt norms on the actual physical coordinates,

$$\begin{aligned} |k_t(y)| \le{}& 2\|C^{1/2}Tm_y\|\,\mathbb{E}_y\|C^{1/2}\nabla h\| \\ &+ \|C^{1/2}TC^{1/2}\|_{\mathrm{HS}}\,\mathbb{E}_y\|C^{1/2}(\nabla^2 h)C^{1/2}\|_{\mathrm{HS}}. \end{aligned} \tag{PF10.24}$$

Integrating this inequality against the original retained mass ρ bounds the ordinary scalar covariance in PF10.5. It neither normalizes that mass nor replaces C by the unconditional covariance. If $h = \widetilde{h}(QA)$, both terms vanish because $CQ^* = 0$, recovering the exact cancellation in Section 1.

For the constant observation, $m = 0$ and $C = \Sigma$. Take the actual D8 coefficient

$$T = 2g_j^{-4}\beta^{-1}w_a H_a^3, \qquad w_a = \begin{cases} 1 & \text{for the physical packet,} \\ a^4 & \text{for the separately recorded own-word packet.} \end{cases}$$

Then PF10.22 reduces to

$$\mathrm{Cov}_\mu(h,t) = 2g_j^{-4}\beta^{-3}w_a\,\mathbb{E}_\mu[H_a : \nabla^2 h]. \tag{PF10.25}$$

This is the precise Gaussian cancellation that a fine derivative estimate can exploit. It does not assign a new w_a, suppress the trace's coordinate count, or prove that the right side is small. The tilt in Section 3 has large enough fine derivatives to be consistent with this identity and its divergent scalar covariance.

For an original sharp selector, the differentiated multiplier in PF10.23 includes its distributional boundary derivatives. Using only the derivative of the interior smooth density loses those partners. For a nonlinear prepared observation, the conditional law is generally not Gaussian; PF10.22 then cannot be substituted for its original joint integration rule. Finally, later differentiation in retained variables also differentiates the conditional mean, the covariance when it varies, and any live reference. This finite identity does not remove those response terms.

Thus a viable derivative route must bound the literal contracted fine jets of the inherited multiplier and its original selector partners on the same support. The strong real or complex coefficient bounds for one terminal current do not by themselves establish that property.

144.6 Consequence for the remaining induction

PF10.4 identifies the missing conditional covariance packet exactly. PF10.13–17 show that the scalar part can be large even when the unweighted outer density comparison is uniformly small. PF10.18–19 specify the actual coefficient and effective-score derivatives needed for an alternative recovery argument. PF10.22–25 exhibit the exact fine derivative cancellation in the Gaussian conditional covariance, including joint constraint relations. All original masses remain visible.

The additional Gaussian contact cancellations developed in the physical-current chapter may remove such terms for their explicitly completed Gaussian source family. They do not establish that the original

nonlinear marked induction already contains those counterterms or their mixed-domain estimates. The actual inherited pair, its selector boundary partners and the required simultaneous complex domains therefore remain open model-specific obligations.

145.2 The complete native insertion can be summed exactly

145.2.1 Keep the original old state before choosing reference coordinates

Fix the full retained datum b. Every old variable on which a reference depends belongs to b, even if its integration occurs later. Write the remaining old numerator as a positive finite measure

$$d\rho_b(\xi), \qquad Q(b) = \int d\rho_b(\xi). \tag{NR10.1}$$

This is the original physical numerator before choosing an auxiliary reference chart. In particular, it includes every old characteristic function, constraint measure, action factor, and independent scalar

which belongs inside that numerator. A separately retained factor $e^{s(b)}$ will be carried explicitly below. It is not absorbed into a probability normalization.

Let G label one of the original allowed preparatory choices. The corresponding output reference fiber is

$$D_G(b) = \int \chi_G(C;b) r_G(C;b)\, d\lambda_{G,b}(C), \qquad K_G(dC \mid b) = \frac{\chi_G(C;b) r_G(C;b)}{D_G(b)}\, d\lambda_{G,b}(C). \tag{NR10.2}$$

Here D_G is the genuine original denominator. The measure $d\lambda_{G,b}$ is the actual relative Haar/tree fiber. Neither is replaced by an unrestricted product measure or a completed-component normalizer. We work on the real admitted support, where $0 < D_G < \infty$. The original source-zero construction supplies that admitted insertion; the following calculation does not prove its analytic-domain theorem.

Let α denote all output small/large labels of this insertion, with their original selectors

$$\pi_{G,\alpha}(C,\xi,b) \geq 0, \qquad \sum_{\alpha} \pi_{G,\alpha}(C,\xi,b) = 1. \tag{NR10.3}$$

The source-zero case of S58b gives its fully integrated selected sum as

$$\mathcal{I}_G(b) = \sum_{\alpha} \int K_G(dC \mid b) \int \pi_{G,\alpha}(C,\xi,b)\, d\rho_b(\xi). \tag{NR10.4}$$

Any original coordinate changes in the old numerator are undone when writing $d\rho_b$ in NR10.1. Their constraints and Jacobians are retained; the unit Haar/tree Jacobian is precisely S54. In a completed-action representation, the compensation factors must first be multiplied back together to obtain NR10.4. Individual completed factors need not be independent of C or of G.

145.2.2 Exact complete-sum identity

For every admitted native choice G,

$$\boxed{\mathcal{I}_G(b) = Q(b).} \tag{NR10.5}$$

Indeed, positivity allows Tonelli. Sum NR10.3 while keeping the original denominator in place. The integrand becomes

$$\frac{\chi_G(C;b) r_G(C;b)}{D_G(b)}\, d\lambda_{G,b}(C)\, d\rho_b(\xi).$$

The integral in C is then exactly one by NR10.2. This proves NR10.5 without a lower bound on D_G, a comparison between two denominators, or a claim that one selected label has unit mass.

Let $w_G(b)$ be the actual normalized finite-choice weights, with $\sum_G w_G(b) = 1$. Then

$$e^{s(b)} \sum_{G} w_G(b) \mathcal{I}_G(b) = e^{s(b)} Q(b). \tag{NR10.6}$$

Thus the absolute scalar $s(b)$ survives unchanged. When $Q(b) > 0$, the logarithm of either side is $s(b) + \log Q(b)$. There is no additional $\log \#\{G\}$, no gauge-volume term, and no loss proportional to

the number of components. The actual finite-choice weights have been summed, rather than bounded by their number.

This proves choice independence for the **complete integrated reference insertion**. It also applies to a finite sequence of such auxiliary insertions if the whole sequence is integrated and every selector label is summed, with all older variables kept in their original slots. Integrate the last new fiber first and iterate NR10.5. This is an exact finite identity and spends none of the shared Wilson reserve.

The calculation is useful when a normalization subdiagram is closed: all new variables introduced by it are integrated within it, all its selector alternatives are summed there, and its outside carrying function depends only on its retained old datum. It does not establish that the proposed local pieces of the complete inherited vacuum have that property.

145.2.3 Why an inherited carrying factor prevents the simplification

If a factor $\Psi_G(C,\xi,b)$ is still carried by a later operation, the same calculation gives instead

$$\sum_{\alpha}\int K_G(dC\mid b)\int \pi_{G,\alpha}\Psi_G\,d\rho_b=\int d\rho_b(\xi)\,K_G\big(\Psi_G(\cdot,\xi,b)\big). \tag{NR10.7}$$

There is no replacement of the right-hand side by $Q(b)$ unless the complete carrying function has the required normalization or is independent of the new variable. Even boundedness or gauge invariance of Ψ_G does not imply that property. Its dependence on the genuine reference kernel remains.

Similarly, restricting NR10.4 to healed labels gives

$$\mathcal{I}_G^{\mathrm{heal}}(b)=\int K_G(dC\mid b)\int \pi_G^{\mathrm{heal}}(C,\xi,b)\,d\rho_b(\xi),\qquad 0\le \pi_G^{\mathrm{heal}}\le 1. \tag{NR10.8}$$

The selector sum here is generally not one. Thus a healed branch is not a closed normalization subdiagram solely because one relative output cutoff has become identically one on its support.

This distinction is particularly relevant to S94. A component's connected carrying function can include an old activity whose full read set crosses that component. Treating that activity as a constant before the new fiber is integrated replaces NR10.7 by a different expression. The exact complete-sum identity supplies no estimate for that error.

145.4 An exact SU(N) test of forest-dependent cutoff denominators

145.4.1 One graph, one Wilson action, two legitimate rooted trees

Use two adjacent unit square plaquettes in a coordinate plane of a four-dimensional lattice. Label their six vertices

$$A = (0,0), \quad B = (1,0), \quad C = (2,0), \quad D = (0,1), \quad E = (1,1), \quad F = (2,1).$$

The seven oriented links are

Link	Endpoints
a	$A \to B$
b	$B \to C$
c	$D \to E$
d	$E \to F$
e	$A \to D$
f	$B \to E$
g	$C \to F$

The plaquette products are

$$P_1 = afc^{-1}e^{-1}, \qquad P_2 = bgd^{-1}f^{-1}. \tag{NR10.11}$$

Pin the gauge at A, use the identity reference on every link, and consider the two rooted spanning trees

$$T_1 = \{a, b, d, e, f\}, \qquad T_2 = \{a, b, c, d, e\}. \tag{NR10.12}$$

Both trees have five edges and fix the same five nonroot gauge variables. Both leave two independent group coordinates. For T_1, put $X = c^{-1}$, $Y = g$; then $(P_1, P_2) = (X, Y)$. For T_2, put $X = f$, $Y = g$; then $(P_1, P_2) = (X, YX^{-1})$. All coordinate measures are the same product of two normalized Haar measures.

Let the group be SU(N), $N \geq 2$, and put

$$t(U) = \frac{1}{N}\operatorname{Re}\operatorname{Tr} U, \qquad A(P_1, P_2) = 2 - t(P_1) - t(P_2). \tag{NR10.13}$$

This is the ordinary two-plaquette Wilson action, with a harmless stated normalization. Choose a conjugation-invariant, inversion-invariant measurable neighborhood $\mathcal{B}$ of the identity such that

$$v = \mu(\mathcal{B}) > 0, \qquad \alpha = \frac{1}{v}\int_{\mathcal{B}} t(U)\, d\mu(U) \in (0,1). \tag{NR10.14}$$

A sufficiently small group ball with positive radius satisfies these conditions. Apply the same relative-link cutoff to every free link. Tree links are already the identity, so the two reference denominators are exactly

$$\begin{aligned} D_1(\beta) &= \int_{\mathcal{B}^2} e^{-\beta[2-t(X)-t(Y)]}\, dX\, dY, \\ D_2(\beta) &= \int_{\mathcal{B}^2} e^{-\beta[2-t(X)-t(YX^{-1})]}\, dX\, dY. \end{aligned} \tag{NR10.15}$$

The action, group, reference, root, number of variables, Haar convention, and scalar coupling are identical. What changes is the physical domain represented by a box of small chord variables.

145.4.2 The denominators are not identically equal

Conjugation invariance and irreducibility of the defining SU(N) representation imply

$$\int_{\mathcal{B}} U\, d\mu(U) = v\alpha I. \tag{NR10.16}$$

To see this directly, the matrix integral commutes with every SU(N) matrix, hence is scalar. Inversion invariance makes that scalar real; its trace fixes it to $v\alpha$. Consequently, for independent uniform variables in $\mathcal{B}$,

$$\mathbb{E}[t(X)] = \mathbb{E}[t(Y)] = \alpha, \qquad \mathbb{E}[t(YX^{-1})] = \alpha^2. \tag{NR10.17}$$

Both denominators equal v^2 at $\beta = 0$. Differentiation of their finite bounded integrals gives

$$\frac{d}{d\beta}\log D_1\Big|_0 = -2 + 2\alpha, \qquad \frac{d}{d\beta}\log D_2\Big|_0 = -2 + \alpha + \alpha^2.$$

Therefore

$$\boxed{\log\frac{D_1(\beta)}{D_2(\beta)} = \beta\alpha(1-\alpha) + O(\beta^2).} \tag{NR10.18}$$

In particular, the denominators differ for every sufficiently small positive β. Their difference is an absolute scalar difference; it is not a missing root count, a Haar Jacobian, or a gauge-group volume.

This calculation does not assert that the displayed trees and cutoff are the particular choices of an original Balaban component. It proves that the proposed inference from only those common structural data is false, even for the same finite nonabelian Wilson action.

If the cutoff is removed, the substitution $(X,Y)\mapsto(X,YX^{-1})$ preserves the full product Haar measure and does identify the two integrals. With the cutoff present, it sends the domain to

$$\{(X,Z): X\in\mathcal{B},\ ZX\in\mathcal{B}\},\tag{NR10.19}$$

which is not generally $\mathcal{B}^2$. This is exactly the support information that a native-record comparison must retain.

145.4.3 The discrepancy can also survive a weak-coupling scaling

The preceding derivative test uses small β. A second calculation shows that the issue is not restricted to that regime. Let $m=N^2-1$, use traceless Hermitian Lie-algebra coordinates with norm $\|X\|^2=N^{-1}\operatorname{Tr}X^2$, and fix $R>0$. For small $g>0$, put

$$\mathcal{B}_g=\{e^{igX}:\|X\|<R\},\qquad \beta=g^{-2}.\tag{NR10.20}$$

On this fixed rescaled ball, the exponential map is injective for small g. Smoothness of Haar measure and the Wilson function gives, uniformly for bounded X,Y,

$$\begin{aligned} d\mu(e^{igX}) &= \kappa_N g^m(1+o(1))\,dX,\\ g^{-2}[1-t(e^{igX})] &= \tfrac12\|X\|^2+o(1),\\ g^{-2}[1-t(e^{igY}e^{-igX})] &= \tfrac12\|Y-X\|^2+o(1). \end{aligned}\tag{NR10.21}$$

The same constant $\kappa_N>0$ occurs on both trees. Let

$$\begin{aligned} A_1(R) &= \left(\int_{\|X\|<R} e^{-\|X\|^2/2}\,dX\right)^2,\\ A_2(R) &= \int_{\|X\|<R} e^{-\|X\|^2/2}\int_{\|Y\|<R} e^{-\|Y-X\|^2/2}\,dY\,dX. \end{aligned}\tag{NR10.22}$$

Uniform convergence on the bounded domain, or dominated convergence, now proves

$$D_i(g^{-2};\mathcal{B}_g)=\kappa_N^2 g^{2m}(A_i(R)+o(1)).\tag{NR10.23}$$

For every finite positive R,

$$0<A_2(R)<A_1(R).\tag{NR10.24}$$

For completeness, fix $X=te_1$, $t>0$, and slice the ball in the e_1 direction. Each nonempty slice is an interval $[-a,a]$. The one-dimensional integral

$$J_a(t)=\int_{-a}^{a} e^{-(u-t)^2/2}\,du$$

satisfies $J_a'(t)=e^{-(a+t)^2/2}-e^{-(a-t)^2/2}<0$ for $a,t>0$. The perpendicular Gaussian factor is positive, so

$$\int_{\|Y\|<R} e^{-\|Y-X\|^2/2}\,dY < \int_{\|Y\|<R} e^{-\|Y\|^2/2}\,dY \quad (X \neq 0).$$

Integrating this strict inequality against the positive outer factor proves NR10.24. Hence

$$\boxed{\log \frac{D_1(g^{-2};\mathcal{B}_g)}{D_2(g^{-2};\mathcal{B}_g)} \longrightarrow \log \frac{A_1(R)}{A_2(R)} > 0.} \tag{NR10.25}$$

The leading Haar/count term $2m \log g + 2 \log \kappa_N$ is exactly the same on both sides; the finite scalar remainder differs. The fixed rescaled radius in this test is not asserted to be the original cutoff radius. If an actual radius grows, its tail estimate must be calculated with that actual growth. NR10.25 supplies no bound on that original tail.

This explains the role of absolute scalar bookkeeping. Equality of the divergent count term, or cancellation in a normalized expectation, does not determine the finite logarithmic reference mass. A bounded unmatched scalar at each fine-scale component is not automatically acceptable in a vacuum density with a growing number of components.

145.6 The remaining original-record assertions

The denominator-copying conclusion RN9.7 is available once the complete fixed-target record has been identified. To apply it to the actual native histories, the following data still have to be checked together.

145.6.1 Fixed-target branch on the entire original fiber

The raw determining set must specify every prescribed average, including its order, endpoints, parallel transports, scale, and interface completion. The full target is the one produced by $CV_{\Lambda}(b)$, with every older datum fixed as in NR10.1. Its relationship to the hierarchy, admissible regularity class, and gauge group in the 1985 theorem must hold for every C admitted by the original χ_G, rather than only at the relaxed saddle.

The action and boundary rule defining the fixed-target problem must also coincide. The cutoff weight ζ_0 in the reference action is an evaluation weight; treating it as the action minimized by the fixed-target branch changes the variational problem. The original localized field must be identified with the selected nonlinear branch, not merely with a function having the same formal first variation.

On a common complex extension, one additionally needs the same germ, admitted simultaneous walls, and logarithm branch. A real tree substitution does not produce a complex nonvanishing theorem.

145.6.2 Native chart and cutoff matching

If the native preparatory forests differ, the actual physical change of coordinates must be written before comparing the cutoffs. The image of a linkwise cutoff can be a coupled condition like NR10.19. Its equality, or a quantitative comparison of the resulting complete integrals, requires proof. The same free-link count is insufficient.

One may instead use NR10.5 on a genuinely closed reference insertion. That route requires all its original selectors to be summed and all its new variables to be integrated before any outside carrying factor is removed. It must be checked on the actual component read set, not just on the visible support of its newest local action.

145.6.3 Inherited fields, scalar atoms, and full carriers

Every old activity and typed seed must be included in the matching record, as must its retained target variables and every later coefficient which reads the new field. An independent scalar atom is part of the absolute mass even when it has zero field derivative. The complete shared reserve and its actual component count must be used once, rather than being reassigned after a change of chart.

The required cross-torus statement has to cover full determining read sets. A visible polymer can fit into a common coordinate box while an old averaging prefix, reference halo, or inherited carrier wraps around one torus. Exact copying of a smaller displayed set does not match that larger record. The wrapped and terminal contributions still require their own uniform comparison.

These are the inputs missing before the completed R-formation comparison VC28 and its ultraviolet summation can be applied to the full source-zero vacuum. NR10.5 removes an auxiliary normalization subdiagram only when it is closed; it supplies no bound on the unmatched inherited remainder.

148.3 The actual record must be fixed before taking a norm

Fix one original history ω, its retained/integrated variables ξ, and the literal preparation

$$\Gamma_{\omega}^{R}(z,\xi). \tag{SG11.1}$$

The parameter z includes whichever outer and wall coordinates actually occur in the asserted common domain. This is the same object as AE8.1 and AE9.1. It is not an independently chosen family of constrained minima.

Let a be the original mesh of the plaquette word under study. A retained identity row means a literal equation

$$U_b = V_b, \tag{SG11.2}$$

in the recorded gauge. Prescribed exterior bonds may be included among such fixed links. If rows are initially expressed with endpoint transports, all transports and site/junction equations must first

be kept in one common recorded gauge. An equality of gauge classes is not silently substituted for SG11.2.

Let $\mathcal{P}$ be a collection of plaquettes in the proposed complete field record for which all four fine links are fixed by these rows. Count every plaquette once. Define the target holonomy

$$H_p(z,\xi) = V_{b_1}(z,\xi)^{\epsilon_1} V_{b_2}(z,\xi)^{\epsilon_2} V_{b_3}(z,\xi)^{\epsilon_3} V_{b_4}(z,\xi)^{\epsilon_4}, \qquad p \in \mathcal{P}, \tag{SG11.3}$$

where the signs, order, base point, and boundary values are those of the actual plaquette. This is a finite group word in the actual retained target array.

There are two different hypotheses.

Endpoint hypothesis. SG11.2 holds for the prepared field at its physical wall vertex. This implies the identity in Section 4 at that vertex only.

Common-domain hypothesis. SG11.2 holds for SG11.1 throughout the specific allowed common complex domain. This is a stronger statement; it is needed in Section 5. If an off-shell localization changes a constraint equation, the changed equation must be inspected rather than treated as SG11.2.

One legitimate way to prove the latter hypothesis is available. Suppose the relevant field branch, target rows, inverses, and raw constraint maps are holomorphic on a connected polydomain. If the original constraint equations hold on an open product of real parameter intervals in that domain, they hold throughout its holomorphic continuation. Indeed, fix all but the first coordinate to real values in those intervals and apply the one-variable identity theorem. Repeat coordinate by coordinate, and then continue on the connected domain. The proof applies entrywise to every matrix equation. Knowing an identity at a single real vertex is not sufficient for this argument.

148.5 Simultaneous complex-wall tests with all radii retained

Assume now the common-domain hypothesis of Section 3. Fix ξ on the original real selector support and a polydisc

$$\mathbb{D}(z^0;r)=\{z:|z_i-z_i^0|<r_i,\ 1\le i\le m\}\subset\mathcal{D}_\omega. \tag{SG11.9}$$

These are the actual available radii. There is no enlargement or renewal of them at a later age. An interior polydisc may be used if the original domain is described by more general inequalities.

For the chosen plaquettes, introduce the finite-dimensional Banach space norm

$$\|(X_p)\|_{a,p_*}=\left(\sum_{p\in\mathcal{P}}a^{4-2p_*}\|X_p\|^{p_*}\right)^{1/p_*}. \tag{SG11.10}$$

If the asserted common-domain curvature bound holds, SG11.7 gives

$$\sup_{z\in\mathbb{D}(z^0;r)}\|(H_p(z,\xi)-I)_p\|_{a,p_*}\le K^2\tau. \tag{SG11.11}$$

Expand the literal holonomy array as

$$(H_p(z,\xi)-I)_p=\sum_{\alpha\in\mathbb{N}^m}c_\alpha(\xi)(z-z^0)^\alpha. \tag{SG11.12}$$

Banach-valued Cauchy integration on any strictly smaller product of circles, followed by its radius limit, proves

$$r^\alpha\|c_\alpha(\xi)\|_{a,p_*}\le K^2\tau\quad\text{for every }\alpha. \tag{SG11.13}$$

This conclusion does not require a holomorphic choice of the gauge in SG11.6. The holonomy itself is holomorphic, and the bounded conjugation already supplies SG11.11 pointwise.

There is also a first-order condition that tests simultaneous directions. For $v\in\mathbb{C}^m$, $\|v\|_\infty\le1$, use the one-variable disc $z=z^0+t(r_1v_1,\dots,r_mv_m)$. Its Cauchy derivative bound gives

$$\left\|D_z(H_p)_p(z^0,\xi)\,\mathrm{diag}(r_i)\right\|_{\ell^\infty_m\longrightarrow(a,p_*)}\le K^2\tau. \tag{SG11.14}$$

It is essential that this is an operator norm. Bounds for its individual columns do not imply SG11.14 with the same constant. For example, if m columns in one fixed target-holonomy component are the

same nonzero vector X, the choice $v_i = 1$ gives the norm $m\|X\|$, whereas each separate column has norm $\|X\|$. If the columns occupy disjoint plaquette components, their simultaneous norm instead contains the actual ℓ^{p_*} aggregation. Neither count is suppressed.

The Jacobian in SG11.14 is explicitly computable from the literal rows. If $H = V_1 \cdots V_4$ with the orientations already absorbed into the factors, put $S_i = (DV_i)V_i^{-1}$. Direct differentiation gives

$$(DH)H^{-1} = \sum_{i=1}^{4} \mathrm{Ad}(V_1 \cdots V_{i-1})S_i. \tag{SG11.15}$$

For an inverse-oriented link, its score is $-\mathrm{Ad}(V^{-1})[(DV)V^{-1}]$. Thus all original endpoint transports, inverse factors, and dependencies of the target on an outer or wall variable occur in SG11.15. At the flat target this reduces to the oriented target curl. A proof that controls only an untransported coordinate derivative has not checked the stated norm.

SG11.13 gives necessary coefficient bounds. To obtain a sufficient supremum bound from coefficient estimates, one needs a summation estimate, for example

$$\sum_{\alpha} r^{\alpha} \|c_\alpha\|_{a,p_*} \le K^2\tau. \tag{SG11.16}$$

This implication follows immediately from the triangle inequality and absolute convergence. Separate bounds of the form SG11.13 are not enough at the original radii. Even on radii $\theta_i r_i$, their naive sum costs

$$\prod_{i=1}^{m} (1-\theta_i)^{-1}. \tag{SG11.17}$$

If the wall count is unbounded, this product is not a uniform constant. Any improved bound must use the actual locality, support, and interaction structure. The original single reserve cannot be spent again by calling SG11.17 a harmless analytic constant.

148.6 Vanishing linearized curl does not settle the nonlinear test

Take noncommuting $X, Y \in \mathfrak{su}(2)$ and consider the four-link target word

$$H(t) = e^{tX}e^{tY}e^{-tX}e^{-tY}. \tag{SG11.18}$$

This is a statement about a literal holonomy word. No claim is made that SG11.18 is generated by a particular original R selector. Multiplying the four convergent exponential series through order two gives

$$H(0) = I, \qquad H'(0) = 0, \qquad \tfrac{1}{2}H''(0) = [X,Y]. \tag{SG11.19}$$

For example,

$$X = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad [X,Y] = \begin{pmatrix} -2i & 0 \\ 0 & 2i \end{pmatrix} \neq 0. \tag{SG11.20}$$

The first-order test SG11.14 is zero on this direction at the flat target. The second coefficient test, on $|t| < r$, nevertheless forces

$$a^{4/p_*-2}r^2\|[X,Y]\| \le K^2\tau, \qquad r \le C_{X,Y}K\tau^{1/2}a^{1-2/p_*}. \tag{SG11.21}$$

The implication follows from Cauchy applied to this one plaquette; the quadratic coefficient is exact. Higher Taylor terms cannot cancel it in the Cauchy coefficient integral.

More generally, if the first nonzero coefficient of an actual one-parameter retained holonomy occurs at order q, with coefficient $B_q \neq 0$, common-domain admission forces

$$a^{4/p_*-2} r^q \|B_q\| \leq K^2 \tau. \tag{SG11.22}$$

The factor a^{-2} belongs to the original curvature word at every derivative order. The derivative does not change it to a positive mesh power. These are ordinary derivatives of the actual parameterized target. They neither reorder noncommuting field derivatives nor remove the contact and boundary terms appearing when those derivatives are converted into the original current calculus.

The example isolates a real issue for an attempted Ward repair: vanishing of the linearized curl, or a first-order constraint identity, does not by itself bound nonlinear target holonomy. An exact pure-gauge target family would have $H(t) = I$ at all orders and would pass this test. That is a stronger nonlinear identity which must be verified for the actual rows.

148.7 A literal source-record cut gives exact restoration

The preceding tests are relevant only to plaquettes actually required by the source's field record. It is equally important not to enlarge that record without justification. Here is a positive, finite-record criterion for an exact reduction.

Let $\mathcal{Q}_X(U)$ be the *complete* original readout record for one fixed old source carrier. It contains every original target row read by that carrier, every prescribed fine boundary value, the endpoint gauges and junction data, and every additional slot actually used by its original source definition. Expand each raw averaging prefix and endpoint transport as its original finite calculation. The result is a directed acyclic calculation graph: its leaves are fine links and prescribed data, and its internal vertices are the actual products, inverses, logarithmic branches, exponentials, and original one-step means.

Fix the actual retained constraint record $C_\omega(U)$. Suppose there is a cut through the graph with the following literal property: every cut vertex is either a retained row of C_ω, with the same scale, path ordering, endpoint frame and branch, or prescribed data in a record b_X. Every input path to an output vertex crosses this cut. In particular, a row with the same name but a different determining path is not an admissible replacement.

Then there is an exact restoration map, formed by evaluating the graph above the cut,

$$\mathcal{Q}_X(U) = \mathcal{R}_{X,\omega}(C_\omega(U), b_X). \tag{SG11.23}$$

To prove it, evaluate vertices in any topological order. At a cut vertex the recorded value equals its original subexpression by hypothesis. At each subsequent vertex, the operation and every input are identical to those in the original calculation, so its output is identical. Finite induction reaches all output slots. No associativity of a substituted averaging map is used. If the original operations are holomorphic on the common branch, so is the restored map on the same admitted data.

Suppose now that the actual prepared field satisfies its original constraint equation

$$C_\omega(\Gamma_\omega^R) = V_\omega \tag{SG11.24}$$

on the parameter set being used. Substitution into SG11.23 yields

$$\mathcal{Q}_X(\Gamma_\omega^R) = \mathcal{R}_{X,\omega}(V_\omega, b_X). \tag{SG11.25}$$

This identity remains true even if the prepared field depends nonlocally on ξ. It reduces the readout because of the original constraint equations, not because a visible support happens to be disjoint from a fine defect.

If $E_{j,X}$ is an already defined analytic coefficient as a function of this complete record, its actual composition is exactly

$$E_{j,X}(\mathcal{Q}_X(\Gamma_\omega^R)) = E_{j,X}(\mathcal{R}_{X,\omega}(V_\omega, b_X)). \tag{SG11.26}$$

Thus an existing bound $\|E_{j,X}\| \le A_j W_{e,j}$ is preserved on any parameter domain whose restored record is contained in the old domain on which that bound was proved. The literal birth label j, original contact-word mesh, and saved factor $W_{e,j}$ are unchanged. This step is substitution into an already bounded function; it provides no new grade and proves no estimate for an unbounded contact partner.

In particular, if a retained-variable variation leaves both the cut data V_ω and all of b_X unchanged, the right side of SG11.26 is independent of that variation. All its ordinary analytic mixed derivatives in those variable directions vanish. If a boundary slot or endpoint frame changes, the conclusion does not follow. The full source current packet must also be a function of the declared record before the same assertion is applied to that packet.

The nonlinear restoration M8.3 is one already constructed instance of this kind of calculation in an auxiliary source hierarchy. Its existence does not certify that the actual three R joins admit the required cut for every inherited source. That is the remaining geometric application of SG11.23, along with old-domain admission of the restored target record.

148.8 What these tests do not permit one to cancel or charge

The fixed-holonomy identity SG11.4 is a constraint statement. A constrained Euler–Lagrange equation vanishes on allowed tangent variations; it does not alter a holonomy whose links are prescribed. Integration by parts, a paired current identity, or a small outer density cannot change SG11.4. They may control a later integral, but cannot certify that a pointwise analytic source has been evaluated inside its old domain.

Similarly, SG11.25 is a source-record identity. It does not eliminate the dependence of an inherited density, actual cutoff, or genuine normalizing denominator on a variable that the source record no longer reads. Its differentiation in a larger integral still retains the original score, boundary and normalization partners. No such partner is set to zero by the source substitution.

All results above are deterministic and spend no large-field reserve. If a violation is to be excluded by the original selectors, that exclusion must be proved with those selectors. If a polynomial loss is to be absorbed after integration instead, its moment bound must use the actual same-measure transition and the single existing reserve. Small density mass alone does not extend a source beyond the domain where its analytic definition and bound are known.